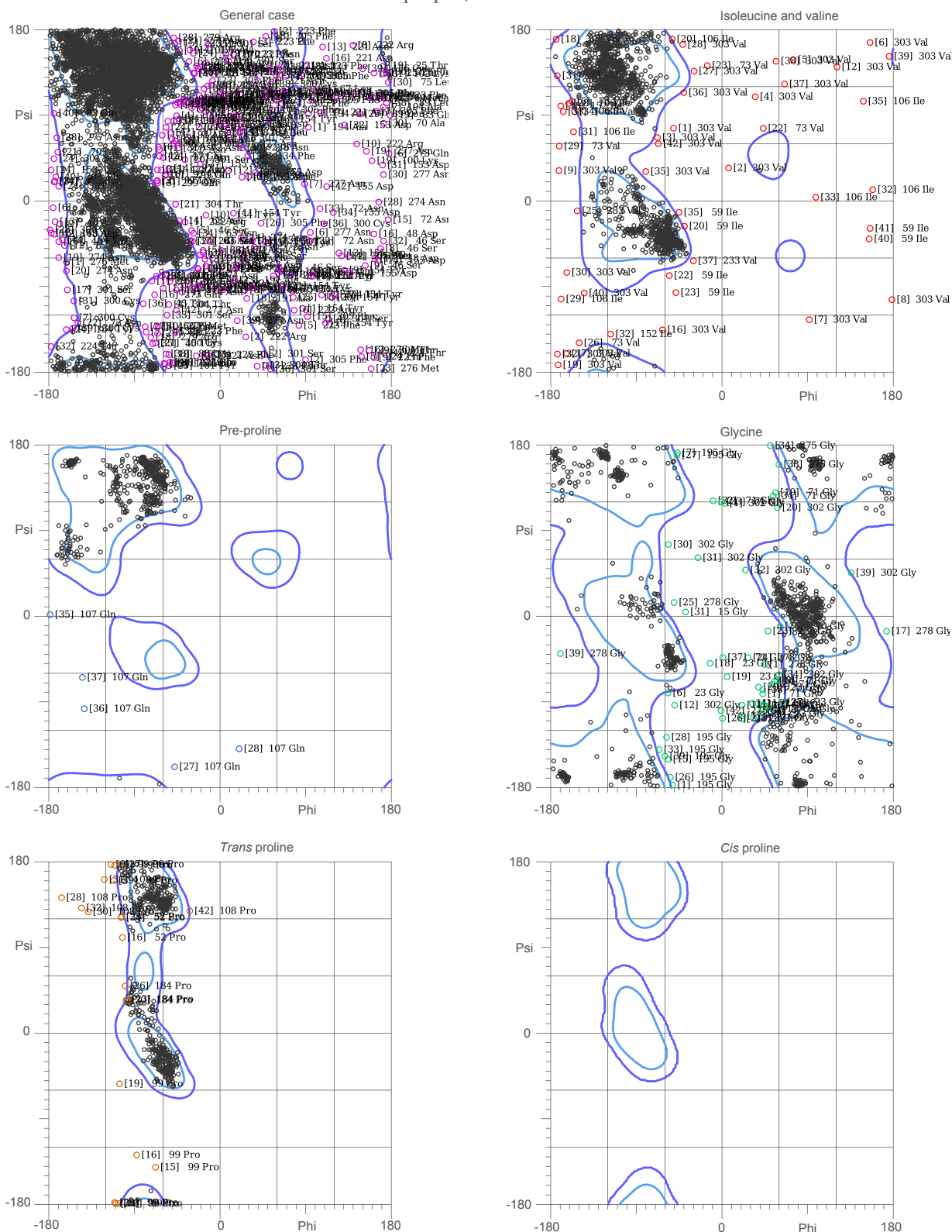


# MolProbity Ramachandran analysis

7mhq.H.pdb, all models



88.3% (11269/12766) of all residues were in favored (98%) regions.  
96.5% (12320/12766) of all residues were in allowed (>99.8%) regions.  
This list is truncated; use the MolProbity multi-chart.html for complete list.

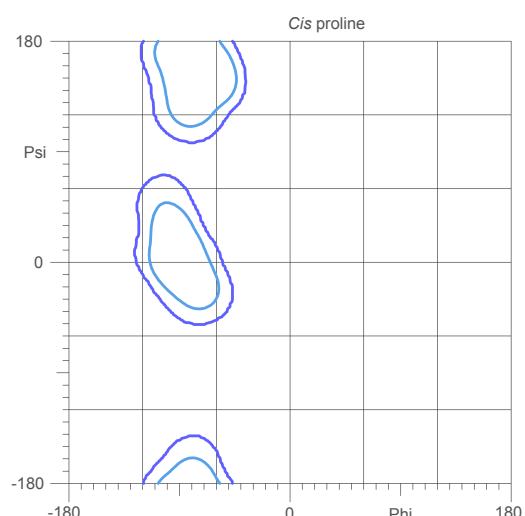
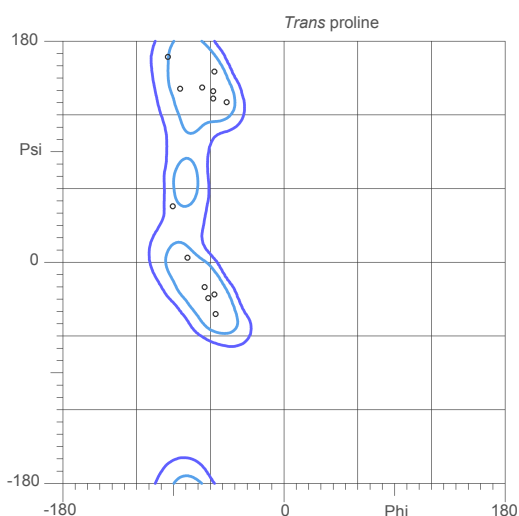
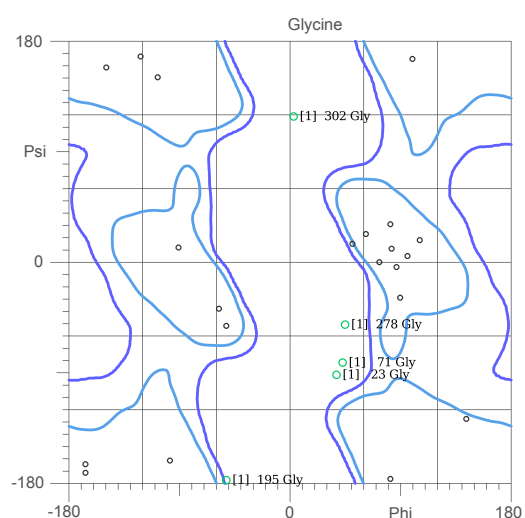
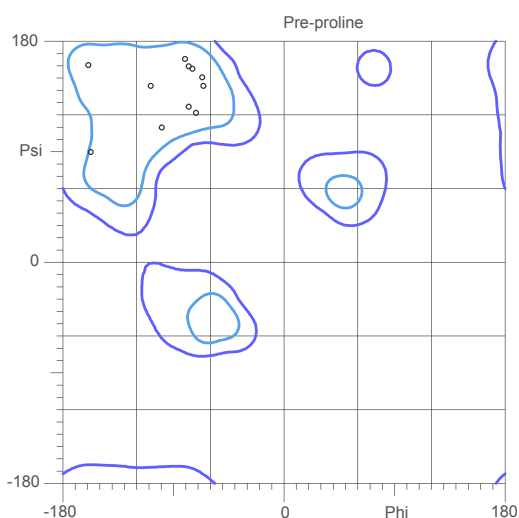
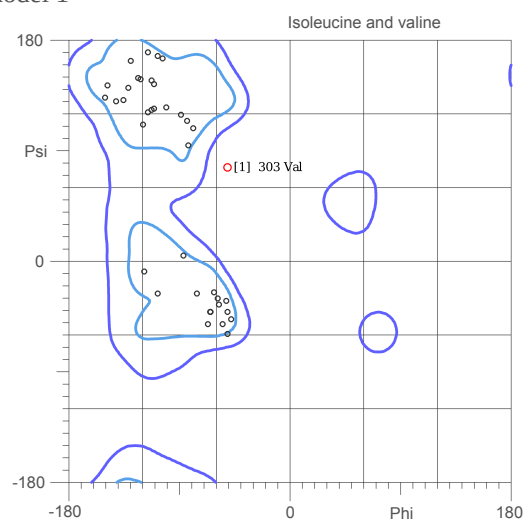
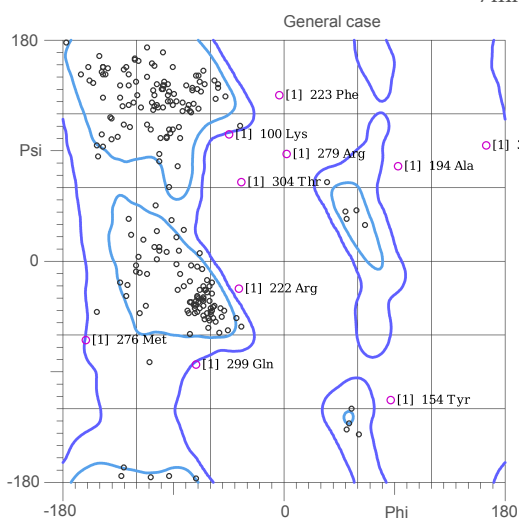
There were 446 outliers (phi, psi):

|                             |                             |                            |
|-----------------------------|-----------------------------|----------------------------|
| [1] 23 Gly (38.2, -92.8)    | [1] 276 Met (-162.7, -64.9) | [2] 23 Gly (21.8, -108.6)  |
| [1] 71 Gly (43.9, -82.9)    | [1] 278 Gly (45.6, -51.1)   | [2] 46 Ser (-169.1, -24.4) |
| [1] 100 Lys (-45.7, 104.4)  | [1] 279 Arg (2.0, 88.5)     | [2] 71 Gly (53.6, -72.7)   |
| [1] 154 Tyr (87.4, -113.3)  | [1] 299 Gln (-72.4, -84.3)  | [2] 97 Lys (-66.4, 22.3)   |
| [1] 194 Ala (93.1, 78.1)    | [1] 302 Gly (3.4, 119.2)    | [2] 194 Ala (18.0, 115.6)  |
| [1] 195 Gly (-52.1, -178.7) | [1] 303 Val (-51.4, 77.2)   | [2] 222 Arg (27.9, -143.3) |
| [1] 222 Arg (-37.0, -22.5)  | [1] 304 Thr (-35.6, 66.0)   | [2] 223 Phe (57.0, 179.6)  |
| [1] 223 Phe (-4.2, 136.2)   | [1] 305 Phe (165.4, 95.7)   | [2] 276 Met (136.4, -58.0) |
|                             |                             | [2] 301 Ser (46.9, -161.2) |
|                             |                             | [2] 303 Val (7.5, 35.1)    |
|                             |                             | [2] 304 Thr (51.6, 138.2)  |
|                             |                             | [2] 305 Phe (90.2, -167.2) |

|                            |                             |                             |
|----------------------------|-----------------------------|-----------------------------|
| [3] 71 Gly (58.9, -68.6)   | [4] 224 Thr (173.3, -160.2) | [5] 154 Tyr (51.5, -44.7)   |
| [3] 154 Tyr (98.3, -79.2)  | [4] 274 Asn (-40.6, -96.8)  | [5] 223 Phe (83.5, -131.5)  |
| [3] 222 Arg (46.0, -91.7)  | [4] 301 Ser (114.5, -125.0) | [5] 224 Thr (149.2, -164.2) |
| [3] 223 Phe (35.1, 168.3)  | [4] 303 Val (35.4, 110.1)   | [5] 279 Arg (0.9, 115.4)    |
| [3] 299 Gln (-66.5, 19.6)  | [4] 304 Thr (-54.4, -109.5) | [5] 303 Val (75.1, 149.8)   |
| [3] 303 Val (-68.4, 67.3)  | [4] 305 Phe (10.0, -60.9)   | [6] 23 Gly (-57.9, -81.4)   |
| [3] 304 Thr (48.6, -173.2) | [5] 46 Ser (-30.8, -31.3)   | [6] 48 Asp (-176.7, -7.8)   |
| [3] 305 Phe (-65.5, 19.6)  | [5] 63 Asn (-18.9, -52.8)   | [6] 67 Leu (98.7, 118.3)    |
| [4] 100 Lys (-46.2, 90.7)  | [5] 66 Phe (-67.1, -132.6)  | [6] 72 Asn (-65.5, 57.4)    |
| [4] 154 Tyr (64.6, -74.7)  | [5] 67 Leu (155.0, 110.7)   | [6] 97 Lys (-57.8, 28.7)    |
| [4] 191 Ala (-25.5, 102.4) | [5] 73 Val (-158.7, 105.7)  | [6] 154 Tyr (74.2, -93.5)   |
| [4] 222 Arg (96.2, -79.1)  | [5] 99 Pro (-111.2, 177.6)  | [6] 222 Arg (78.6, -115.3)  |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 1



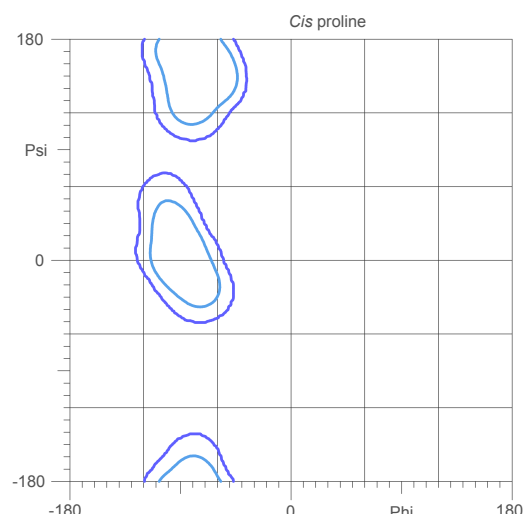
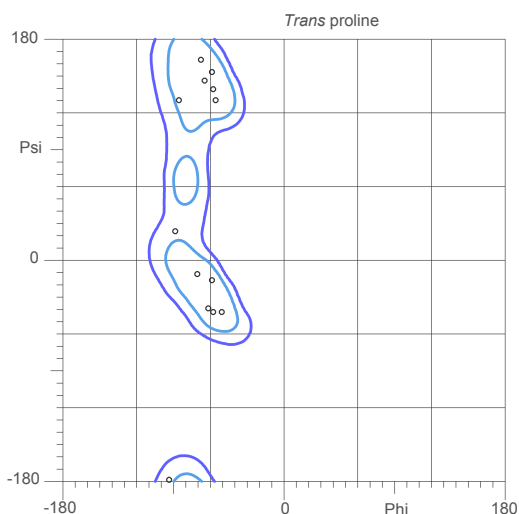
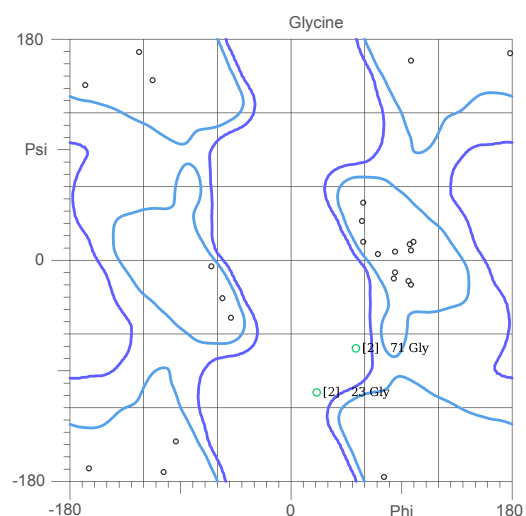
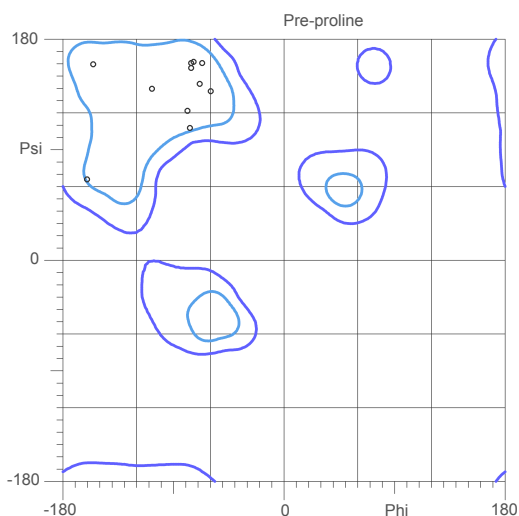
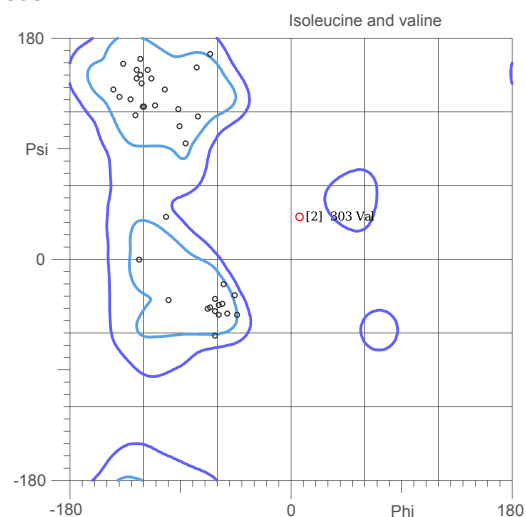
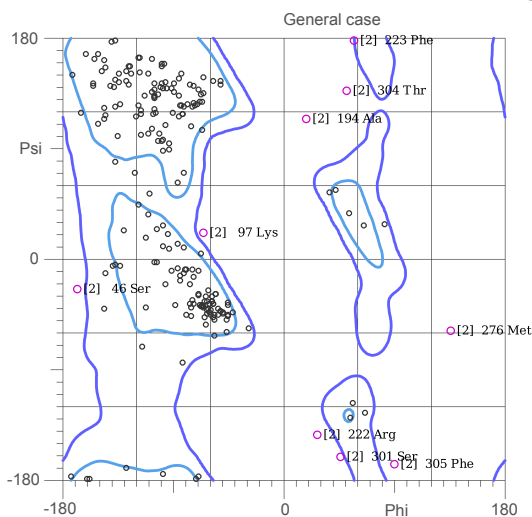
87.8% (267/304) of all residues were in favored (98%) regions.  
94.7% (288/304) of all residues were in allowed (>99.8%) regions.

There were 16 outliers (phi, psi):

|                             |                             |
|-----------------------------|-----------------------------|
| [1] 23 Gly (38.2, -92.8)    | [1] 276 Met (-162.7, -64.9) |
| [1] 71 Gly (43.9, -82.9)    | [1] 278 Gly (45.6, -51.1)   |
| [1] 100 Lys (-45.7, 104.4)  | [1] 279 Arg (2.0, 88.5)     |
| [1] 154 Tyr (87.4, -113.3)  | [1] 299 Gln (-72.4, -84.3)  |
| [1] 194 Ala (93.1, 78.1)    | [1] 302 Gly (3.4, 119.2)    |
| [1] 195 Gly (-52.1, -178.7) | [1] 303 Val (-51.4, 77.2)   |
| [1] 222 Arg (-37.0, -22.5)  | [1] 304 Thr (-35.6, 66.0)   |
| [1] 223 Phe (-4.2, 136.2)   | [1] 305 Phe (165.4, 95.7)   |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 2



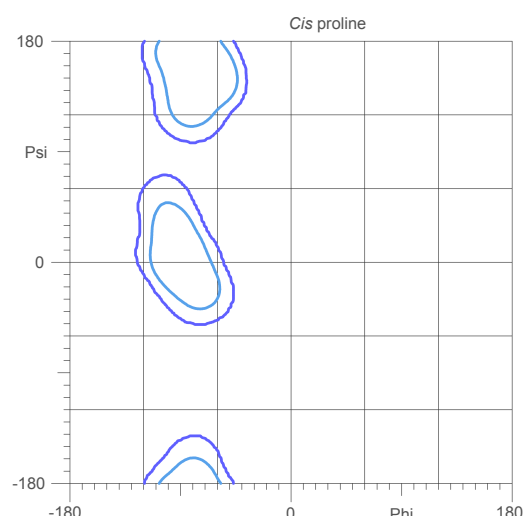
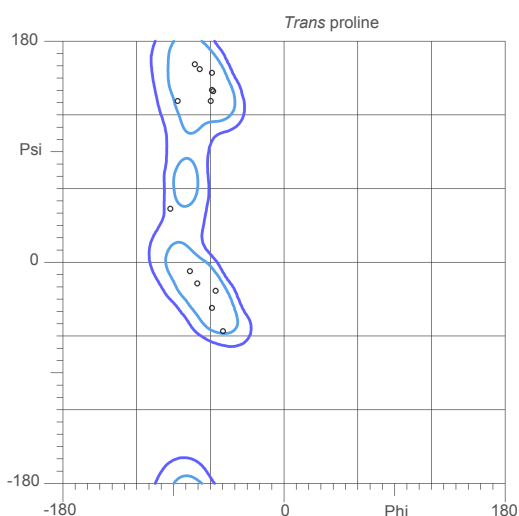
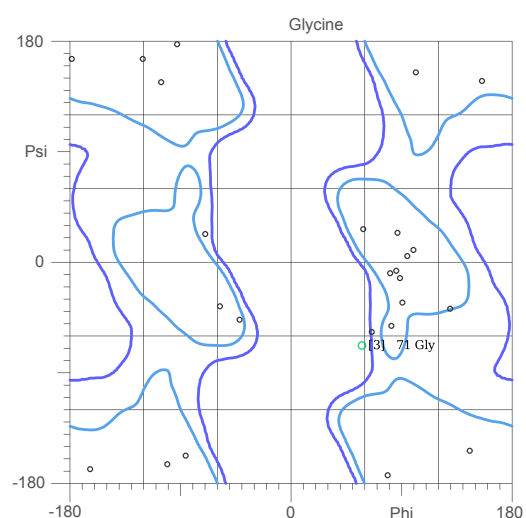
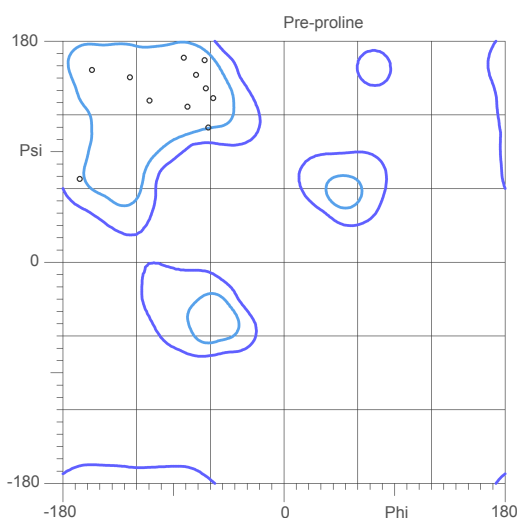
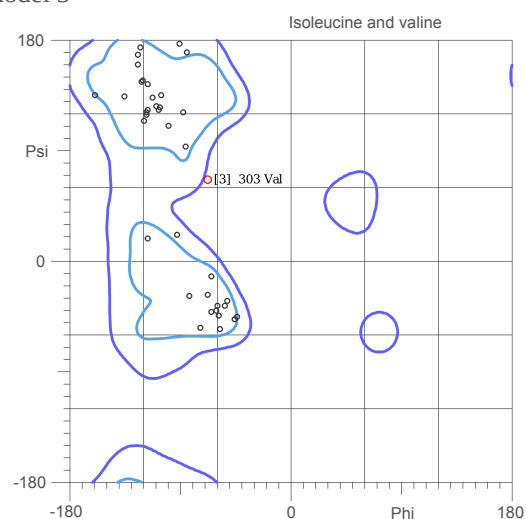
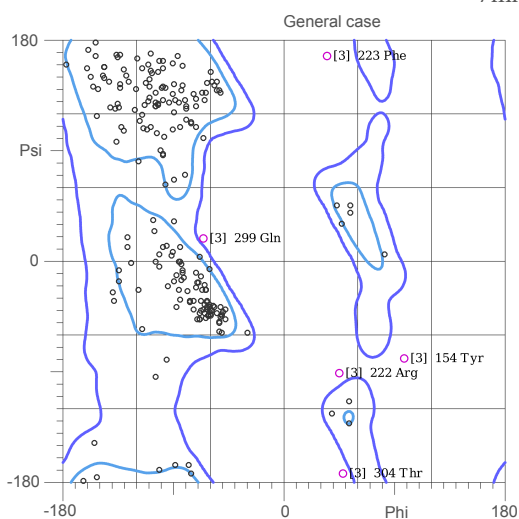
87.5% (266/304) of all residues were in favored (98%) regions.  
96.1% (292/304) of all residues were in allowed (>99.8%) regions.

There were 12 outliers (phi, psi):

|                            |                            |
|----------------------------|----------------------------|
| [2] 23 Gly (21.8, -108.6)  | [2] 301 Ser (46.9, -161.2) |
| [2] 46 Ser (-169.1, -24.4) | [2] 303 Val (7.5, 35.1)    |
| [2] 71 Gly (53.6, -72.7)   | [2] 304 Thr (51.6, 138.2)  |
| [2] 97 Lys (-66.4, 22.3)   | [2] 305 Phe (90.2, -167.2) |
| [2] 194 Ala (18.0, 115.6)  |                            |
| [2] 222 Arg (27.9, -143.3) |                            |
| [2] 223 Phe (57.0, 179.6)  |                            |
| [2] 276 Met (136.4, -58.0) |                            |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 3



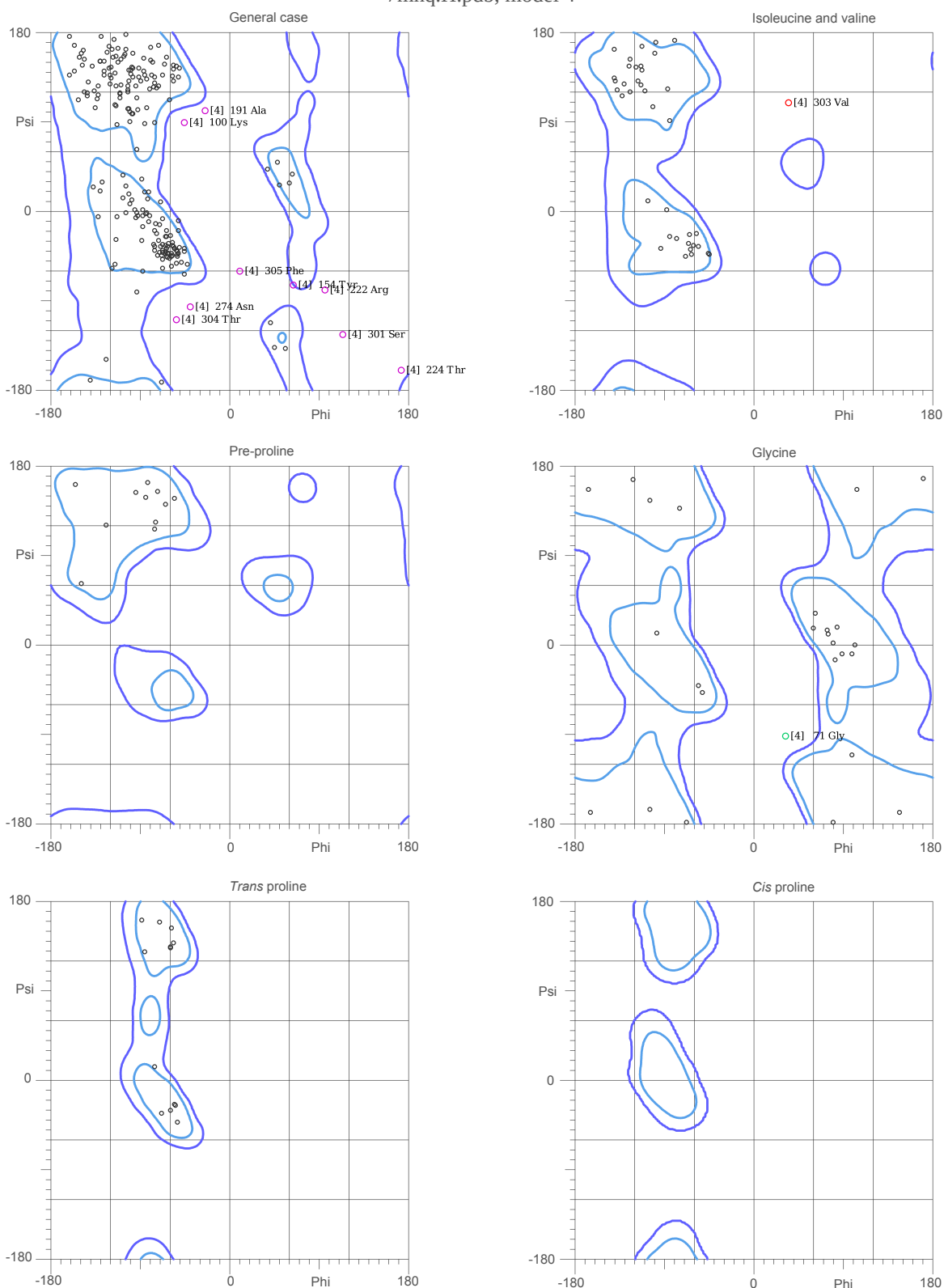
87.8% (267/304) of all residues were in favored (98%) regions.  
97.7% (297/304) of all residues were in allowed (>99.8%) regions.

There were 7 outliers (phi, psi):

[3] 71 Gly (58.9, -68.6)  
[3] 154 Tyr (98.3, -79.2)  
[3] 222 Arg (46.0, -91.7)  
[3] 223 Phe (35.1, 168.3)  
[3] 299 Gln (-66.5, 19.6)  
[3] 303 Val (-68.4, 67.3)  
[3] 304 Thr (48.6, -173.2)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 4



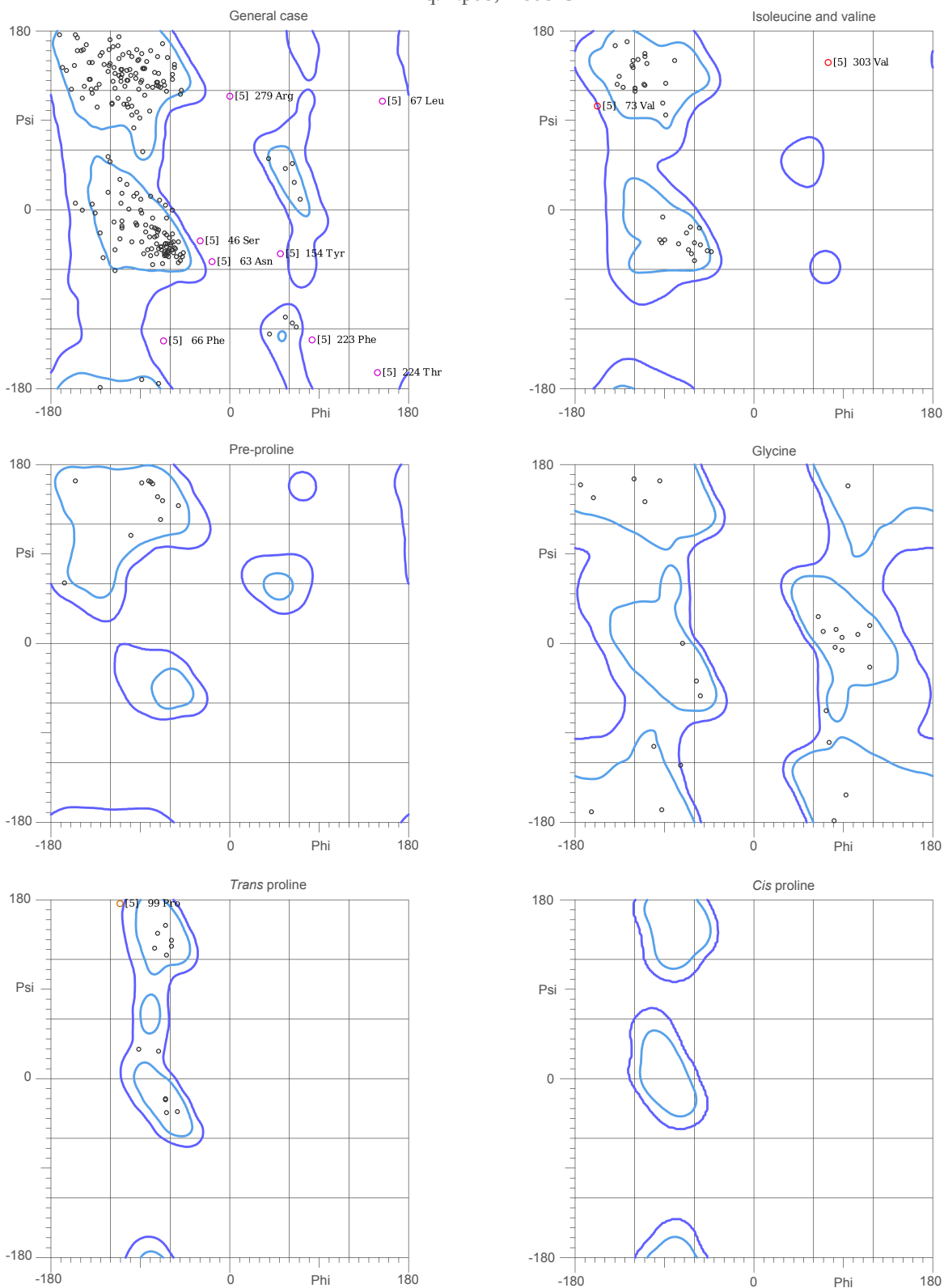
90.8% (276/304) of all residues were in favored (98%) regions.  
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

|                             |                             |
|-----------------------------|-----------------------------|
| [4] 71 Gly (33.0, -92.7)    | [4] 303 Val (35.4, 110.1)   |
| [4] 100 Lys (-46.2, 90.7)   | [4] 304 Thr (-54.4, -109.5) |
| [4] 154 Tyr (64.6, -74.7)   | [4] 305 Phe (10.0, -60.9)   |
| [4] 191 Ala (-25.5, 102.4)  |                             |
| [4] 222 Arg (96.2, -79.1)   |                             |
| [4] 224 Thr (173.3, -160.2) |                             |
| [4] 274 Asn (-40.6, -96.8)  |                             |
| [4] 301 Ser (114.5, -125.0) |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 5



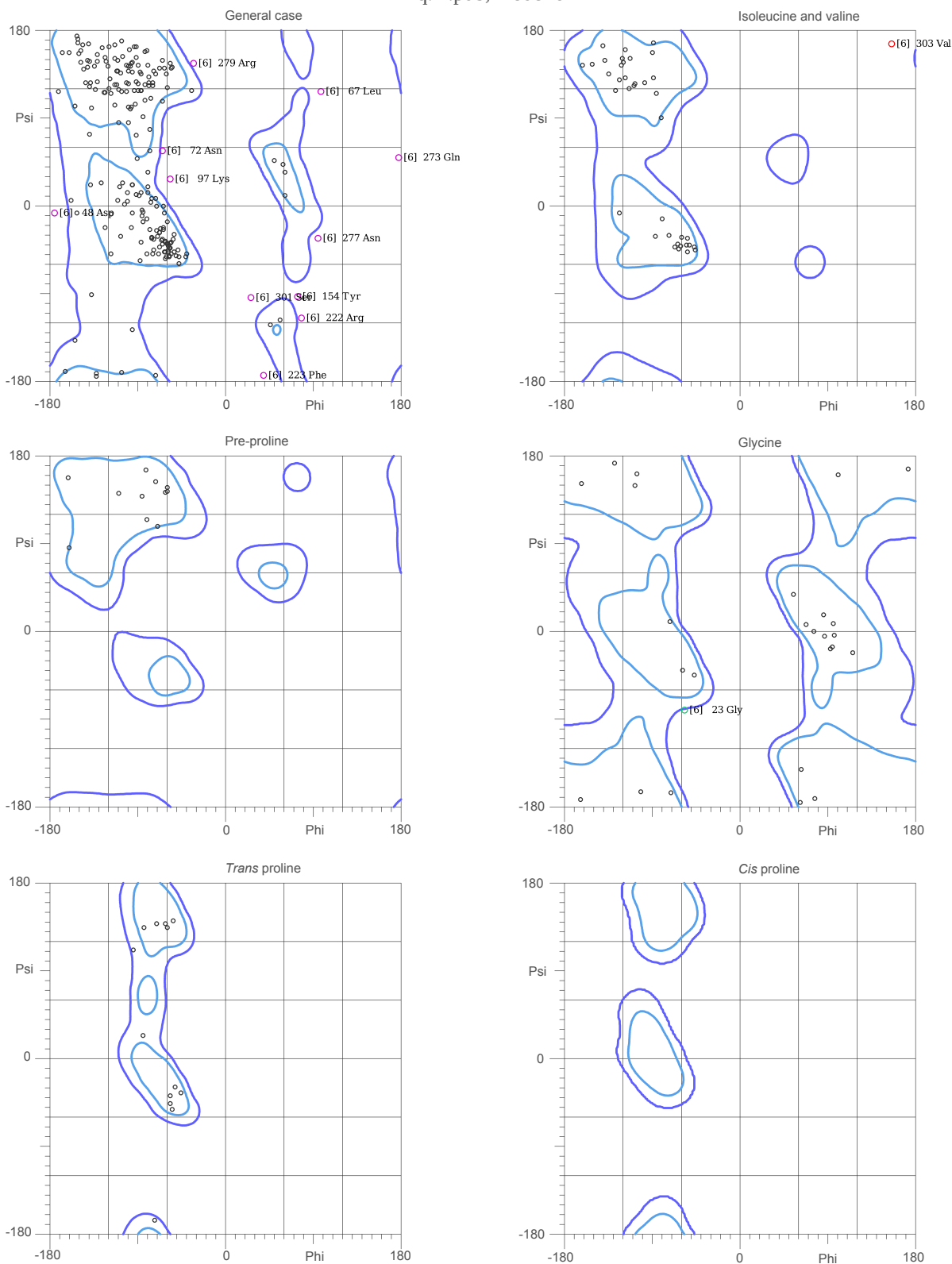
89.5% (272/304) of all residues were in favored (98%) regions.  
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

|                            |                             |
|----------------------------|-----------------------------|
| [5] 46 Ser (-30.8, -31.3)  | [5] 224 Thr (149.2, -164.2) |
| [5] 63 Asn (-18.9, -52.8)  | [5] 279 Arg (0.9, 115.4)    |
| [5] 66 Phe (-67.1, -132.6) | [5] 303 Val (75.1, 149.8)   |
| [5] 67 Leu (155.0, 110.7)  |                             |
| [5] 73 Val (-158.7, 105.7) |                             |
| [5] 99 Pro (-111.2, 177.6) |                             |
| [5] 154 Tyr (51.5, -44.7)  |                             |
| [5] 223 Phe (83.5, -131.5) |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 6



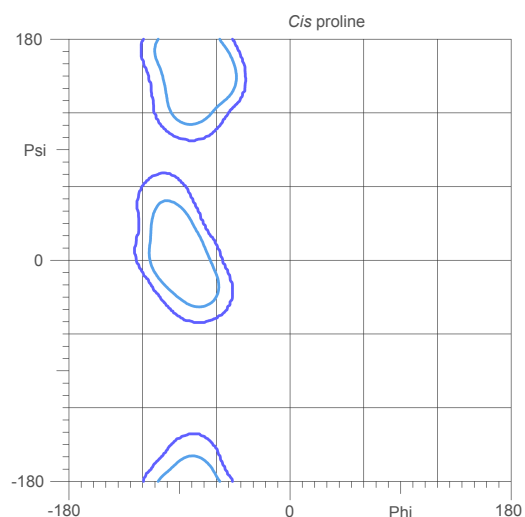
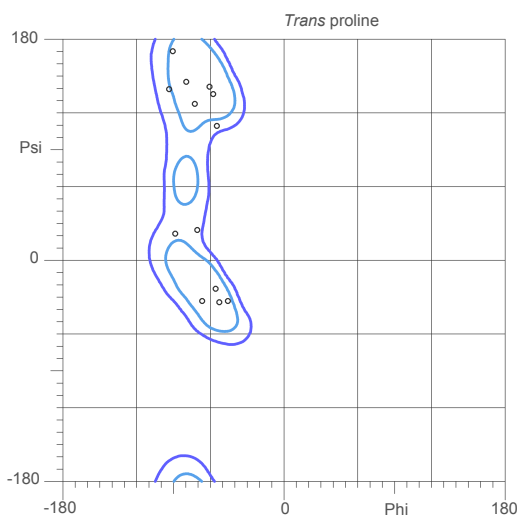
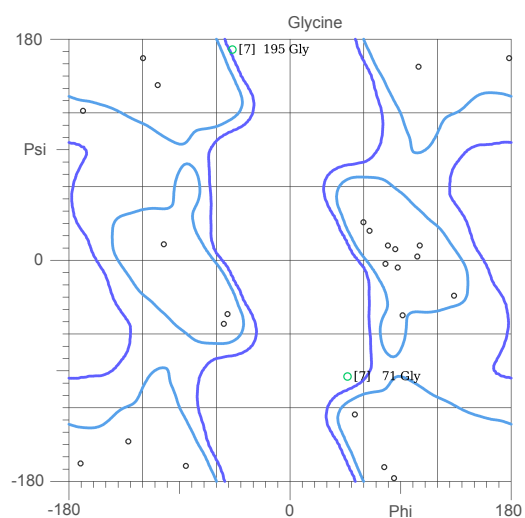
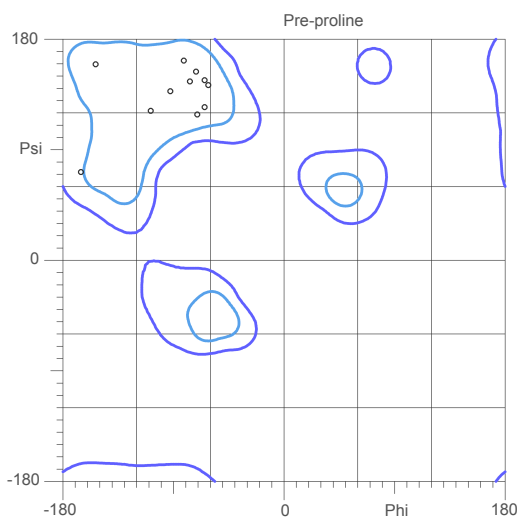
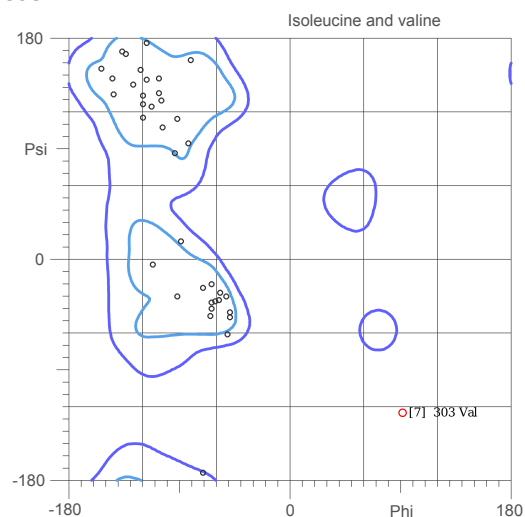
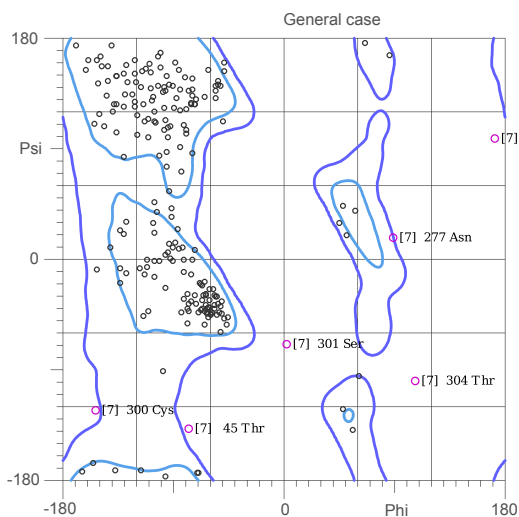
88.2% (268/304) of all residues were in favored (98%) regions.  
95.7% (291/304) of all residues were in allowed (>99.8%) regions.

There were 13 outliers (phi, psi):

|                            |                            |
|----------------------------|----------------------------|
| [6] 23 Gly (-57.9, -81.4)  | [6] 273 Gln (178.5, 50.5)  |
| [6] 48 Asp (-176.7, -7.8)  | [6] 277 Asn (95.1, -33.1)  |
| [6] 67 Leu (98.7, 118.3)   | [6] 279 Arg (-33.4, 147.7) |
| [6] 72 Asn (-65.5, 57.4)   | [6] 301 Ser (26.4, -94.4)  |
| [6] 97 Lys (-57.8, 28.7)   | [6] 303 Val (157.0, 167.1) |
| [6] 154 Tyr (74.2, -93.5)  |                            |
| [6] 222 Arg (78.6, -115.3) |                            |
| [6] 223 Phe (39.8, -174.1) |                            |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 7



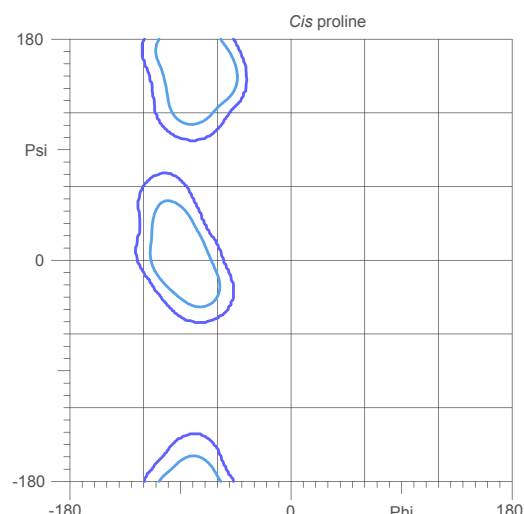
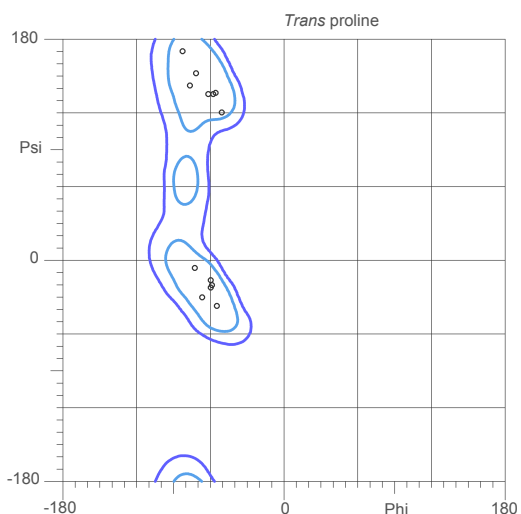
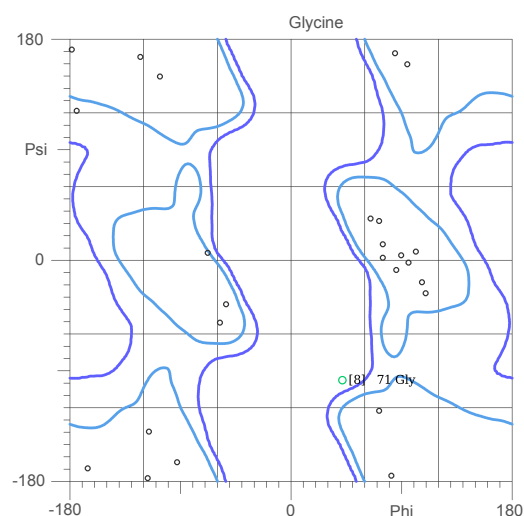
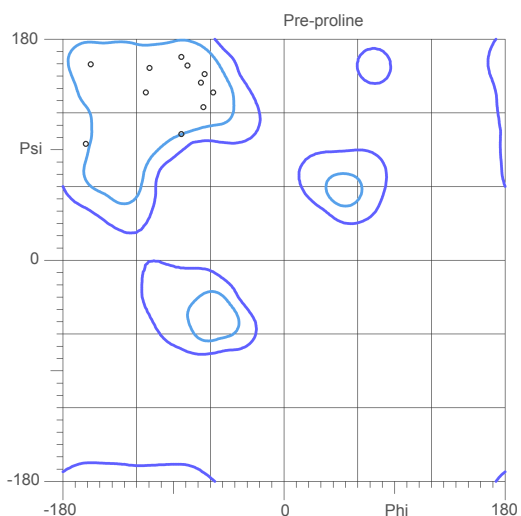
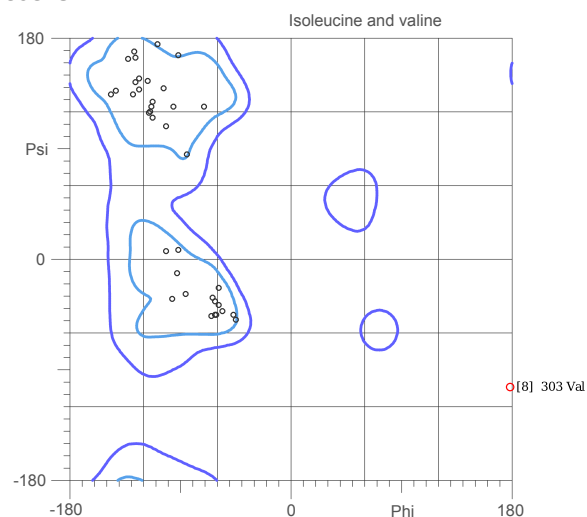
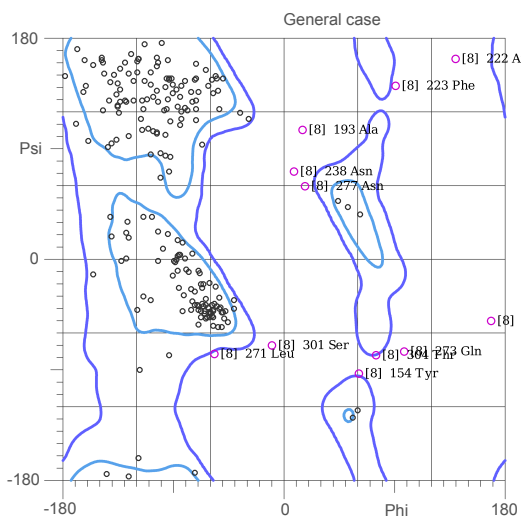
86.8% (264/304) of all residues were in favored (98%) regions.  
97.0% (295/304) of all residues were in allowed (>99.8%) regions.

There were 9 outliers (phi, psi):

|                              |                            |
|------------------------------|----------------------------|
| [7] 45 Thr (-79.0, -138.1)   | [7] 304 Thr (107.1, -99.9) |
| [7] 71 Gly (47.5, -95.0)     |                            |
| [7] 194 Ala (172.2, 99.4)    |                            |
| [7] 195 Gly (-47.2, 172.6)   |                            |
| [7] 277 Asn (89.4, 18.7)     |                            |
| [7] 300 Cys (-154.0, -124.0) |                            |
| [7] 301 Ser (2.3, -69.2)     |                            |
| [7] 303 Val (92.4, -125.2)   |                            |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 8



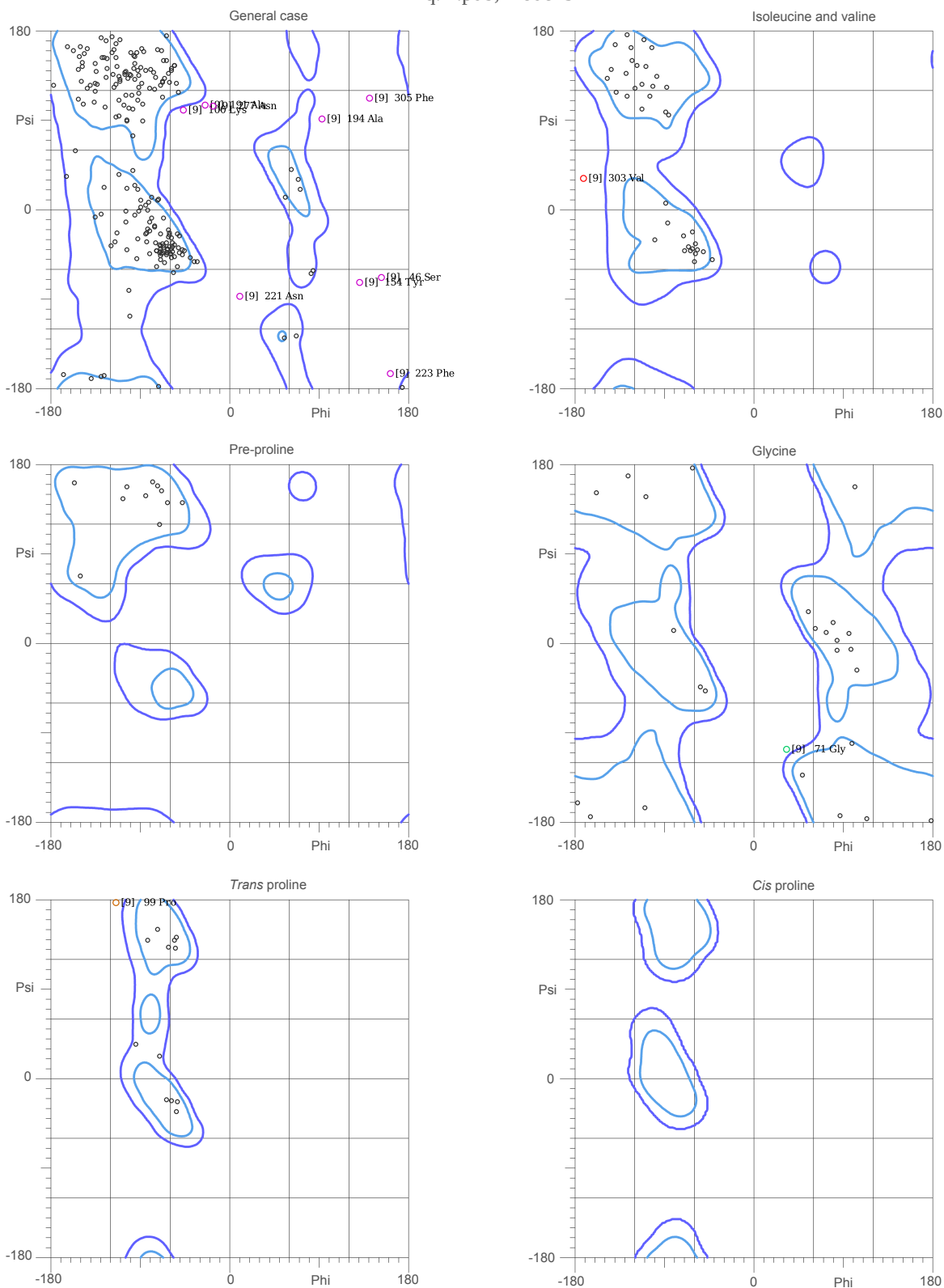
88.8% (270/304) of all residues were in favored (98%) regions.  
95.7% (291/304) of all residues were in allowed (>99.8%) regions.

There were 13 outliers (phi, psi):

|                            |                             |
|----------------------------|-----------------------------|
| [8] 46 Ser (169.2, -50.1)  | [8] 273 Gln (98.2, -76.0)   |
| [8] 71 Gly (42.7, -98.7)   | [8] 277 Asn (17.8, 60.6)    |
| [8] 154 Tyr (61.0, -93.5)  | [8] 301 Ser (-10.7, -70.2)  |
| [8] 193 Ala (15.6, 106.3)  | [8] 303 Val (179.8, -104.1) |
| [8] 222 Arg (140.7, 164.9) | [8] 304 Thr (75.6, -78.5)   |
| [8] 223 Phe (91.8, 142.7)  |                             |
| [8] 238 Asn (8.1, 72.7)    |                             |
| [8] 271 Leu (-57.9, -77.3) |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 9



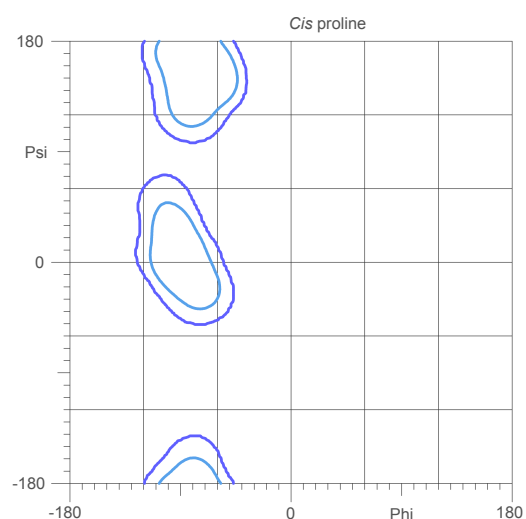
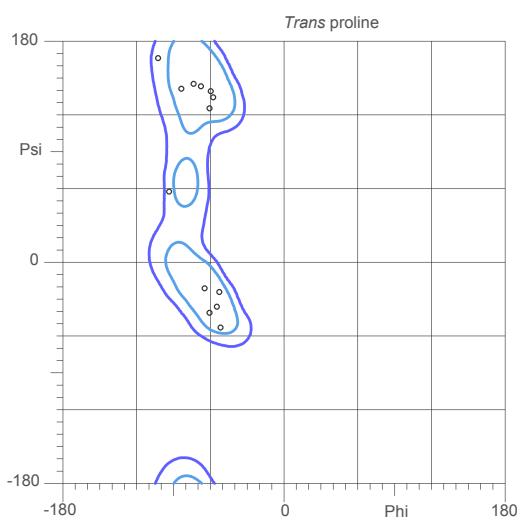
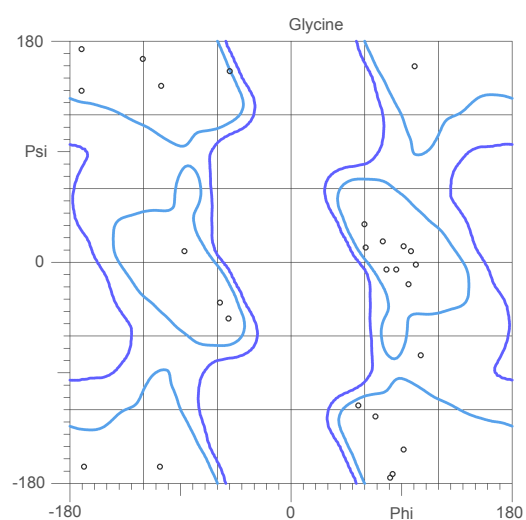
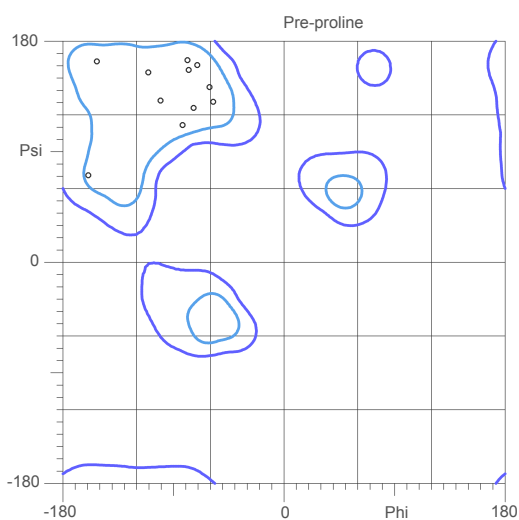
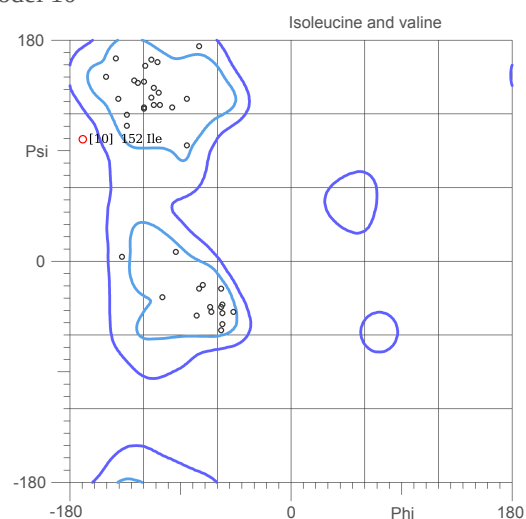
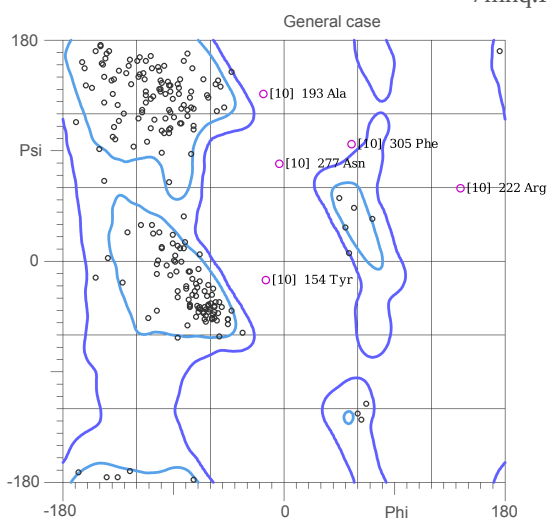
87.8% (267/304) of all residues were in favored (98%) regions.  
96.1% (292/304) of all residues were in allowed (>99.8%) regions.

There were 12 outliers (phi, psi):

|                            |                             |
|----------------------------|-----------------------------|
| [9] 46 Ser (153.1, -68.7)  | [9] 223 Phe (162.2, -165.2) |
| [9] 71 Gly (33.3, -107.6)  | [9] 277 Asn (-17.4, 105.8)  |
| [9] 99 Pro (-115.8, 178.4) | [9] 303 Val (-172.4, 32.6)  |
| [9] 100 Lys (-47.6, 101.6) | [9] 305 Phe (141.3, 113.0)  |
| [9] 154 Tyr (131.9, -73.3) |                             |
| [9] 191 Ala (-25.9, 106.1) |                             |
| [9] 194 Ala (93.7, 92.0)   |                             |
| [9] 221 Asn (10.6, -87.5)  |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 10



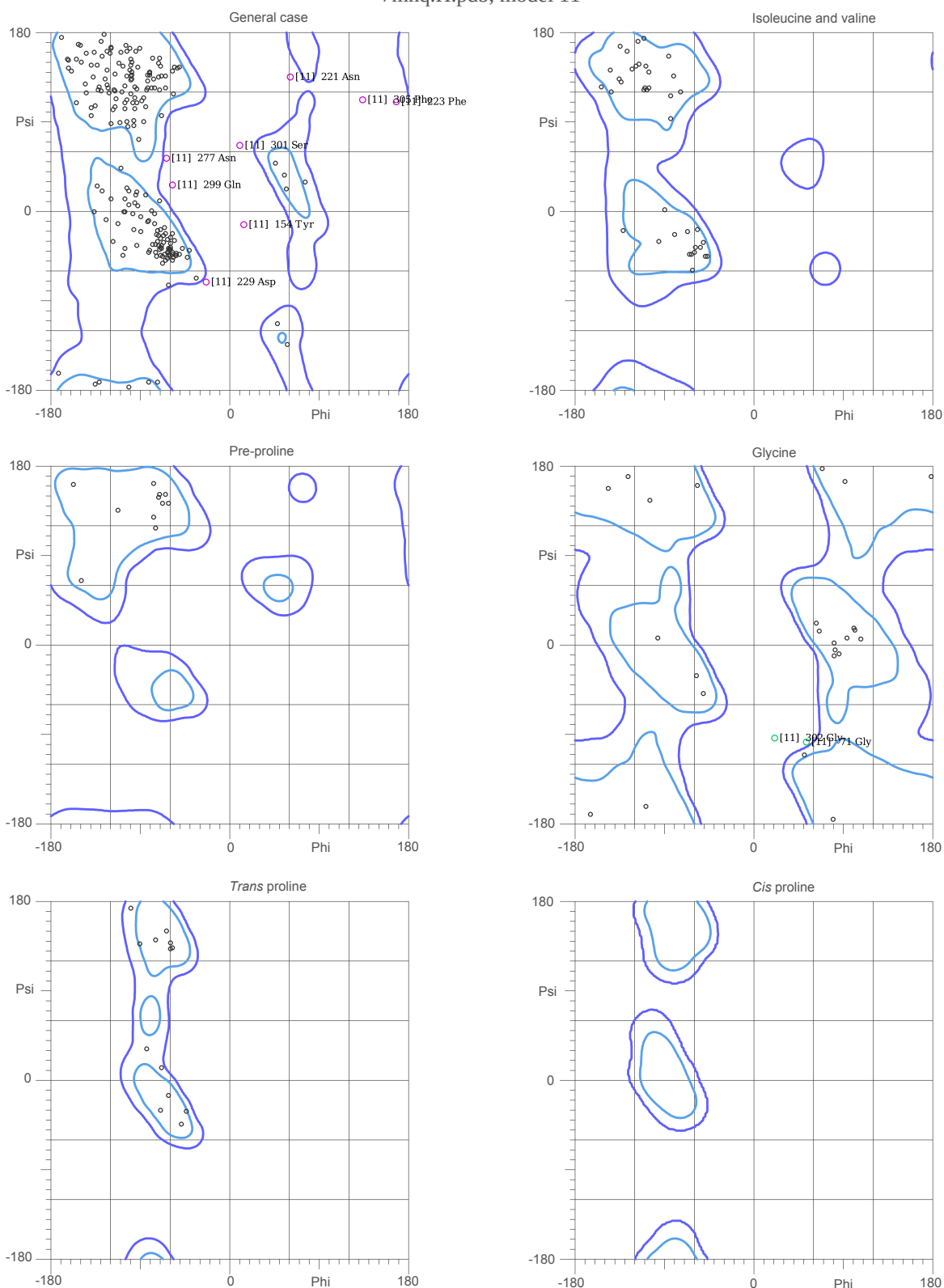
90.1% (274/304) of all residues were in favored (98%) regions.  
98.0% (298/304) of all residues were in allowed (>99.8%) regions.

There were 6 outliers (phi, psi):

[10] 152 Ile (-170.7, 101.0)  
[10] 154 Tyr (-15.5, -15.4)  
[10] 193 Ala (-17.7, 137.6)  
[10] 222 Arg (144.8, 61.0)  
[10] 277 Asn (-4.1, 80.2)  
[10] 305 Phe (55.4, 96.8)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 11



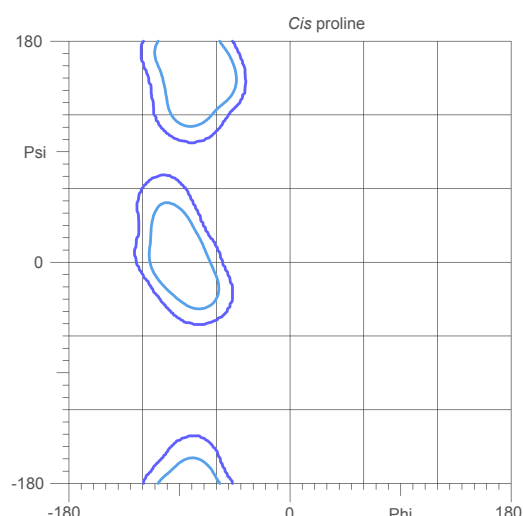
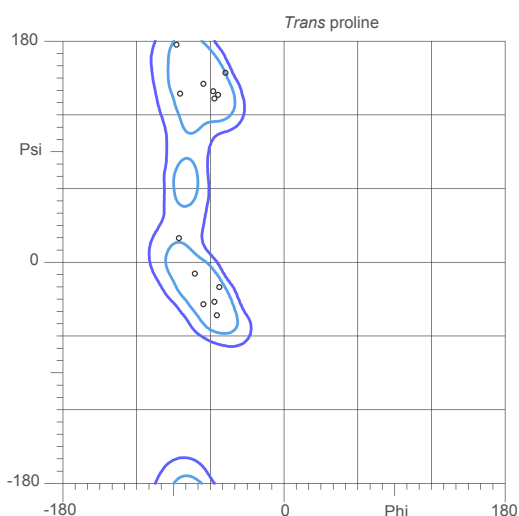
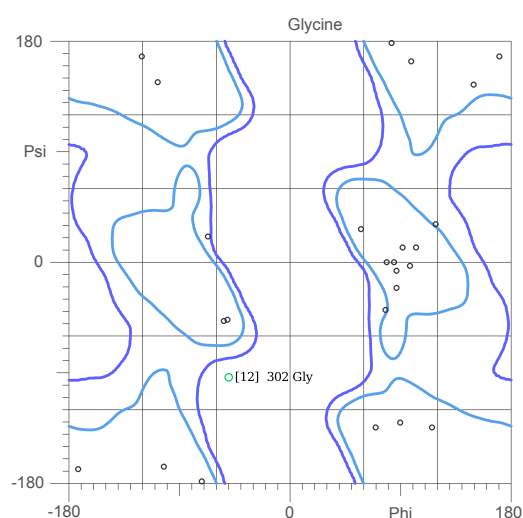
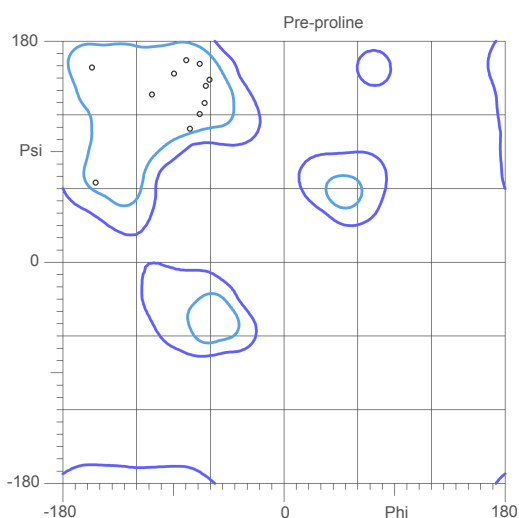
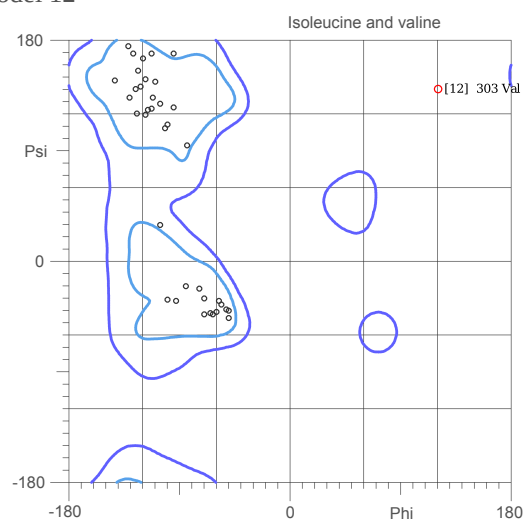
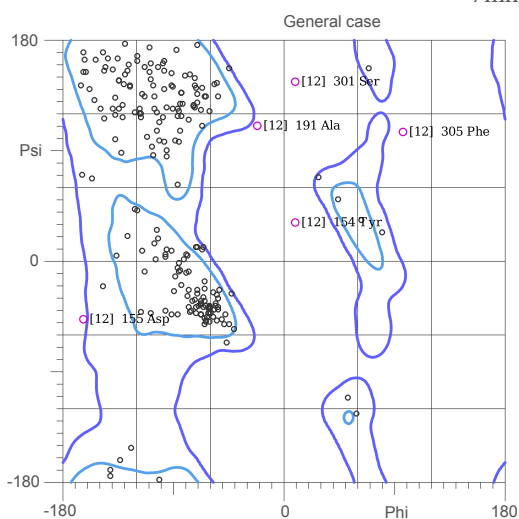
90.8% (276/304) of all residues were in favored (98%) regions.  
96.7% (294/304) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

|                             |                             |
|-----------------------------|-----------------------------|
| [11] 71 Gly (53.2, -98.1)   | [11] 302 Gly (21.1, -94.8)  |
| [11] 154 Tyr (14.0, -13.2)  | [11] 305 Phe (134.9, 113.5) |
| [11] 221 Asn (61.9, 136.3)  |                             |
| [11] 223 Phe (168.9, 111.9) |                             |
| [11] 229 Asp (-24.8, -71.7) |                             |
| [11] 277 Asn (-64.7, 54.8)  |                             |
| [11] 299 Gln (-58.5, 27.1)  |                             |
| [11] 301 Ser (10.5, 67.8)   |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 12



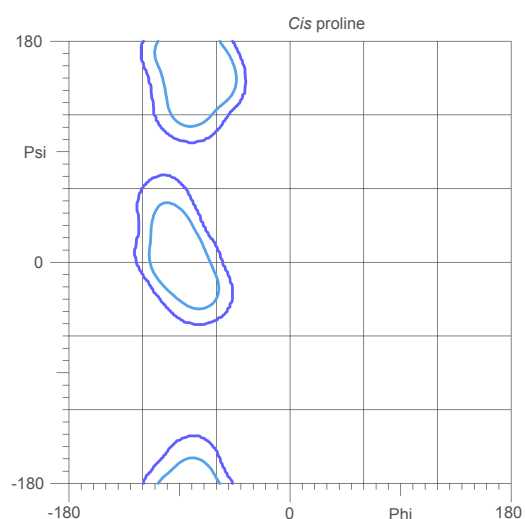
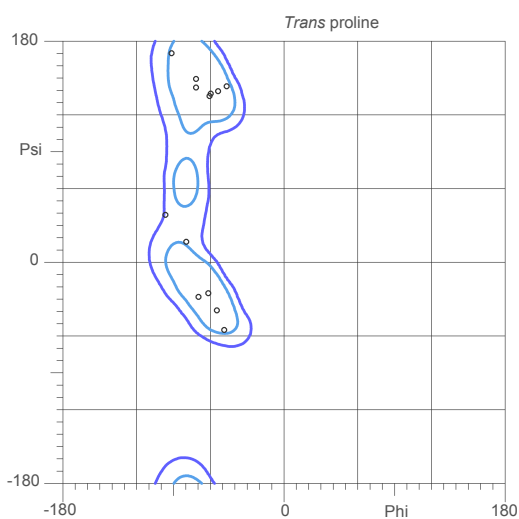
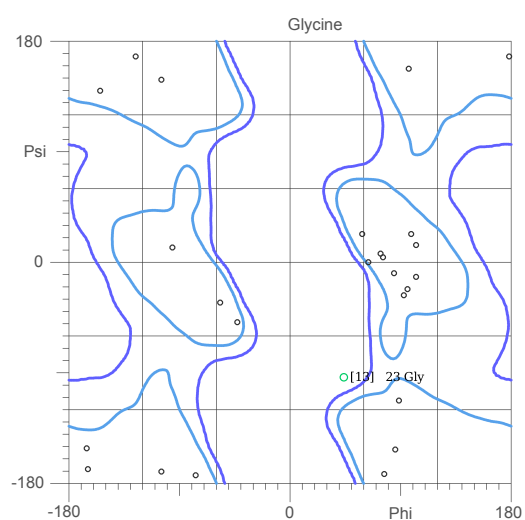
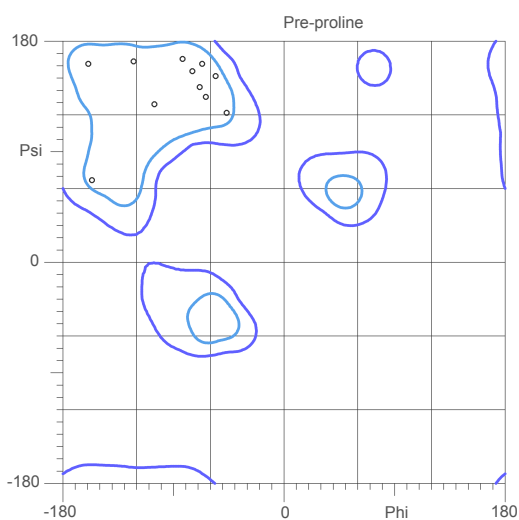
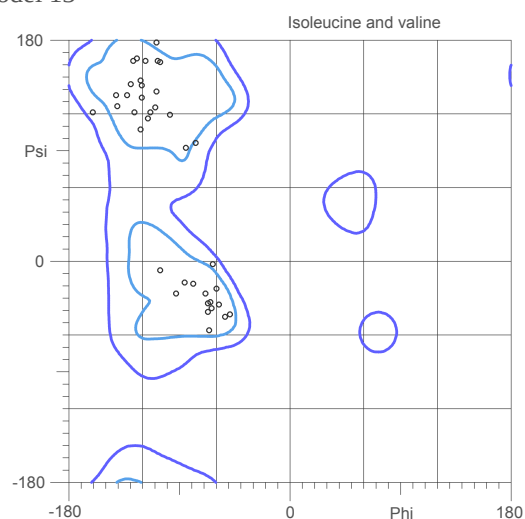
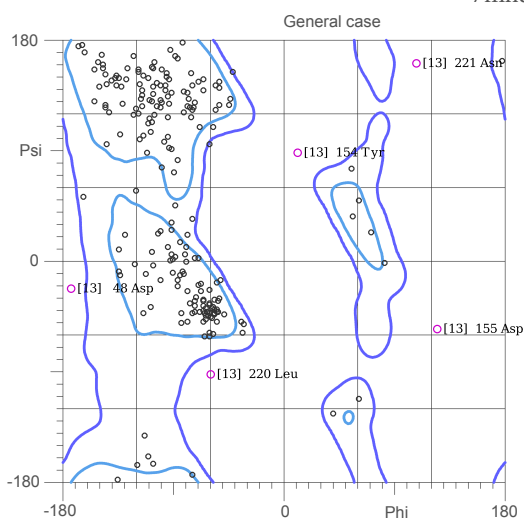
89.8% (273/304) of all residues were in favored (98%) regions.  
97.7% (297/304) of all residues were in allowed (>99.8%) regions.

There were 7 outliers (phi, psi):

- [12] 154 Tyr (9.5, 32.8)
- [12] 155 Asp (-164.7, -47.5)
- [12] 191 Ala (-22.2, 111.5)
- [12] 301 Ser (9.1, 148.0)
- [12] 302 Gly (-50.8, -94.8)
- [12] 303 Val (121.4, 141.4)
- [12] 305 Phe (97.0, 106.1)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 13



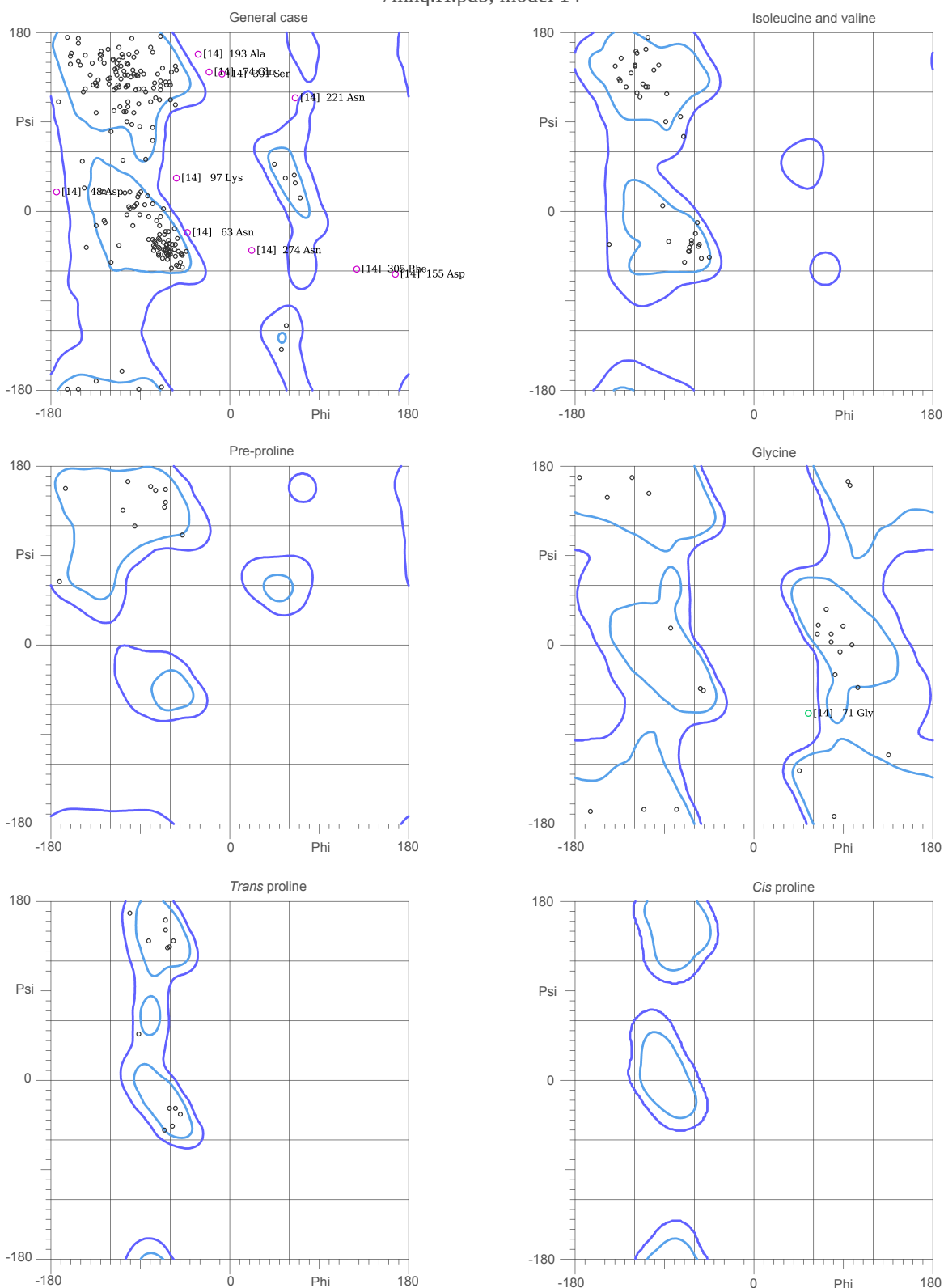
88.8% (270/304) of all residues were in favored (98%) regions.  
98.0% (298/304) of all residues were in allowed (>99.8%) regions.

There were 6 outliers (phi, psi):

- [13] 23 Gly (44.6, -94.7)
- [13] 48 Asp (-174.2, -22.8)
- [13] 154 Tyr (11.2, 89.7)
- [13] 155 Asp (125.3, -56.0)
- [13] 220 Leu (-60.2, -92.2)
- [13] 221 Asn (108.5, 162.1)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 14



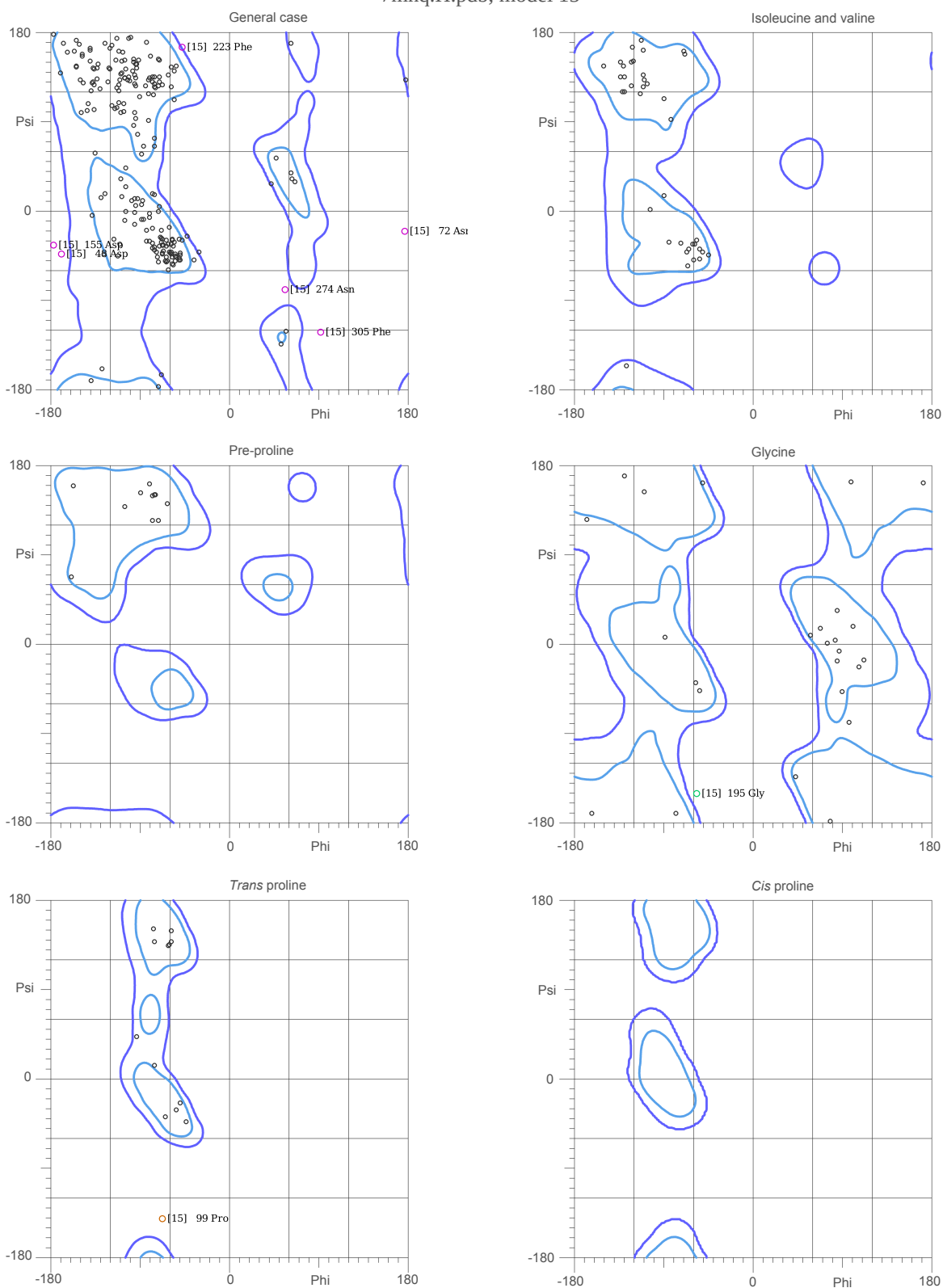
88.5% (269/304) of all residues were in favored (98%) regions.  
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

|                             |                             |
|-----------------------------|-----------------------------|
| [14] 48 Asp (-175.2, 20.8)  | [14] 274 Asn (23.0, -39.0)  |
| [14] 63 Asn (-43.5, -21.6)  | [14] 301 Ser (-8.2, 139.6)  |
| [14] 71 Gly (55.1, -69.7)   | [14] 305 Phe (128.4, -58.5) |
| [14] 74 Gln (-21.9, 141.9)  |                             |
| [14] 97 Lys (-54.3, 34.2)   |                             |
| [14] 155 Asp (167.5, -63.5) |                             |
| [14] 193 Ala (-32.3, 159.4) |                             |
| [14] 221 Asn (66.9, 115.8)  |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 15



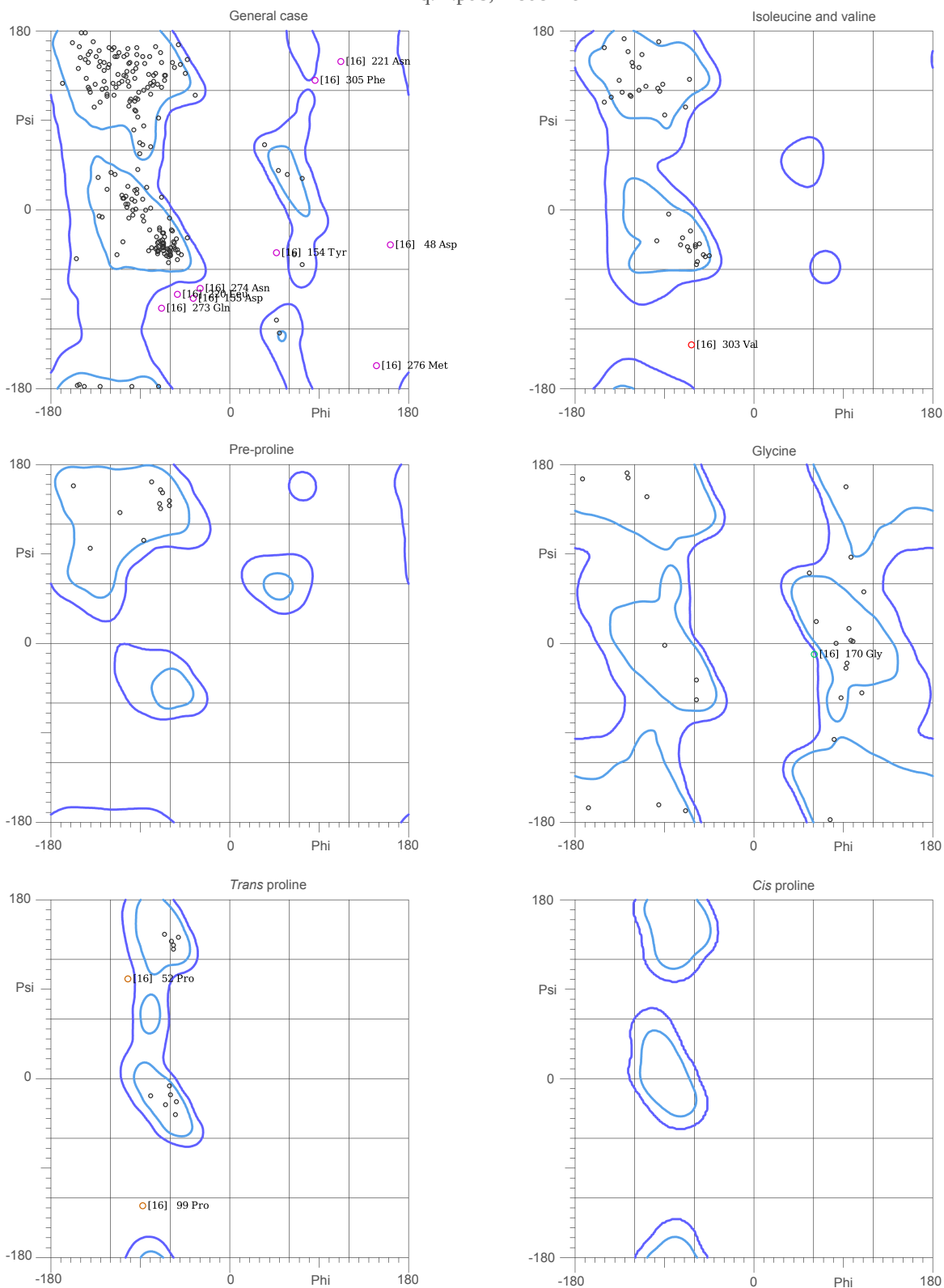
88.5% (269/304) of all residues were in favored (98%) regions.  
97.4% (296/304) of all residues were in allowed (>99.8%) regions.

There were 8 outliers (phi, psi):

[15] 48 Asp (-170.5, -43.9)  
[15] 72 Asn (177.0, -20.5)  
[15] 99 Pro (-68.0, -141.0)  
[15] 155 Asp (-178.8, -34.5)  
[15] 195 Gly (-57.7, -151.7)  
[15] 223 Phe (-48.1, 166.3)  
[15] 274 Asn (57.0, -79.8)  
[15] 305 Phe (92.9, -122.9)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 16



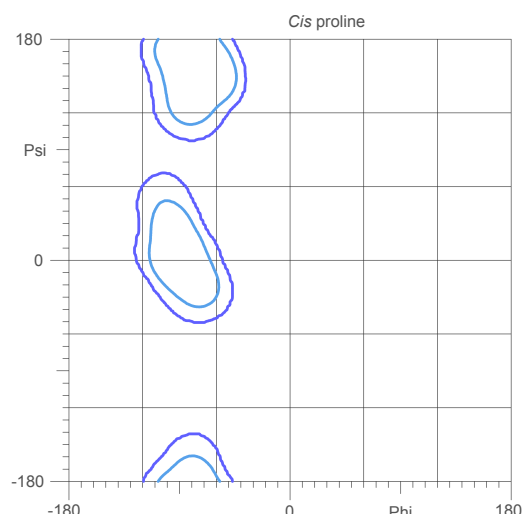
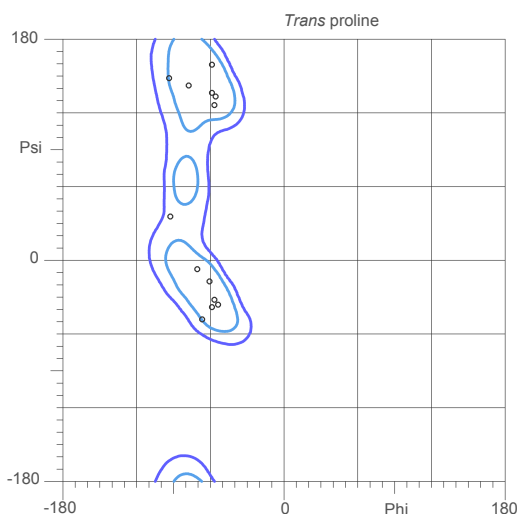
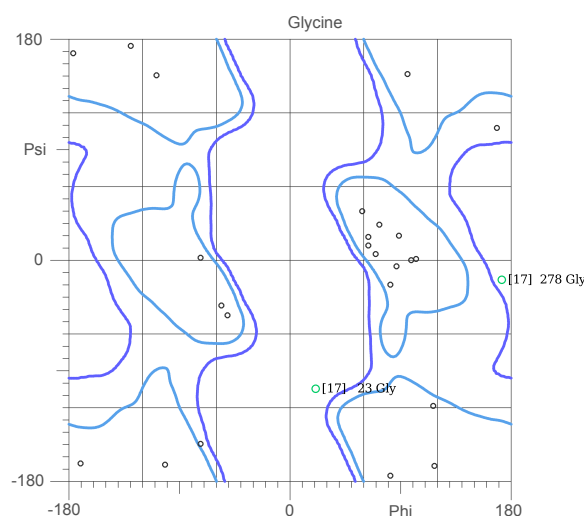
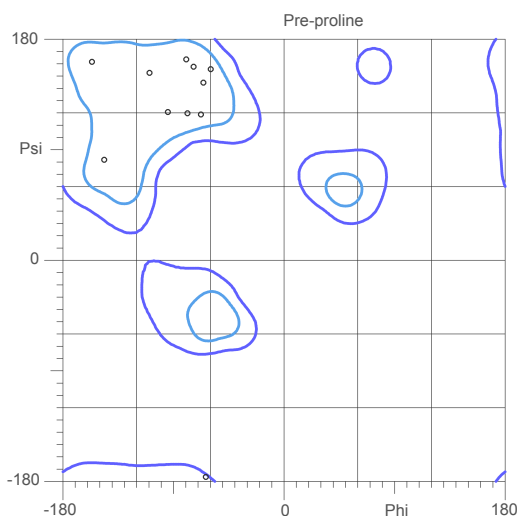
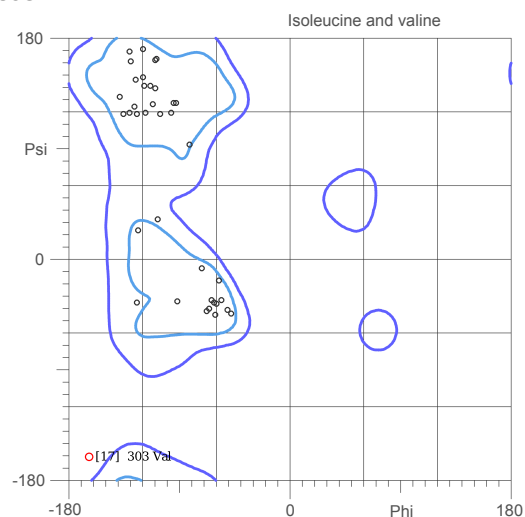
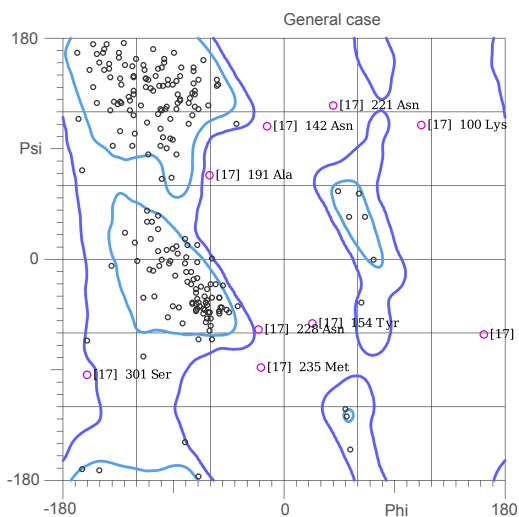
87.8% (267/304) of all residues were in favored (98%) regions.  
95.7% (291/304) of all residues were in allowed (>99.8%) regions.

There were 13 outliers (phi, psi):

|                             |                              |
|-----------------------------|------------------------------|
| [16] 48 Asp (162.8, -35.0)  | [16] 273 Gln (-69.9, -99.3)  |
| [16] 52 Pro (-103.0, 101.5) | [16] 274 Asn (-30.6, -79.7)  |
| [16] 99 Pro (-89.0, -128.5) | [16] 276 Met (148.3, -157.4) |
| [16] 154 Tyr (47.7, -43.9)  | [16] 303 Val (-63.1, -136.8) |
| [16] 155 Asp (-37.0, -89.1) | [16] 305 Phe (86.5, 131.1)   |
| [16] 170 Gly (61.0, -11.6)  |                              |
| [16] 220 Leu (-53.7, -85.9) |                              |
| [16] 221 Asn (112.2, 150.5) |                              |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 17



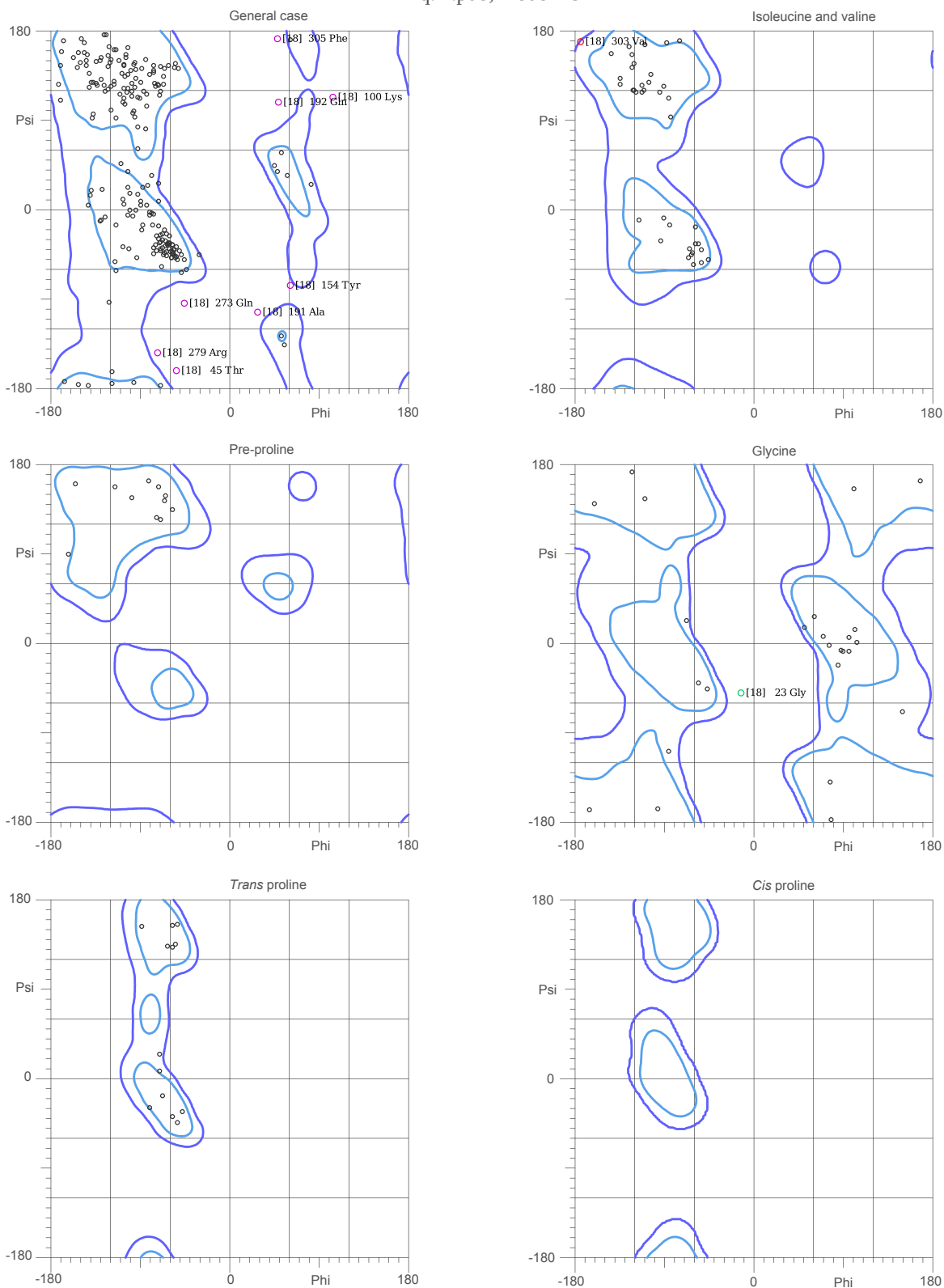
87.2% (265/304) of all residues were in favored (98%) regions.  
96.1% (292/304) of all residues were in allowed (>99.8%) regions.

There were 12 outliers (phi, psi):

|                             |                               |
|-----------------------------|-------------------------------|
| [17] 23 Gly (21.3, -105.5)  | [17] 235 Met (-19.0, -89.0)   |
| [17] 48 Asp (163.7, -62.0)  | [17] 278 Gly (173.7, -16.4)   |
| [17] 100 Lys (112.1, 110.6) | [17] 301 Ser (-161.6, -94.9)  |
| [17] 142 Asn (-14.4, 109.8) | [17] 303 Val (-164.8, -161.7) |
| [17] 154 Tyr (23.7, -52.4)  |                               |
| [17] 191 Ala (-61.3, 69.2)  |                               |
| [17] 221 Asn (40.4, 126.9)  |                               |
| [17] 228 Asn (-21.5, -57.6) |                               |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 18



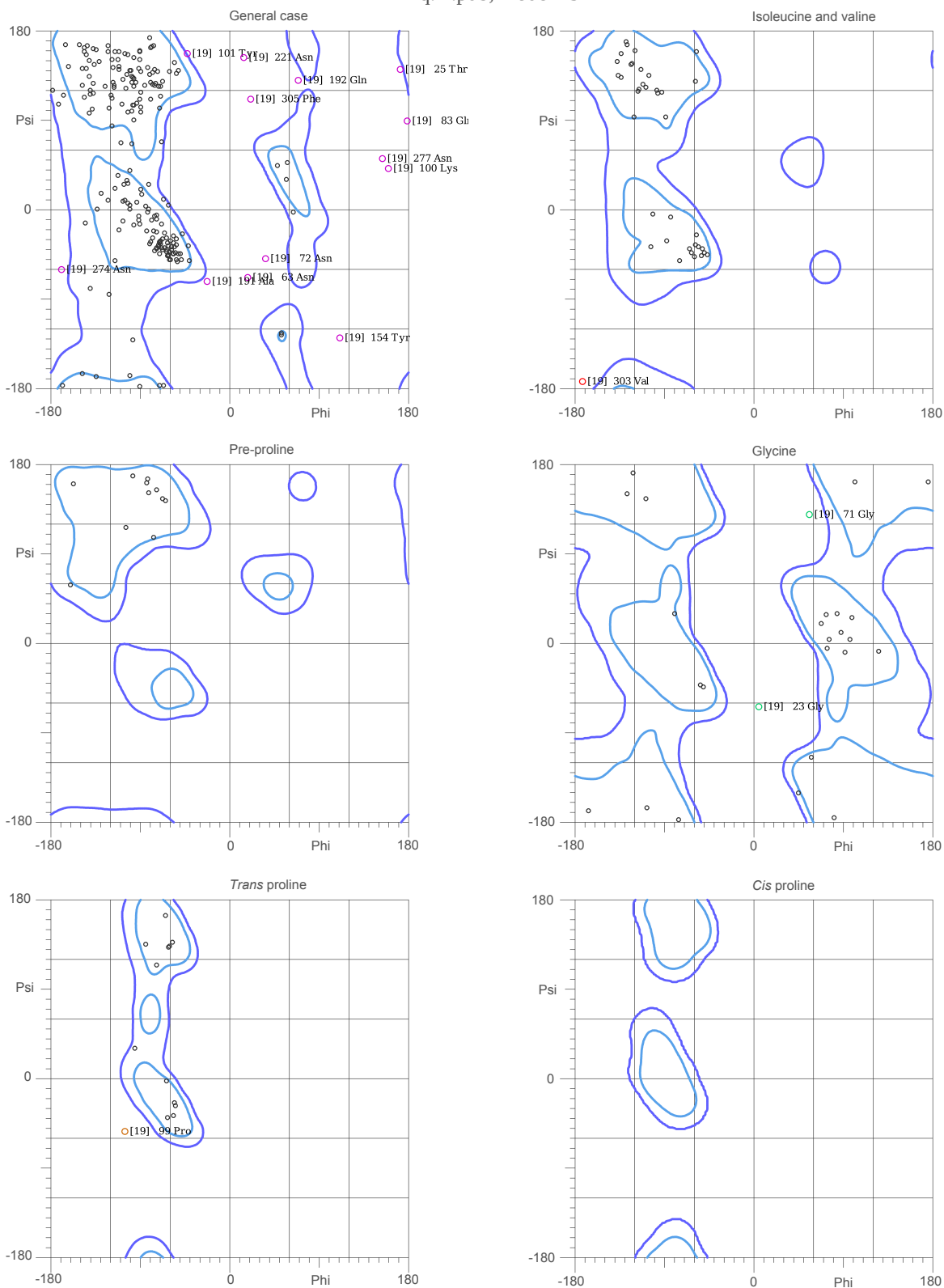
87.8% (267/304) of all residues were in favored (98%) regions.  
96.7% (294/304) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

|                              |                              |
|------------------------------|------------------------------|
| [18] 23 Gly (-13.8, -50.9)   | [18] 303 Val (-175.6, 170.5) |
| [18] 45 Thr (-54.7, -162.5)  | [18] 305 Phe (48.9, 173.0)   |
| [18] 100 Lys (104.0, 115.0)  |                              |
| [18] 154 Tyr (62.0, -76.2)   |                              |
| [18] 191 Ala (28.6, -103.0)  |                              |
| [18] 192 Gln (49.1, 109.8)   |                              |
| [18] 273 Gln (-46.7, -94.2)  |                              |
| [18] 279 Arg (-73.8, -144.7) |                              |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 19



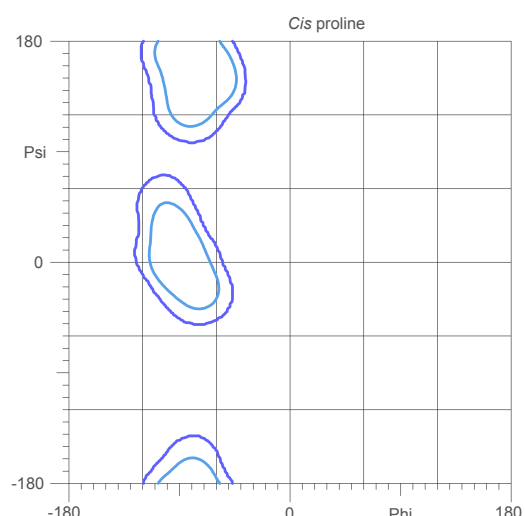
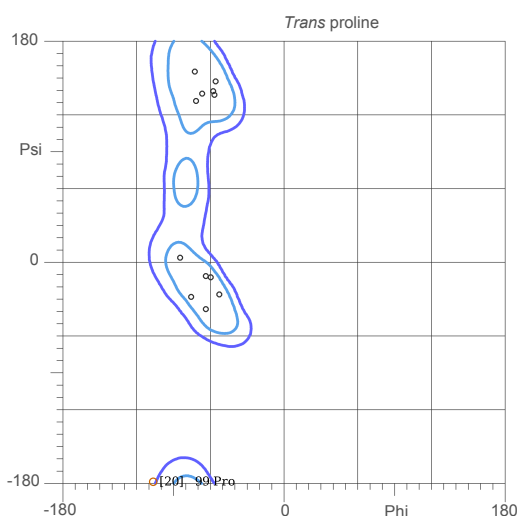
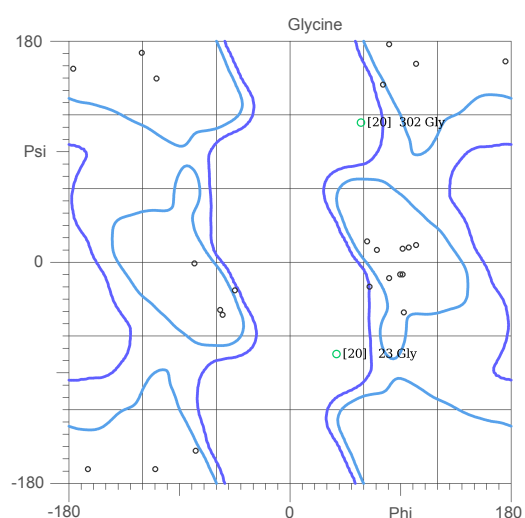
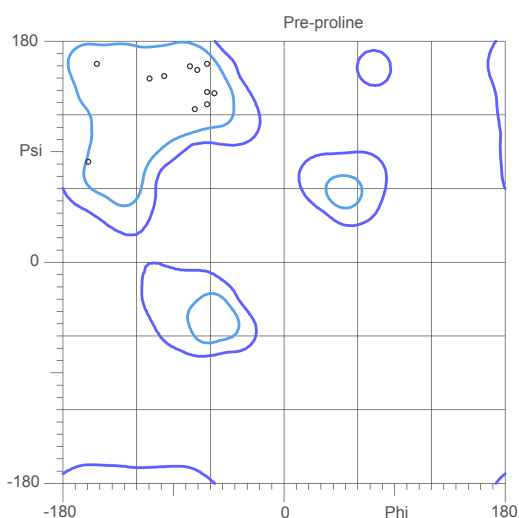
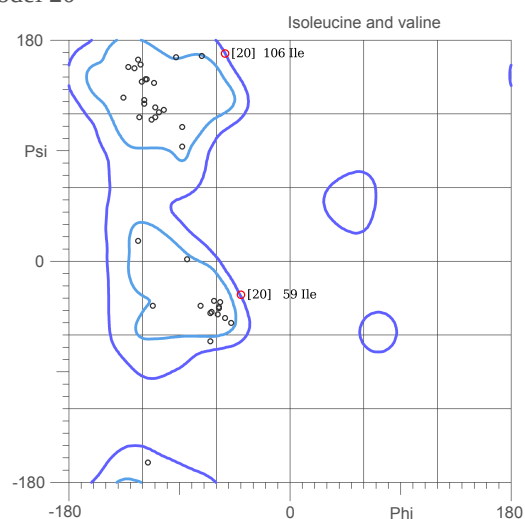
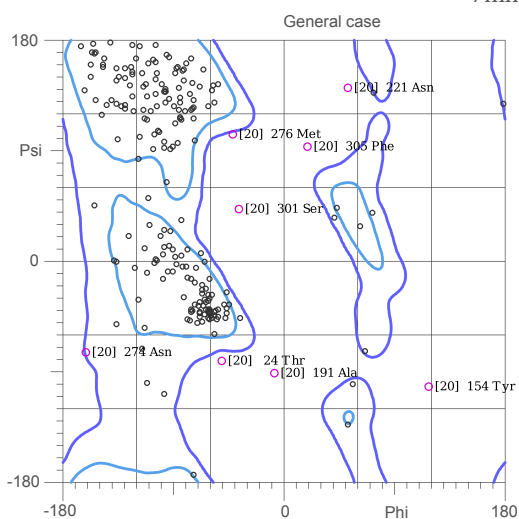
86.5% (263/304) of all residues were in favored (98%) regions.  
94.4% (287/304) of all residues were in allowed (>99.8%) regions.

There were 17 outliers (phi, psi):

|                             |                               |
|-----------------------------|-------------------------------|
| [19] 23 Gly (5.7, -64.4)    | [19] 101 Tyr (-43.7, 158.7)   |
| [19] 25 Thr (172.0, 142.5)  | [19] 154 Tyr (111.9, -129.5)  |
| [19] 63 Asn (18.1, -68.7)   | [19] 191 Ala (-23.0, -72.1)   |
| [19] 71 Gly (56.9, 130.5)   | [19] 192 Gln (69.6, 131.2)    |
| [19] 72 Asn (36.1, -49.2)   | [19] 221 Asn (14.1, 154.2)    |
| [19] 83 Gln (180.0, 90.0)   | [19] 274 Asn (-170.7, -60.4)  |
| [19] 99 Pro (-106.8, -53.8) | [19] 277 Asn (154.8, 52.4)    |
| [19] 100 Lys (160.8, 42.1)  | [19] 303 Val (-173.8, -173.3) |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 20



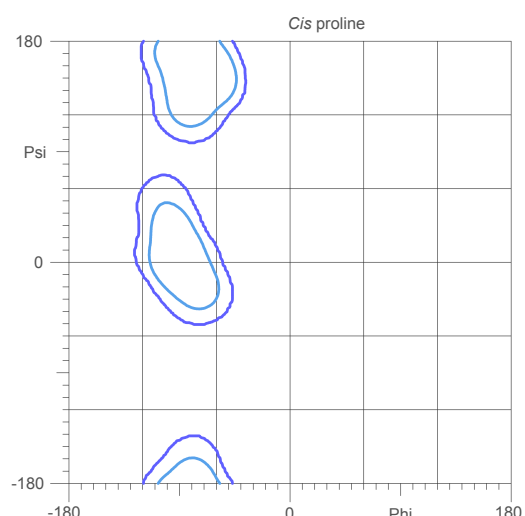
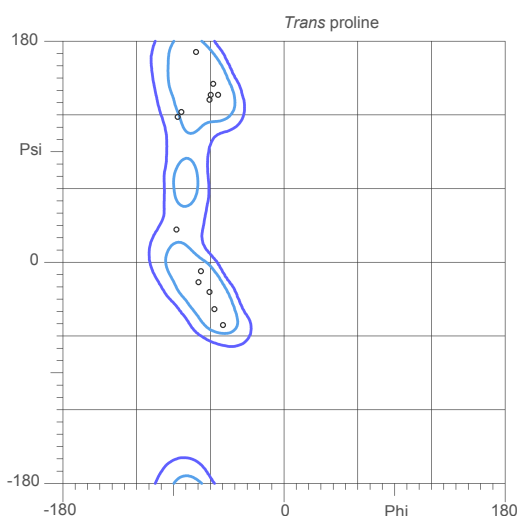
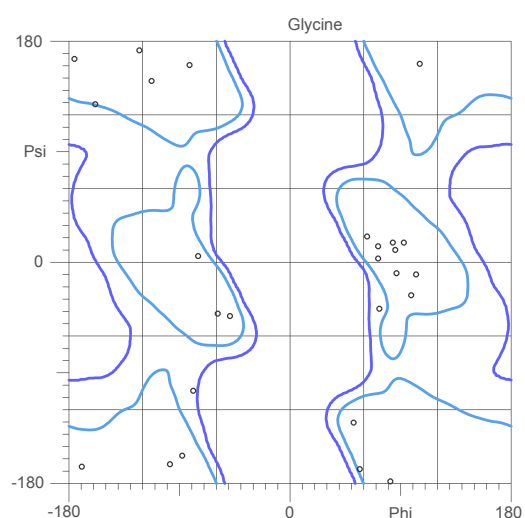
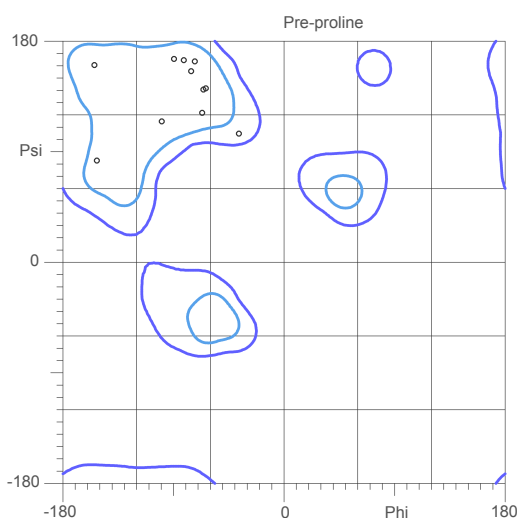
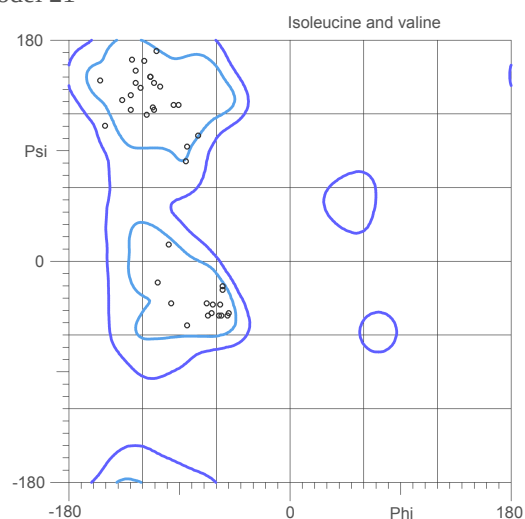
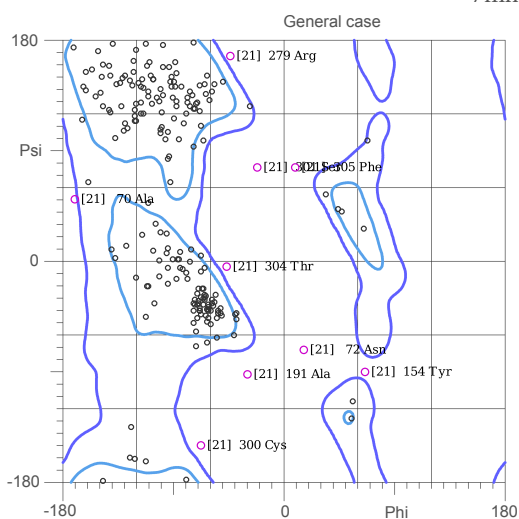
87.2% (265/304) of all residues were in favored (98%) regions.  
95.7% (291/304) of all residues were in allowed (>99.8%) regions.

There were 13 outliers (phi, psi):

|                              |                              |
|------------------------------|------------------------------|
| [20] 23 Gly (38.7, -75.6)    | [20] 274 Asn (-162.8, -74.4) |
| [20] 24 Thr (-51.5, -81.9)   | [20] 276 Met (-42.8, 104.7)  |
| [20] 59 Ile (-40.8, -27.4)   | [20] 301 Ser (-37.9, 43.4)   |
| [20] 99 Pro (-107.4, -179.7) | [20] 302 Gly (58.6, 115.0)   |
| [20] 106 Ile (-53.3, 170.8)  | [20] 305 Phe (19.8, 94.6)    |
| [20] 154 Tyr (118.8, -102.1) |                              |
| [20] 191 Ala (-8.1, -91.3)   |                              |
| [20] 221 Asn (52.6, 142.1)   |                              |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 21



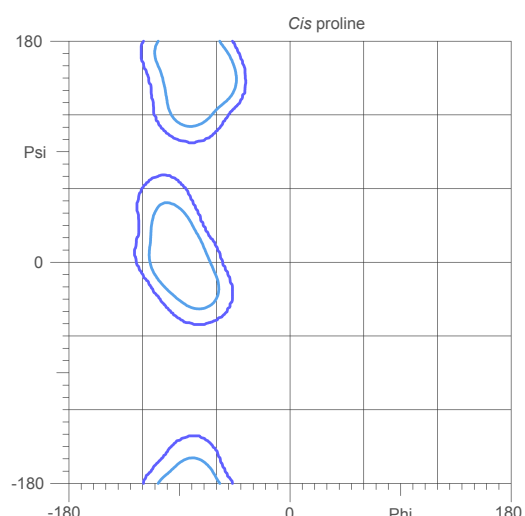
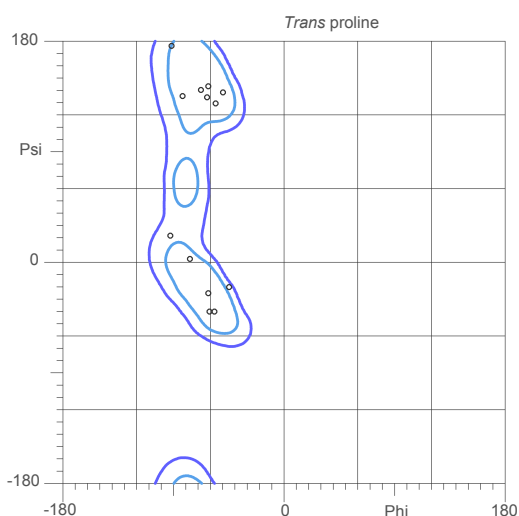
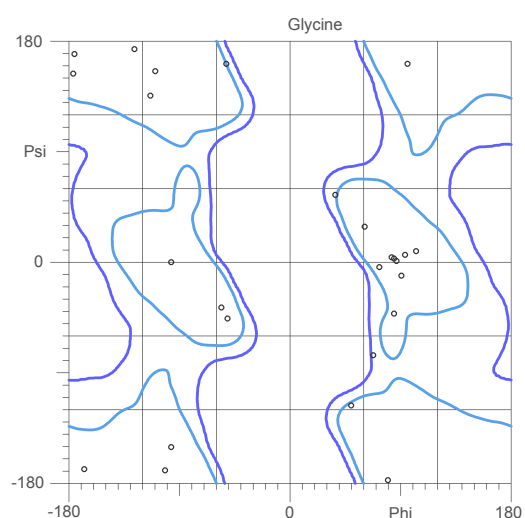
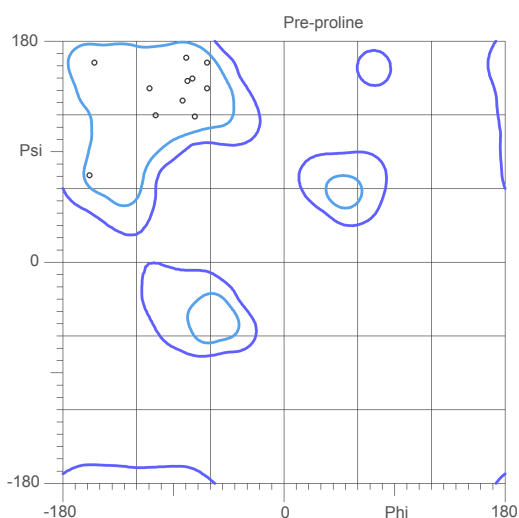
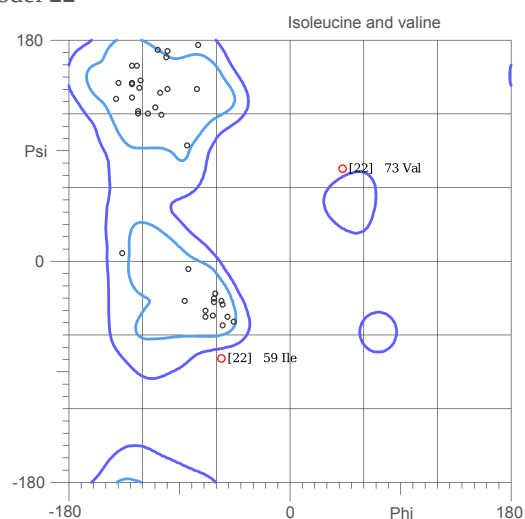
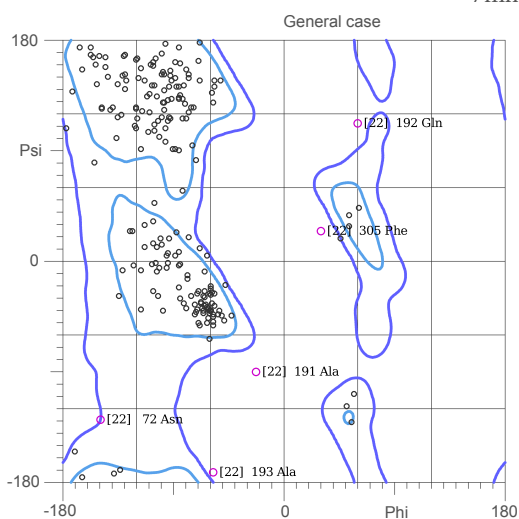
88.2% (268/304) of all residues were in favored (98%) regions.  
97.0% (295/304) of all residues were in allowed (>99.8%) regions.

There were 9 outliers (phi, psi):

|                              |                          |
|------------------------------|--------------------------|
| [21] 70 Ala (-172.0, 51.6)   | [21] 305 Phe (9.6, 77.4) |
| [21] 72 Asn (16.5, -73.0)    |                          |
| [21] 154 Tyr (67.0, -90.8)   |                          |
| [21] 191 Ala (-30.0, -93.0)  |                          |
| [21] 279 Arg (-44.2, 168.1)  |                          |
| [21] 300 Cys (-68.3, -150.3) |                          |
| [21] 301 Ser (-22.6, 77.2)   |                          |
| [21] 304 Thr (-47.5, -4.1)   |                          |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 22



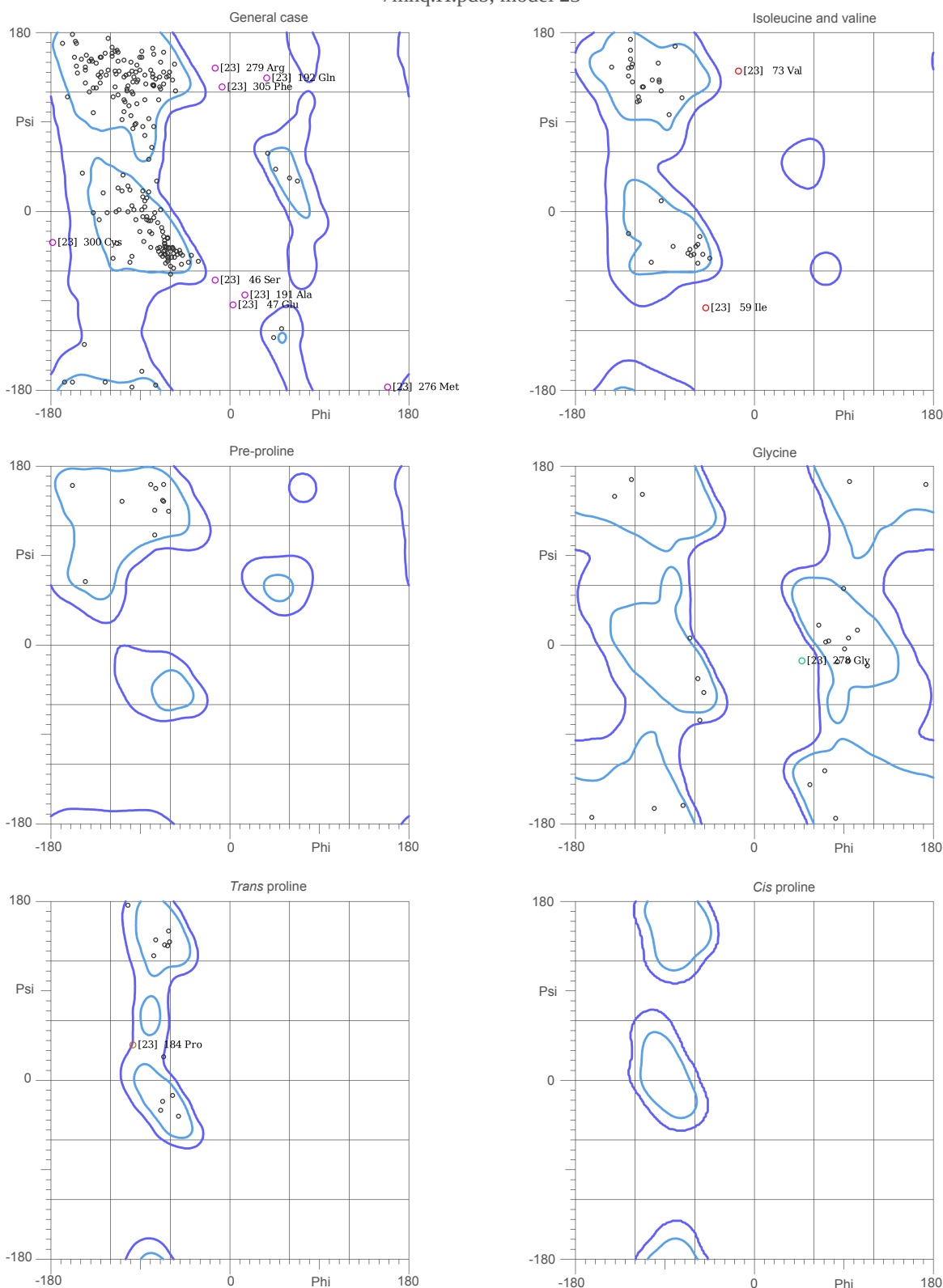
88.8% (270/304) of all residues were in favored (98%) regions.  
97.7% (297/304) of all residues were in allowed (>99.8%) regions.

There were 7 outliers (phi, psi):

[22] 59 Ile (-56.5, -79.0)  
[22] 72 Asn (-150.7, -129.0)  
[22] 73 Val (43.6, 76.2)  
[22] 191 Ala (-23.5, -90.4)  
[22] 192 Gln (60.2, 113.8)  
[22] 193 Ala (-58.6, -172.6)  
[22] 305 Phe (30.5, 25.6)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 23



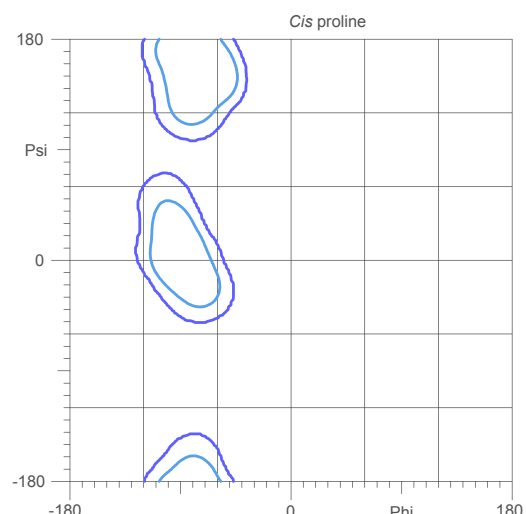
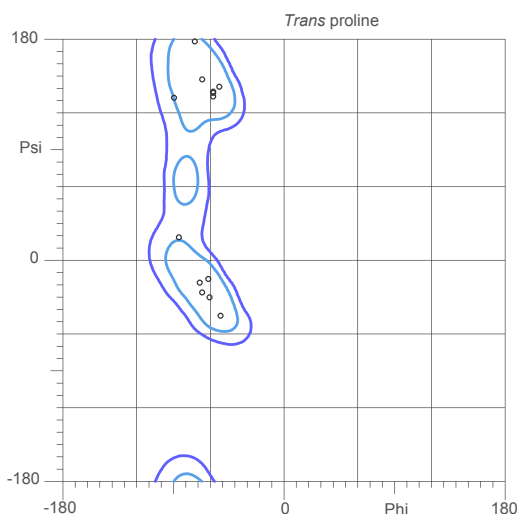
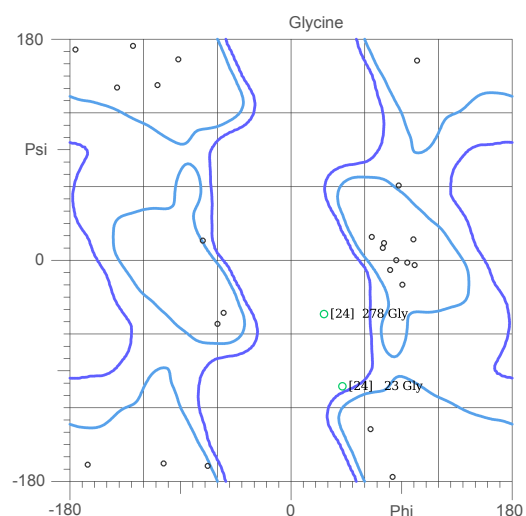
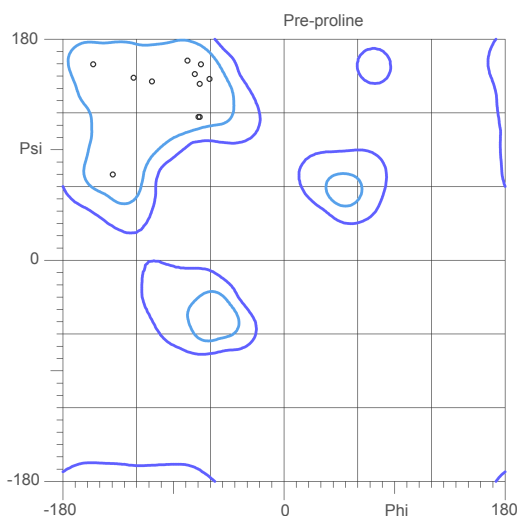
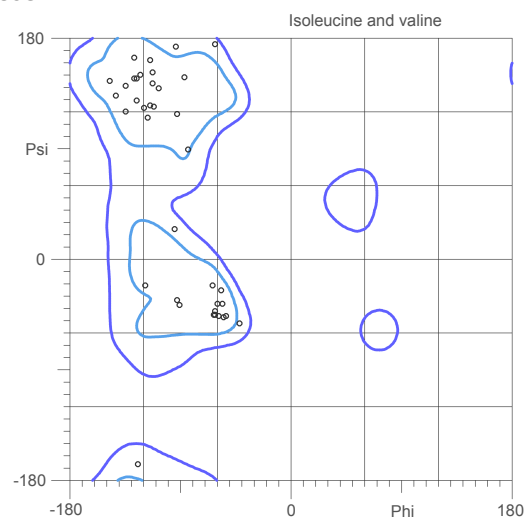
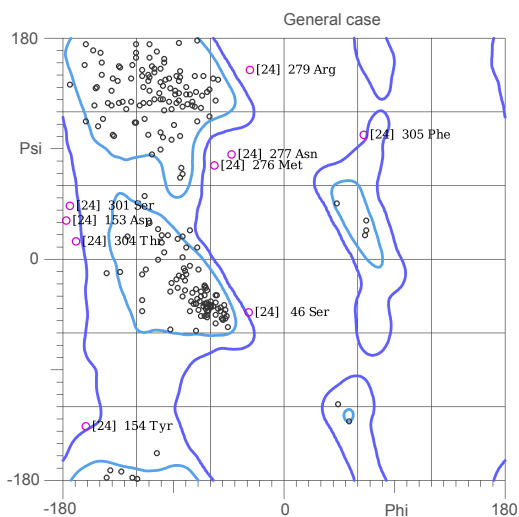
89.8% (273/304) of all residues were in favored (98%) regions.  
96.1% (292/304) of all residues were in allowed (>99.8%) regions.

There were 12 outliers (phi, psi):

|                              |                              |
|------------------------------|------------------------------|
| [23] 46 Ser (-15.6, -69.6)   | [23] 278 Gly (48.4, -16.6)   |
| [23] 47 Glu (3.1, -94.8)     | [23] 279 Arg (-15.6, 145.6)  |
| [23] 59 Ile (-49.4, -97.4)   | [23] 300 Cys (-179.7, -31.3) |
| [23] 73 Val (-16.3, 142.4)   | [23] 305 Phe (-8.4, 126.6)   |
| [23] 184 Pro (-98.8, 36.2)   |                              |
| [23] 191 Ala (15.9, -84.5)   |                              |
| [23] 192 Gln (38.0, 135.4)   |                              |
| [23] 276 Met (159.6, -177.1) |                              |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 24



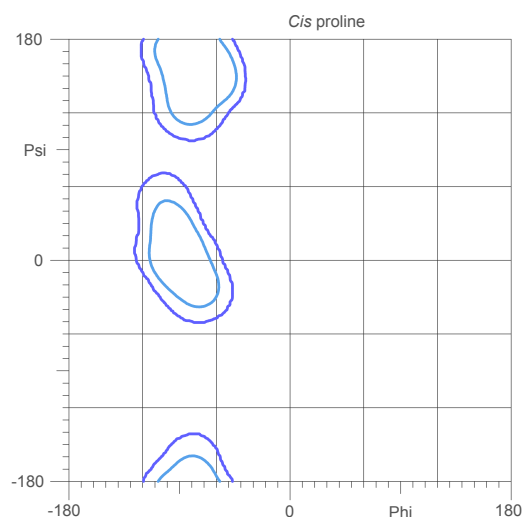
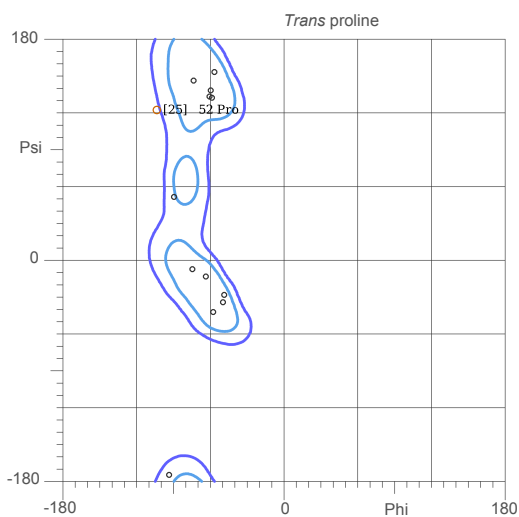
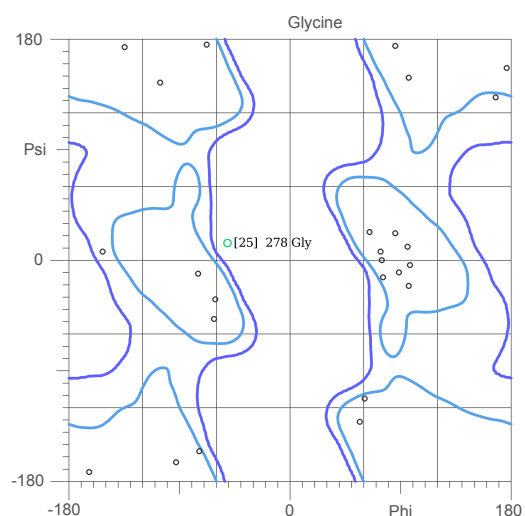
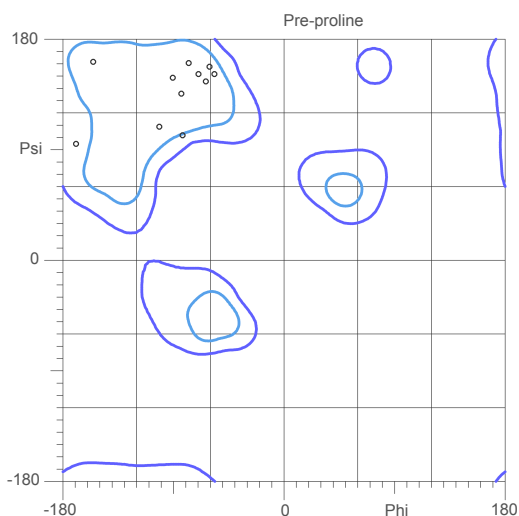
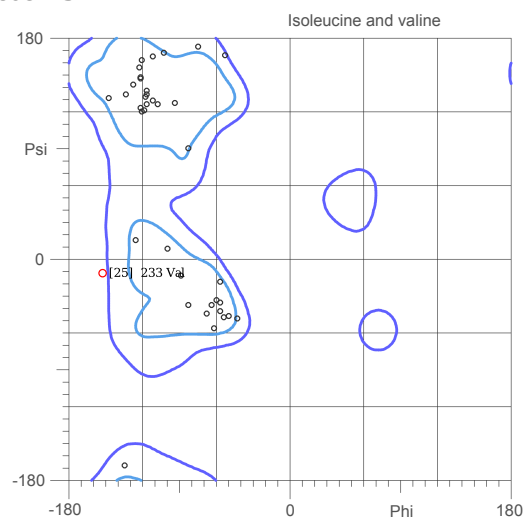
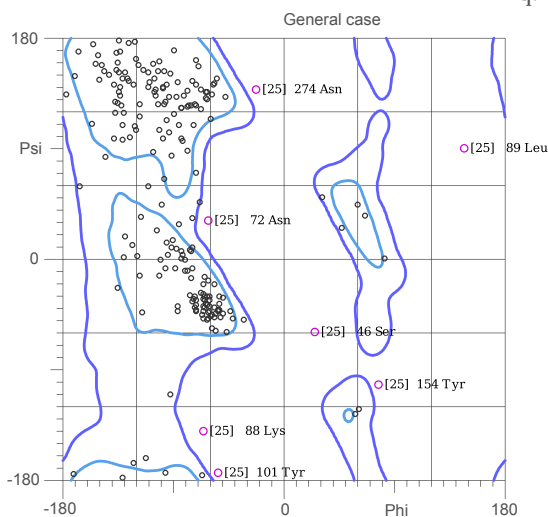
90.8% (276/304) of all residues were in favored (98%) regions.  
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

|                               |                             |
|-------------------------------|-----------------------------|
| [24] 23 Gly (42.4, -103.4)    | [24] 301 Ser (-175.0, 44.1) |
| [24] 46 Ser (-29.9, -43.7)    | [24] 304 Thr (-170.4, 15.6) |
| [24] 153 Asp (-178.8, 32.6)   | [24] 305 Phe (65.5, 102.3)  |
| [24] 154 Tyr (-162.4, -136.6) |                             |
| [24] 276 Met (-57.2, 77.2)    |                             |
| [24] 277 Asn (-43.6, 86.2)    |                             |
| [24] 278 Gly (27.9, -44.6)    |                             |
| [24] 279 Arg (-28.8, 155.8)   |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 25



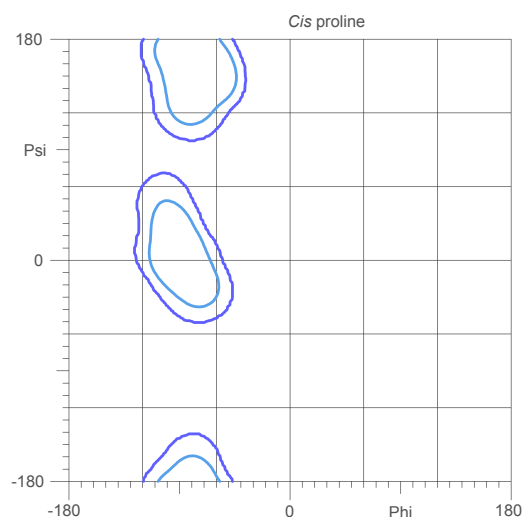
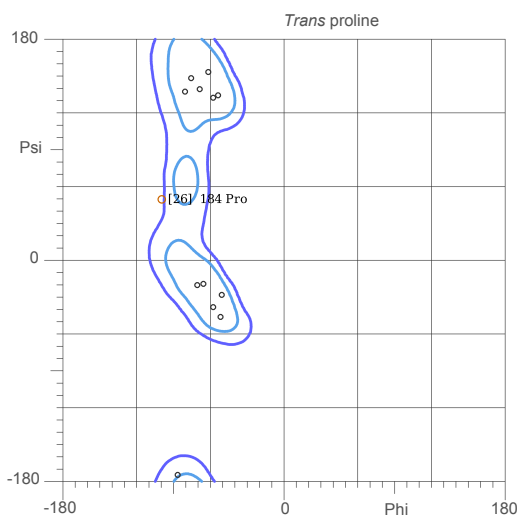
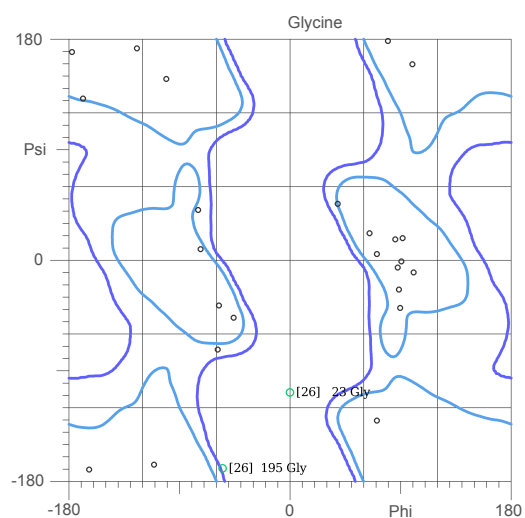
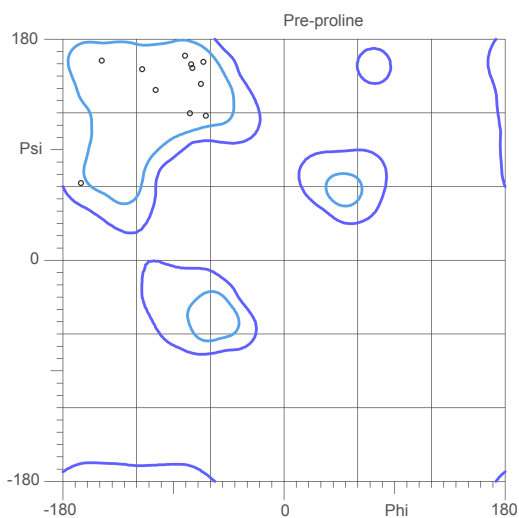
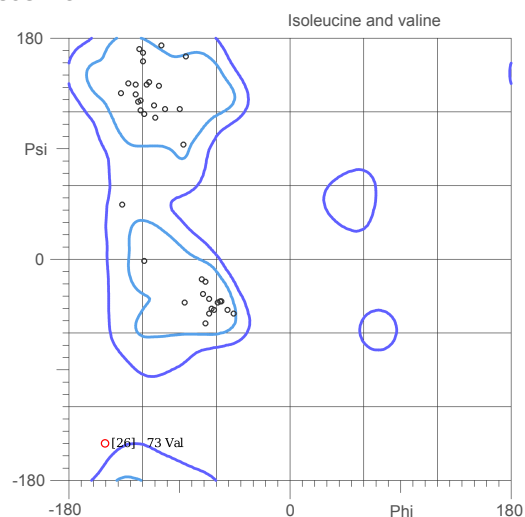
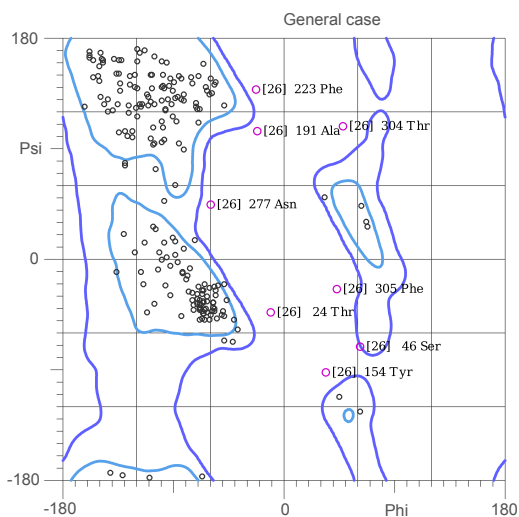
85.9% (261/304) of all residues were in favored (98%) regions.  
96.7% (294/304) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

|                              |                             |
|------------------------------|-----------------------------|
| [25] 46 Ser (25.5, -59.3)    | [25] 274 Asn (-23.3, 139.7) |
| [25] 52 Pro (-105.0, 123.2)  | [25] 278 Gly (-51.2, 14.8)  |
| [25] 72 Asn (-62.6, 32.8)    |                             |
| [25] 88 Lys (-66.0, -140.7)  |                             |
| [25] 89 Leu (147.4, 91.8)    |                             |
| [25] 101 Tyr (-54.7, -174.7) |                             |
| [25] 154 Tyr (77.5, -102.7)  |                             |
| [25] 233 Val (-153.5, -11.9) |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 26



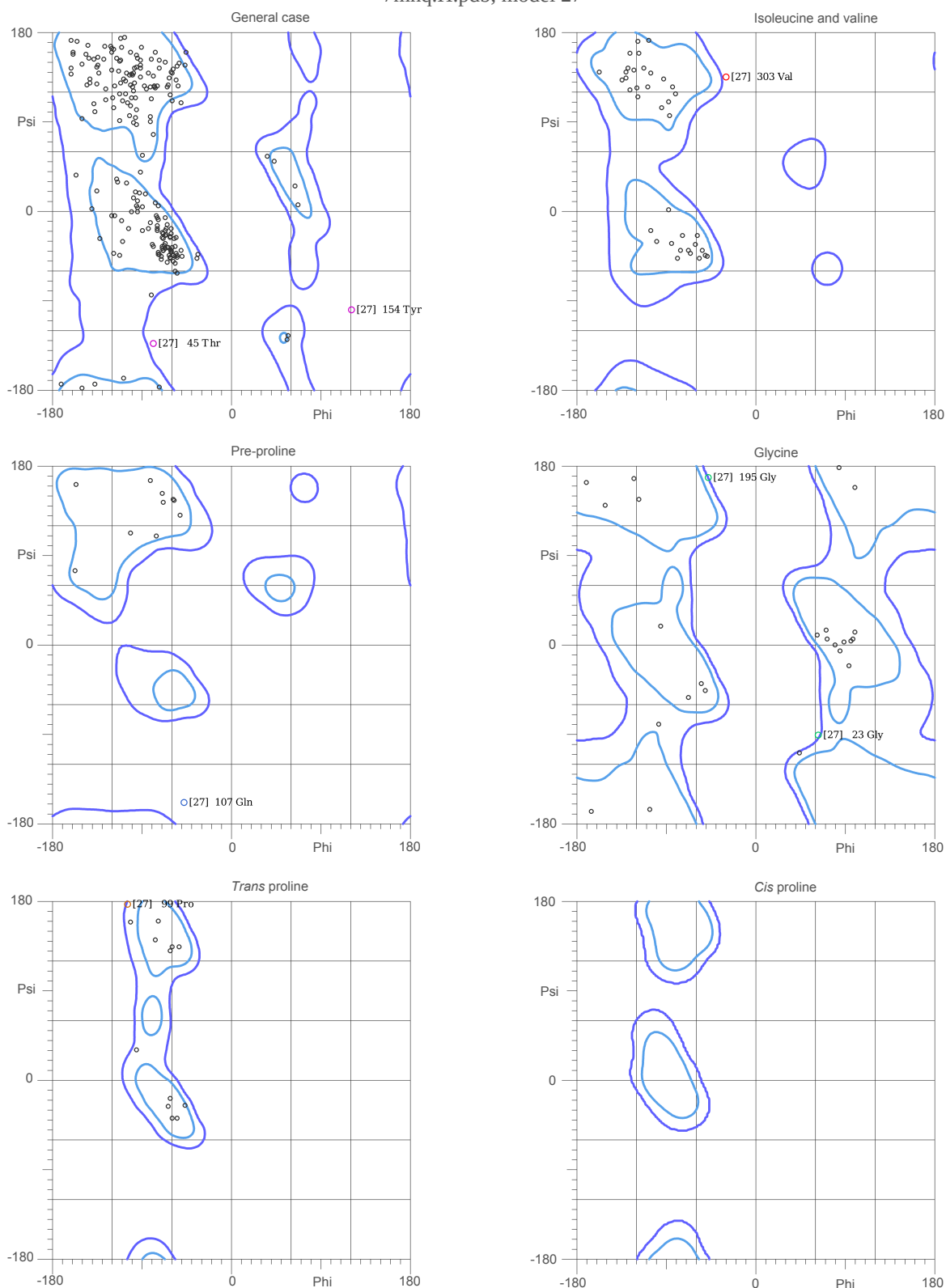
89.1% (271/304) of all residues were in favored (98%) regions.  
96.1% (292/304) of all residues were in allowed (>99.8%) regions.

There were 12 outliers (phi, psi):

|                              |                             |
|------------------------------|-----------------------------|
| [26] 23 Gly (0.9, -108.4)    | [26] 223 Phe (-23.9, 139.3) |
| [26] 24 Thr (-11.4, -43.9)   | [26] 277 Asn (-61.0, 45.4)  |
| [26] 46 Ser (62.3, -71.9)    | [26] 304 Thr (48.7, 109.6)  |
| [26] 73 Val (-151.1, -150.1) | [26] 305 Phe (43.5, -24.7)  |
| [26] 154 Tyr (34.5, -92.7)   |                             |
| [26] 184 Pro (-100.5, 50.1)  |                             |
| [26] 191 Ala (-22.3, 105.0)  |                             |
| [26] 195 Gly (-55.6, -170.1) |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 27



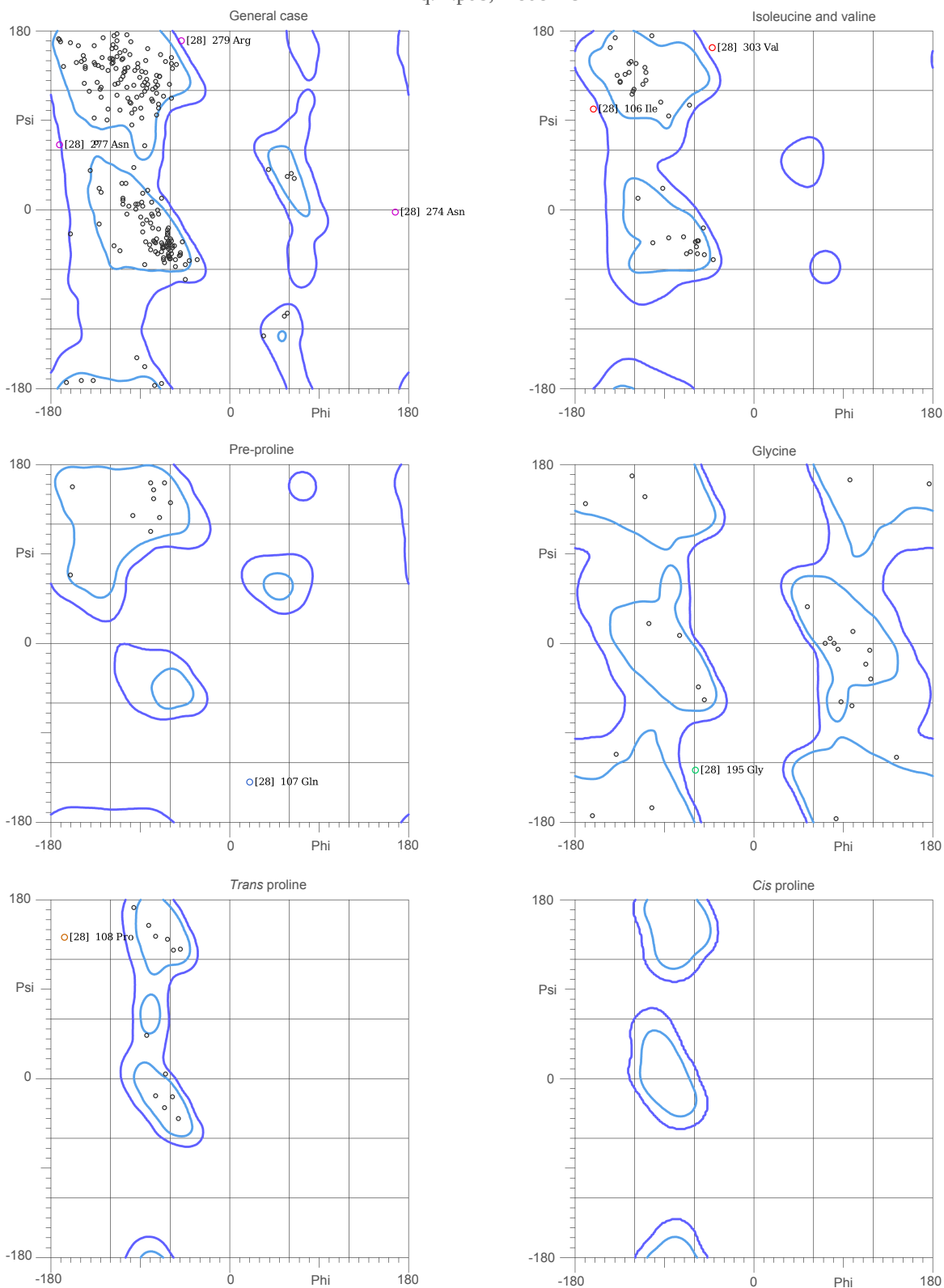
90.5% (275/304) of all residues were in favored (98%) regions.  
97.7% (297/304) of all residues were in allowed (>99.8%) regions.

There were 7 outliers (phi, psi):

- [27] 23 Gly (63.2, -91.5)
- [27] 45 Thr (-79.1, -133.5)
- [27] 99 Pro (-105.7, 178.8)
- [27] 107 Gln (-48.8, -159.6)
- [27] 154 Tyr (121.8, -99.2)
- [27] 195 Gly (-48.2, 169.2)
- [27] 303 Val (-30.0, 136.8)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 28



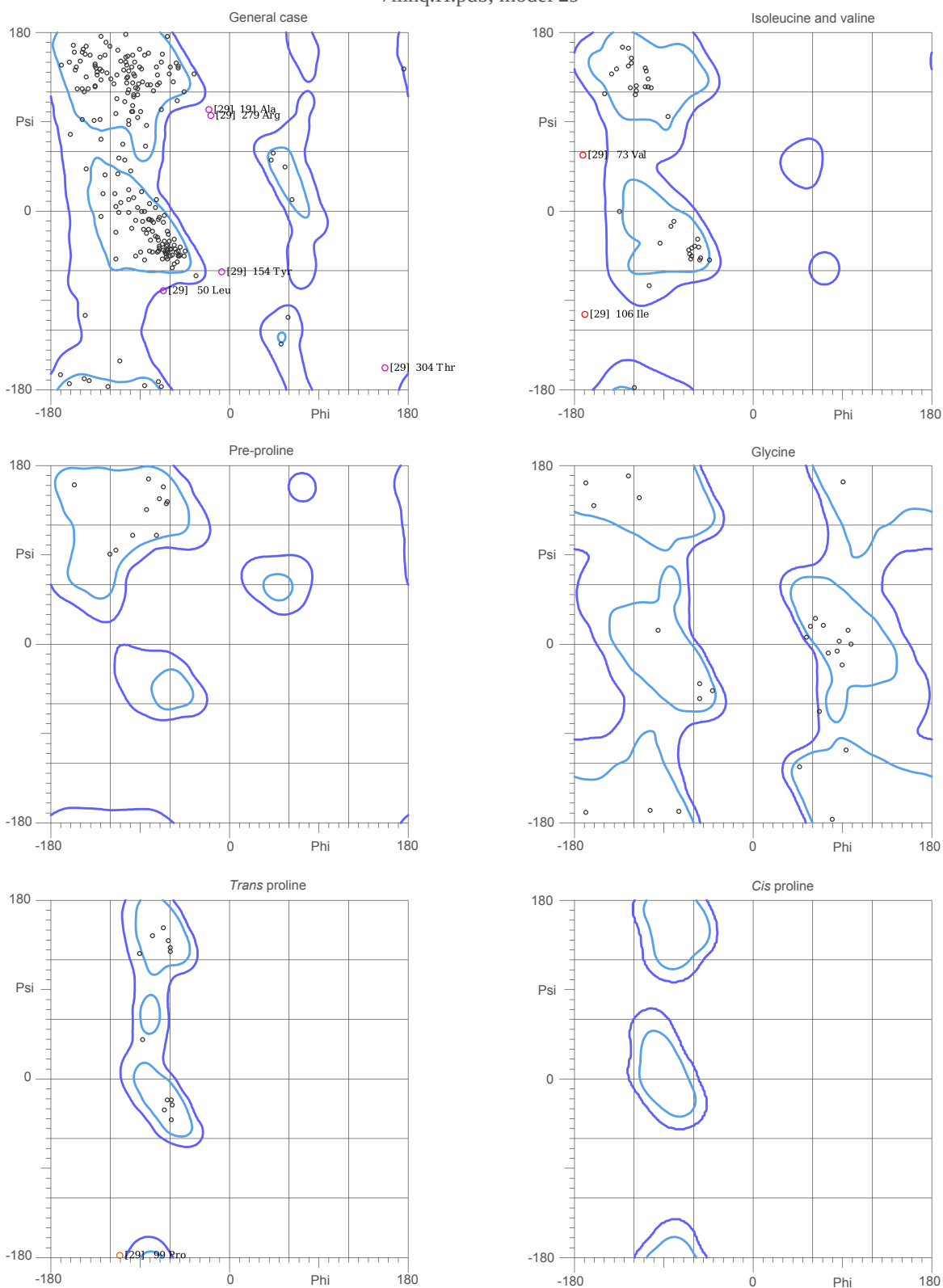
88.8% (270/304) of all residues were in favored (98%) regions.  
97.4% (296/304) of all residues were in allowed (>99.8%) regions.

There were 8 outliers (phi, psi):

[28] 106 Ile (-162.2, 102.4)  
[28] 107 Gln (20.3, -140.4)  
[28] 108 Pro (-167.3, 143.4)  
[28] 195 Gly (-59.7, -128.1)  
[28] 274 Asn (167.8, -2.2)  
[28] 277 Asn (-172.5, 67.0)  
[28] 279 Arg (-49.3, 171.6)  
[28] 303 Val (-42.1, 164.1)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 29



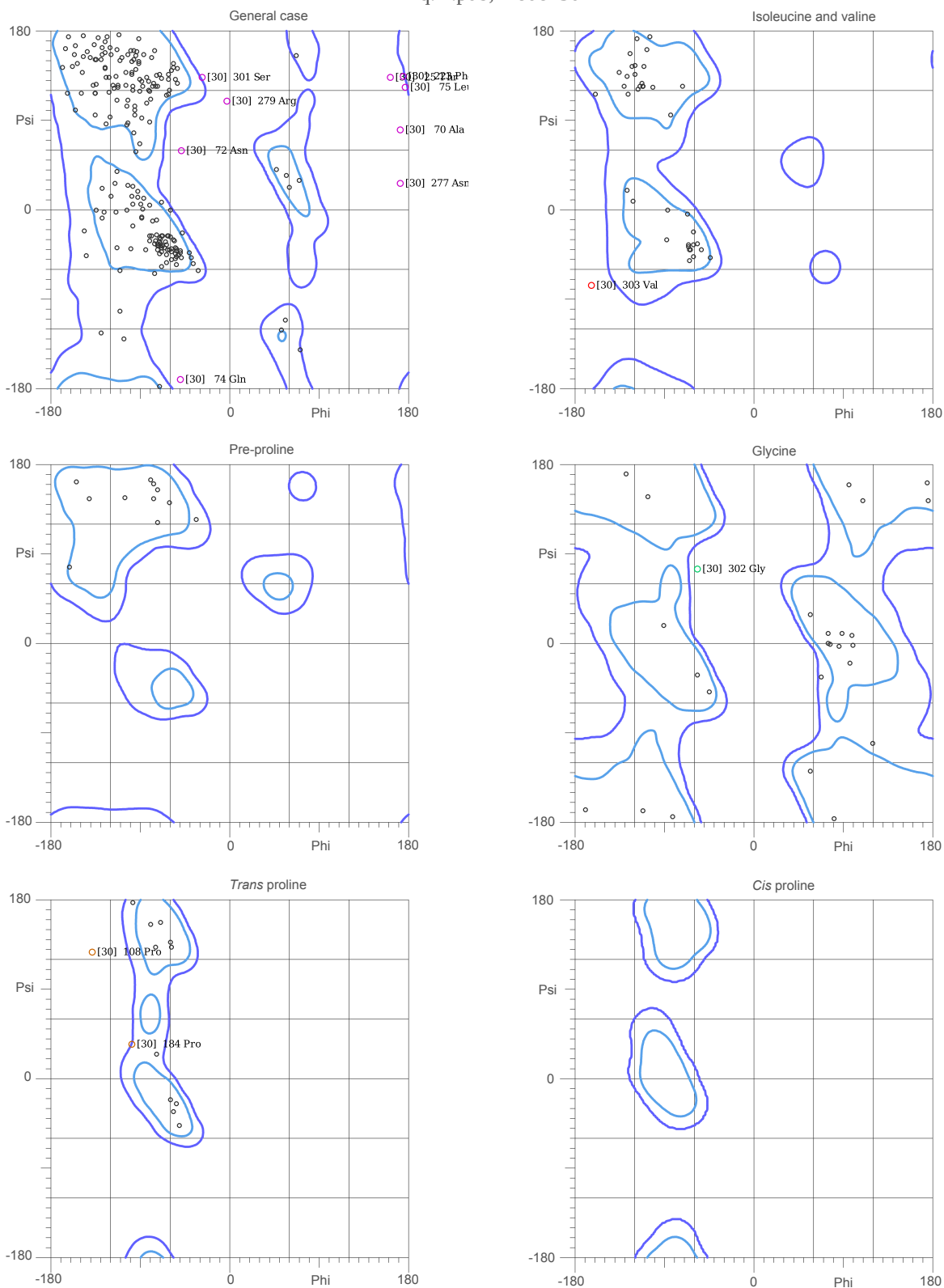
88.2% (268/304) of all residues were in favored (98%) regions.  
97.4% (296/304) of all residues were in allowed (>99.8%) regions.

There were 8 outliers (phi, psi):

[29] 50 Leu (-67.4, -80.5)  
[29] 73 Val (-173.0, 57.6)  
[29] 99 Pro (-111.5, -178.5)  
[29] 106 Ile (-170.3, -104.9)  
[29] 154 Tyr (-9.0, -61.1)  
[29] 191 Ala (-21.2, 103.2)  
[29] 279 Arg (-19.5, 97.3)  
[29] 304 Thr (157.6, -158.5)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 30



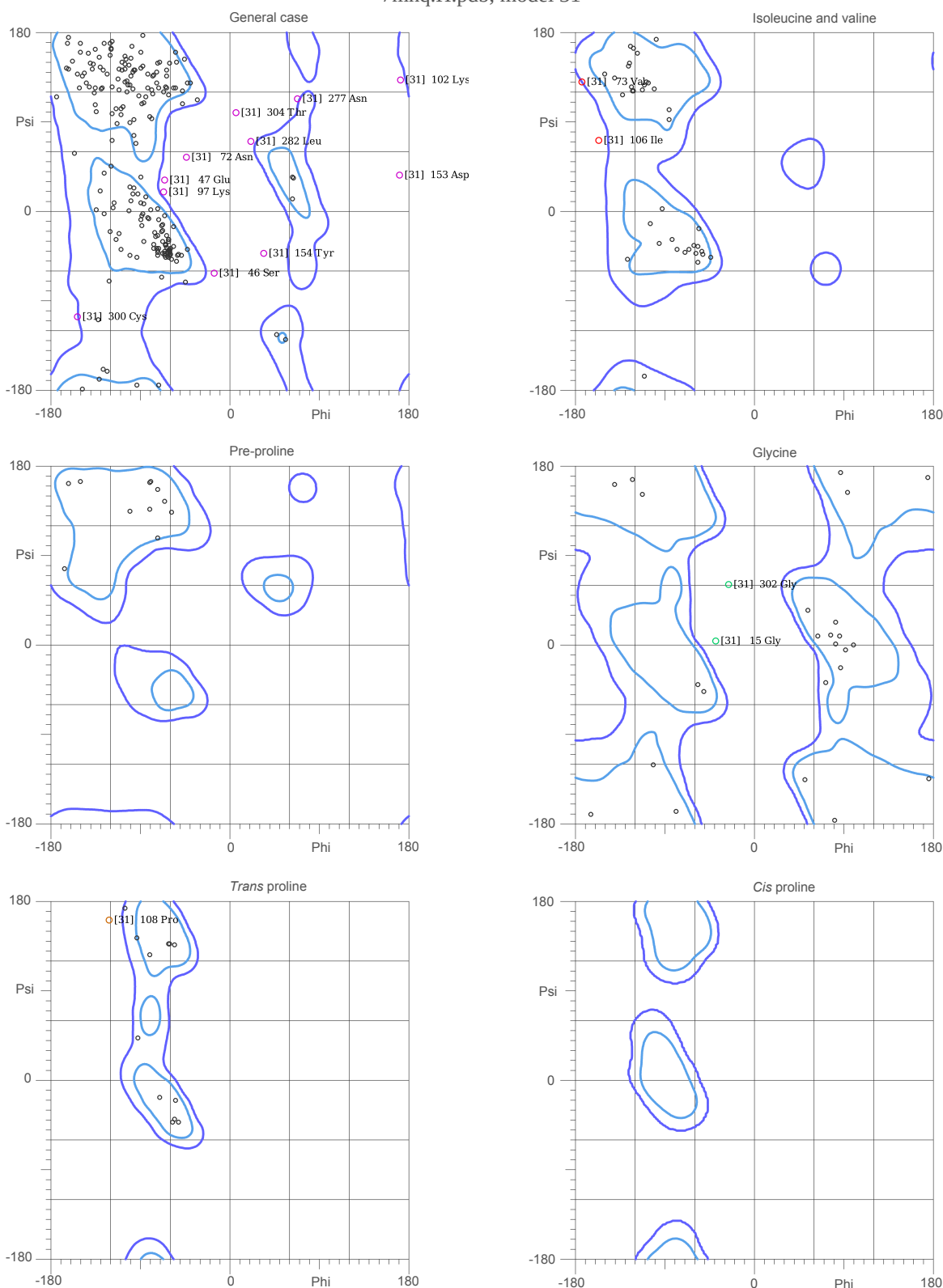
87.2% (265/304) of all residues were in favored (98%) regions.  
95.7% (291/304) of all residues were in allowed (>99.8%) regions.

There were 13 outliers (phi, psi):

|                              |                              |
|------------------------------|------------------------------|
| [30] 25 Thr (162.8, 134.9)   | [30] 277 Asn (172.3, 27.0)   |
| [30] 70 Ala (172.6, 81.4)    | [30] 279 Arg (-3.7, 110.9)   |
| [30] 72 Asn (-50.0, 60.4)    | [30] 301 Ser (-28.8, 134.9)  |
| [30] 74 Gln (-50.2, -171.7)  | [30] 302 Gly (-57.7, 75.0)   |
| [30] 75 Leu (177.1, 124.9)   | [30] 303 Val (-164.7, -76.7) |
| [30] 108 Pro (-139.1, 128.4) |                              |
| [30] 184 Pro (-99.3, 35.1)   |                              |
| [30] 223 Phe (175.5, 135.8)  |                              |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 31



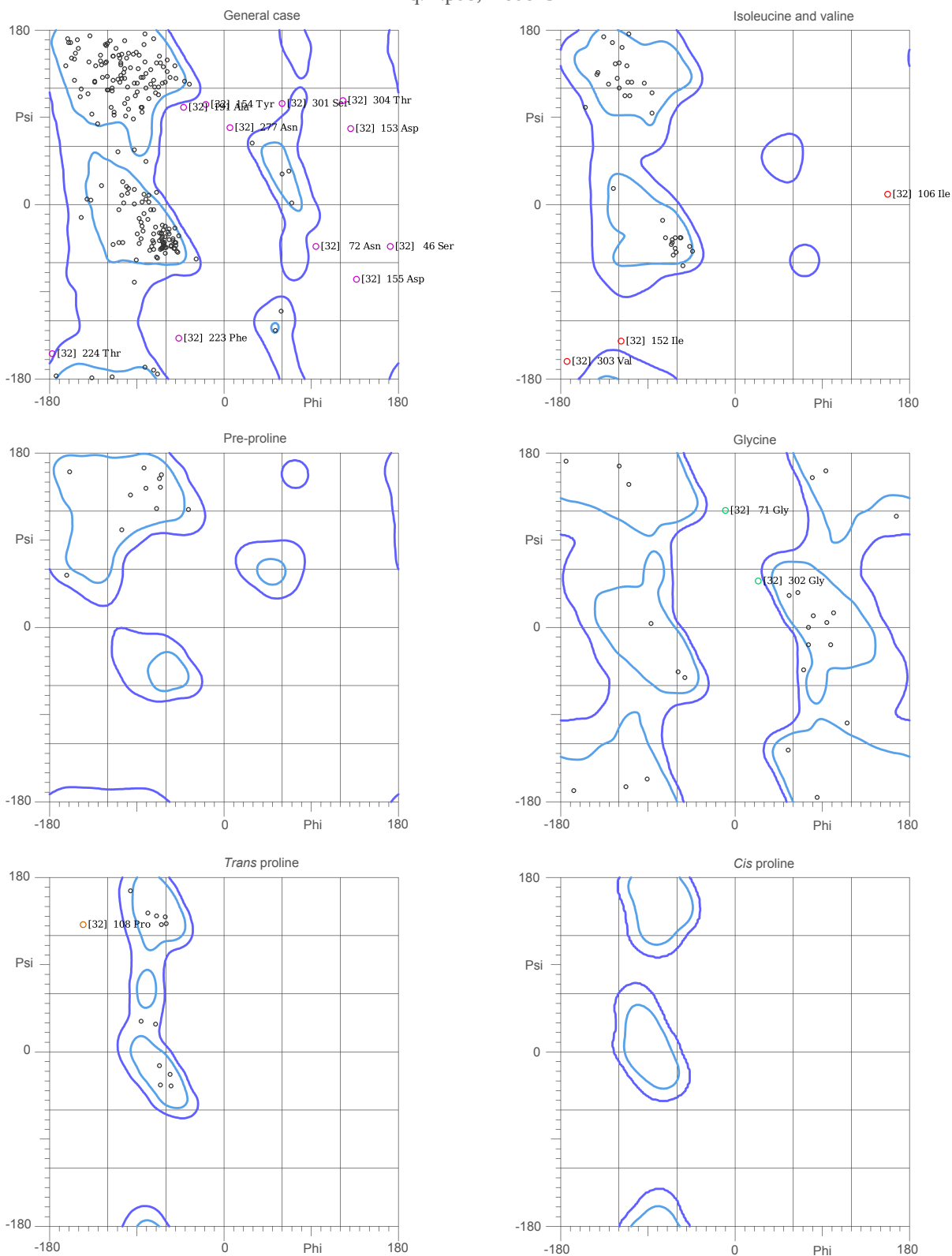
86.5% (263/304) of all residues were in favored (98%) regions.  
94.7% (288/304) of all residues were in allowed (>99.8%) regions.

There were 16 outliers (phi, psi):

|                             |                               |
|-----------------------------|-------------------------------|
| [31] 15 Gly (-39.4, 4.4)    | [31] 108 Pro (-122.5, 162.8)  |
| [31] 46 Ser (-16.2, -62.0)  | [31] 153 Asp (171.8, 37.9)    |
| [31] 47 Glu (-66.1, 32.7)   | [31] 154 Tyr (34.3, -42.5)    |
| [31] 72 Asn (-44.5, 55.2)   | [31] 277 Asn (68.2, 114.9)    |
| [31] 73 Val (-174.8, 131.2) | [31] 282 Leu (21.5, 72.0)     |
| [31] 97 Lys (-67.3, 20.6)   | [31] 300 Cys (-154.7, -106.5) |
| [31] 102 Lys (172.2, 133.2) | [31] 302 Gly (-26.9, 61.1)    |
| [31] 106 Ile (-157.5, 72.6) | [31] 304 Thr (6.2, 100.9)     |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 32



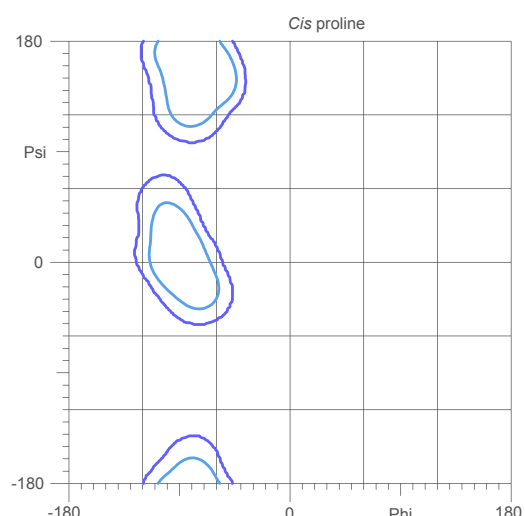
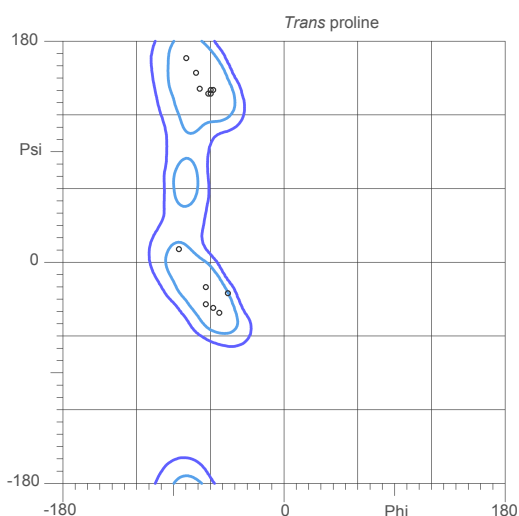
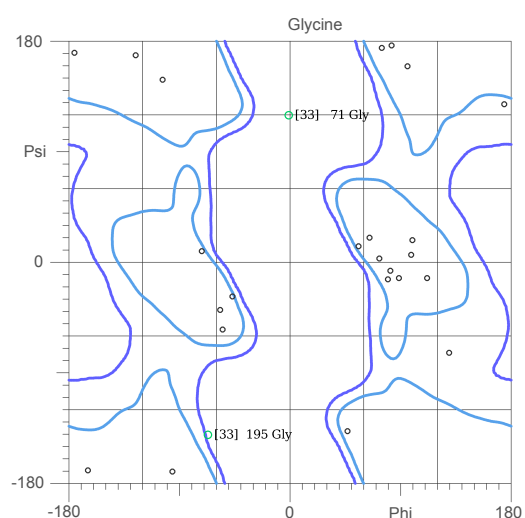
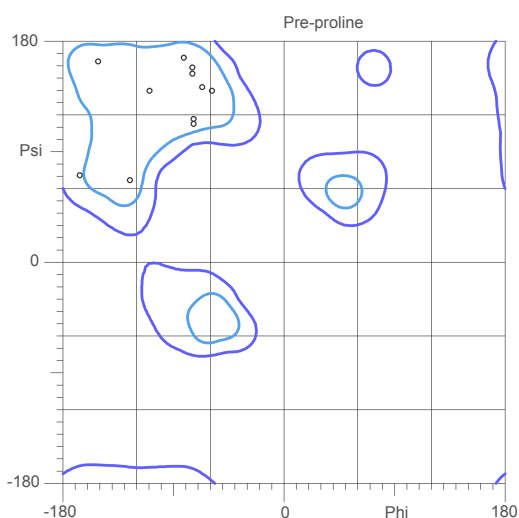
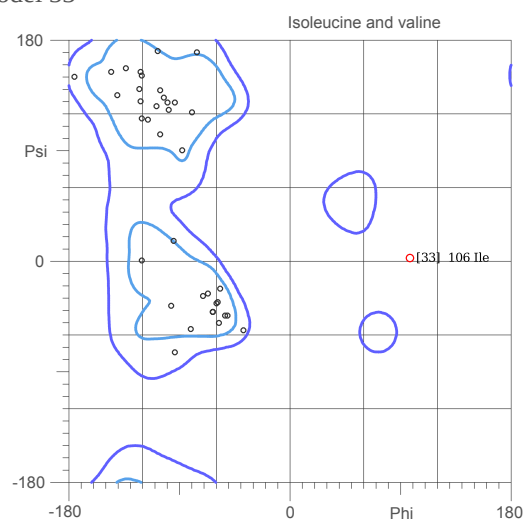
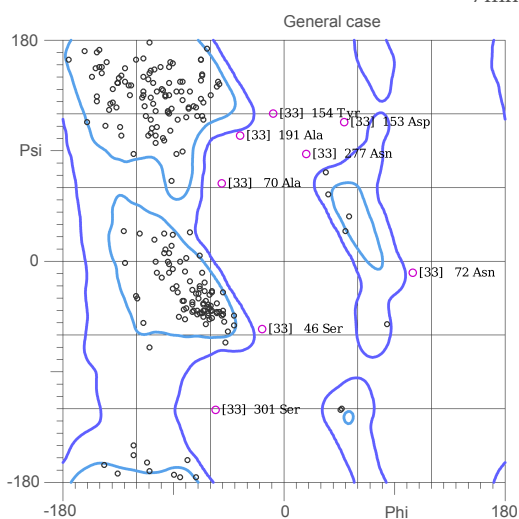
86.5% (263/304) of all residues were in favored (98%) regions. [32] 304 Thr (123.2, 108.6)  
 94.4% (287/304) of all residues were in allowed (>99.8%) regions.

There were 17 outliers (phi, psi):

|                               |                               |
|-------------------------------|-------------------------------|
| [32] 46 Ser (172.7, -43.3)    | [32] 155 Asp (138.0, -77.8)   |
| [32] 71 Gly (-10.0, 121.8)    | [32] 191 Ala (-42.5, 101.4)   |
| [32] 72 Asn (95.4, -43.3)     | [32] 223 Phe (-47.2, -138.8)  |
| [32] 106 Ile (158.4, 11.2)    | [32] 224 Thr (-179.0, -154.8) |
| [32] 108 Pro (-146.0, 132.3)  | [32] 277 Asn (6.8, 80.8)      |
| [32] 152 Ile (-118.0, -141.0) | [32] 301 Ser (60.7, 105.7)    |
| [32] 153 Asp (131.3, 79.2)    | [32] 302 Gly (24.9, 48.6)     |
| [32] 154 Tyr (-19.1, 104.2)   | [32] 303 Val (-174.7, -162.7) |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 33



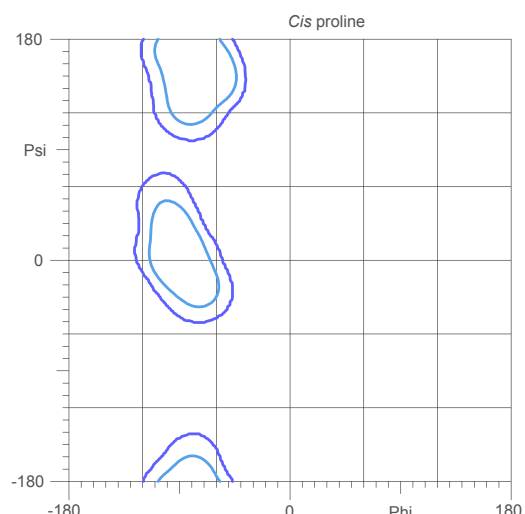
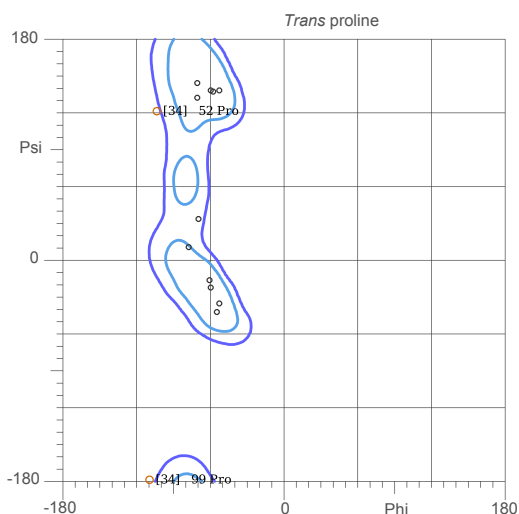
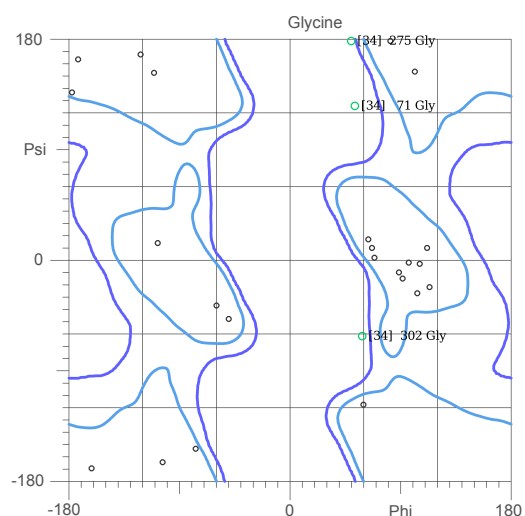
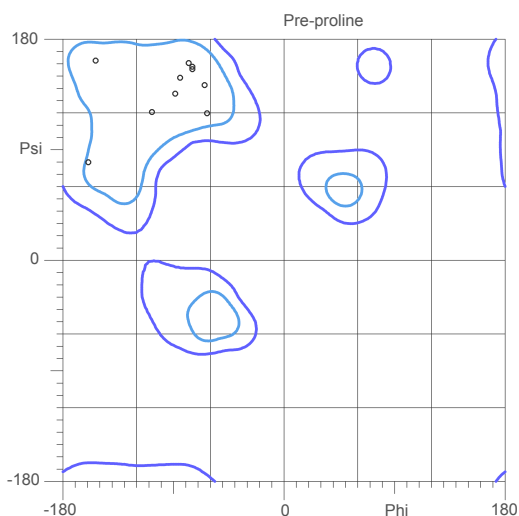
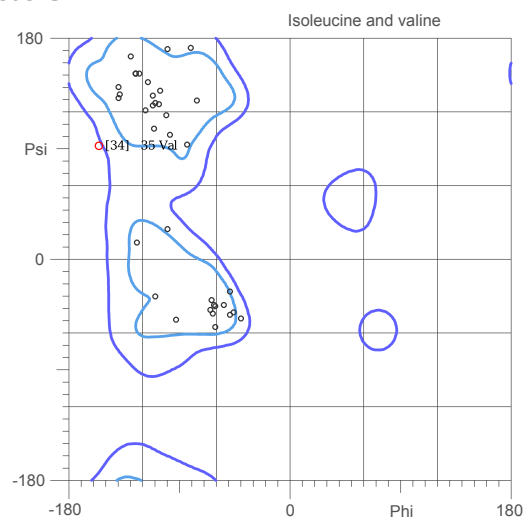
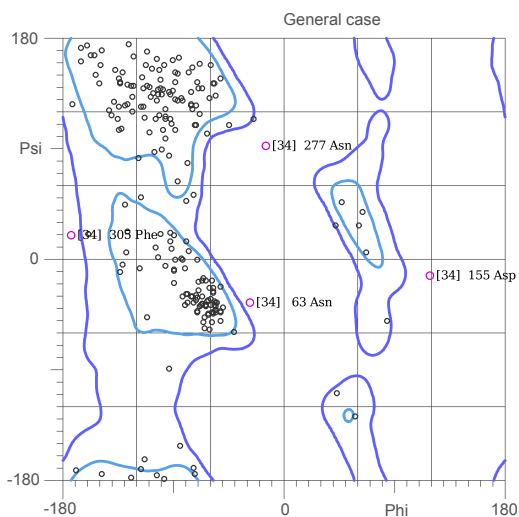
88.2% (268/304) of all residues were in favored (98%) regions.  
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

|                             |                              |
|-----------------------------|------------------------------|
| [33] 46 Ser (-18.3, -55.3)  | [33] 195 Gly (-67.5, -141.2) |
| [33] 70 Ala (-51.5, 64.3)   | [33] 277 Asn (18.1, 88.9)    |
| [33] 71 Gly (-1.3, 120.8)   | [33] 301 Ser (-56.4, -121.3) |
| [33] 72 Asn (105.2, -9.4)   |                              |
| [33] 106 Ile (98.9, 3.5)    |                              |
| [33] 153 Asp (49.9, 114.3)  |                              |
| [33] 154 Tyr (-9.3, 121.2)  |                              |
| [33] 191 Ala (-36.6, 103.1) |                              |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 34



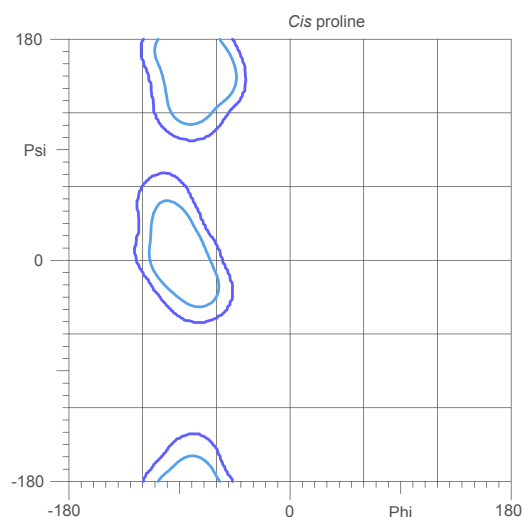
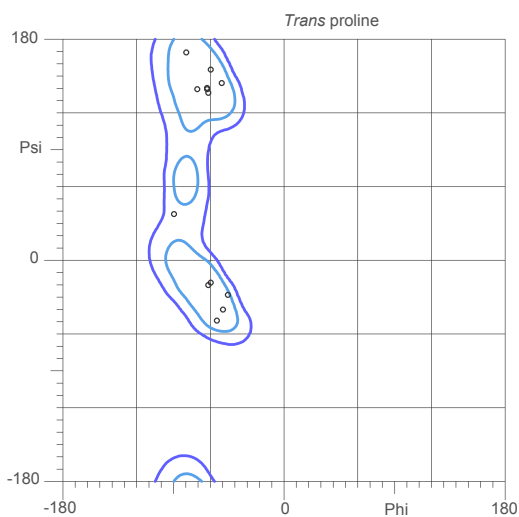
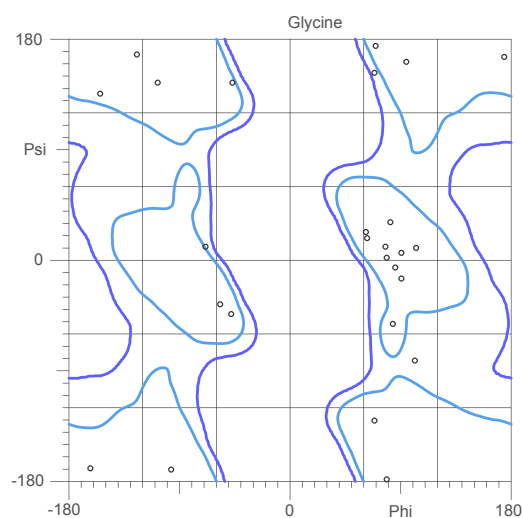
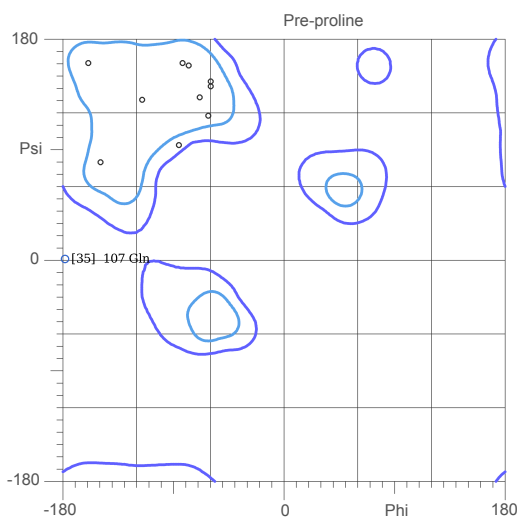
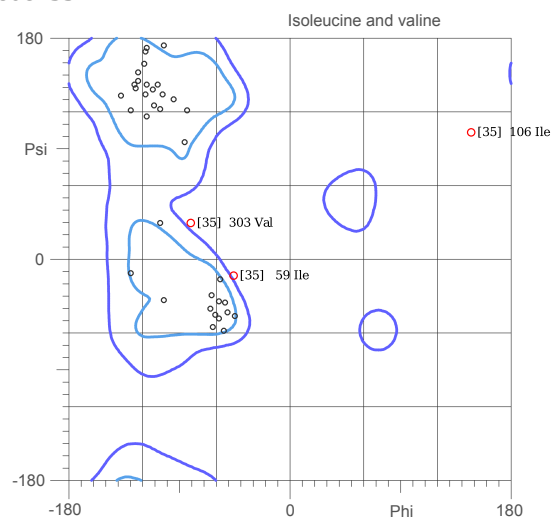
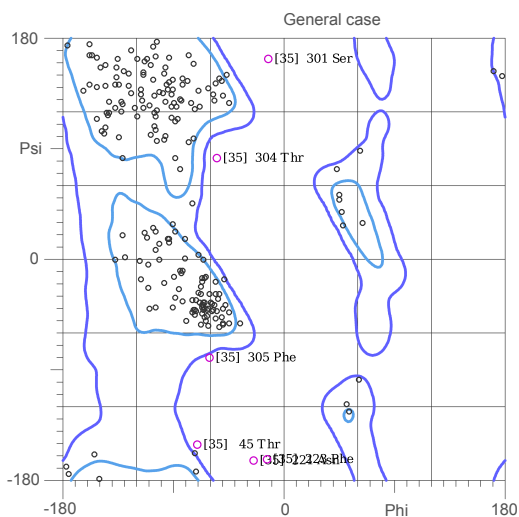
88.4% (267/302) of all residues were in favored (98%) regions.  
96.7% (292/302) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

|                              |                             |
|------------------------------|-----------------------------|
| [34] 35 Val (-156.1, 93.4)   | [34] 302 Gly (59.2, -62.2)  |
| [34] 52 Pro (-104.3, 122.9)  | [34] 305 Phe (-174.6, 20.1) |
| [34] 63 Asn (-28.6, -35.7)   |                             |
| [34] 71 Gly (53.4, 126.3)    |                             |
| [34] 99 Pro (-110.1, -179.1) |                             |
| [34] 155 Asp (119.2, -13.1)  |                             |
| [34] 275 Gly (50.8, 179.2)   |                             |
| [34] 277 Asn (-15.1, 93.3)   |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 35



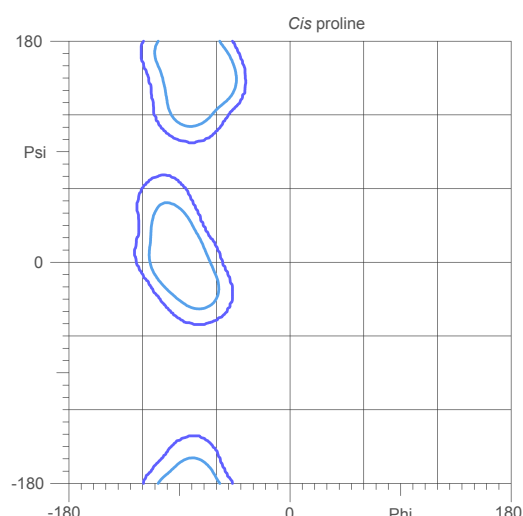
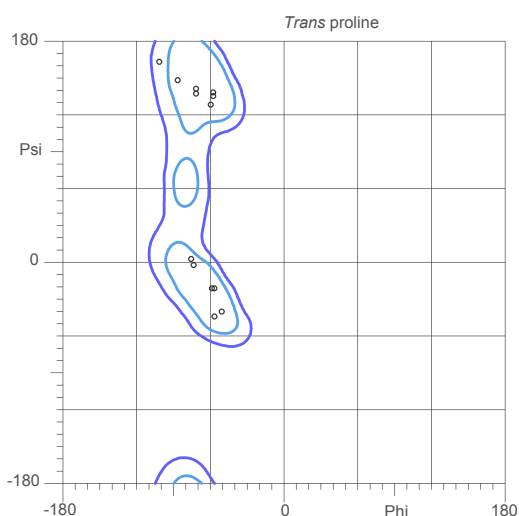
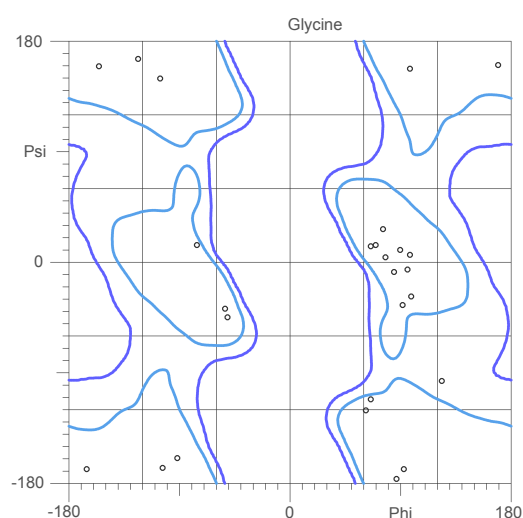
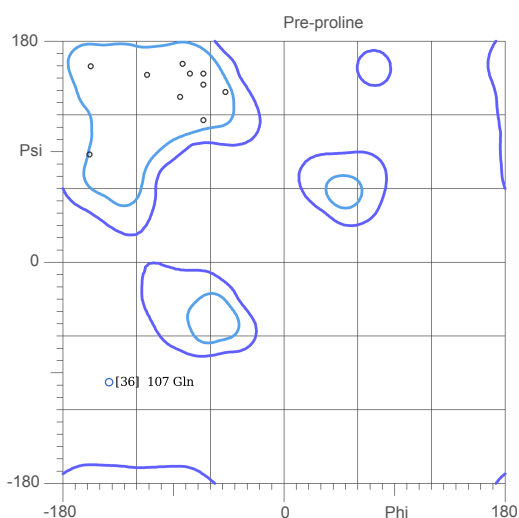
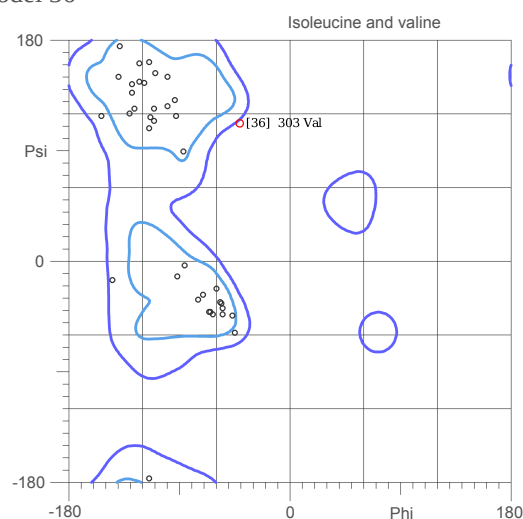
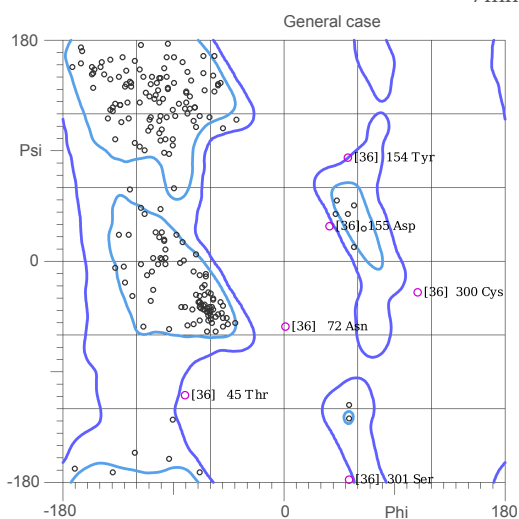
87.8% (267/304) of all residues were in favored (98%) regions.  
96.7% (294/304) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

|                              |                             |
|------------------------------|-----------------------------|
| [35] 45 Thr (-71.9, -151.4)  | [35] 304 Thr (-55.7, 83.5)  |
| [35] 59 Ile (-46.1, -13.8)   | [35] 305 Phe (-61.5, -80.4) |
| [35] 106 Ile (148.1, 104.5)  |                             |
| [35] 107 Gln (-179.3, 1.0)   |                             |
| [35] 221 Asn (-25.4, -164.3) |                             |
| [35] 223 Phe (-14.0, -163.8) |                             |
| [35] 301 Ser (-13.8, 164.8)  |                             |
| [35] 303 Val (-81.8, 30.4)   |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 36



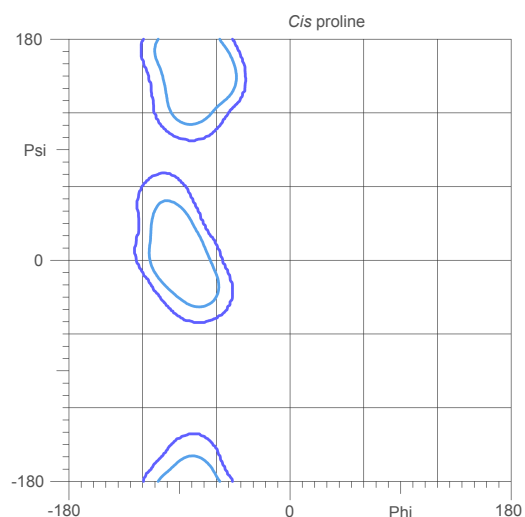
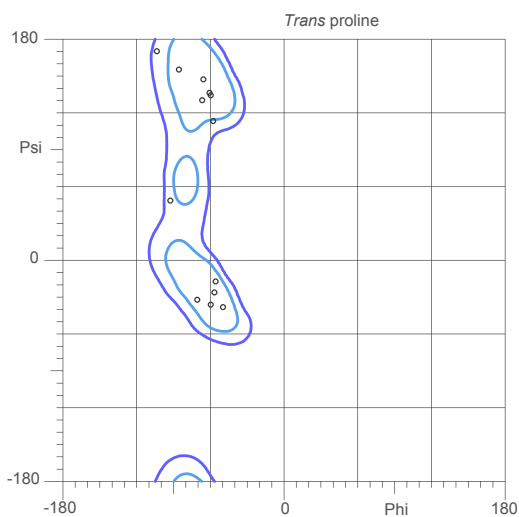
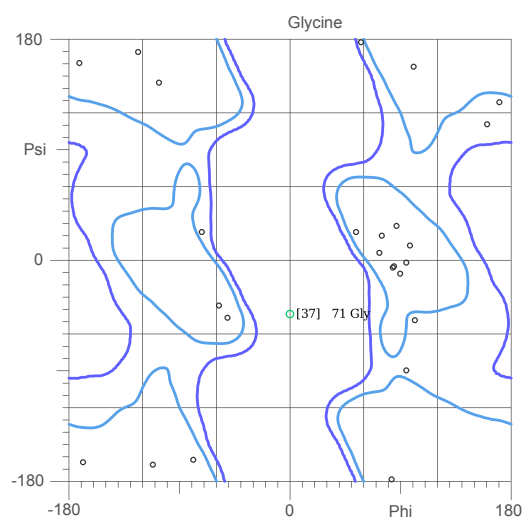
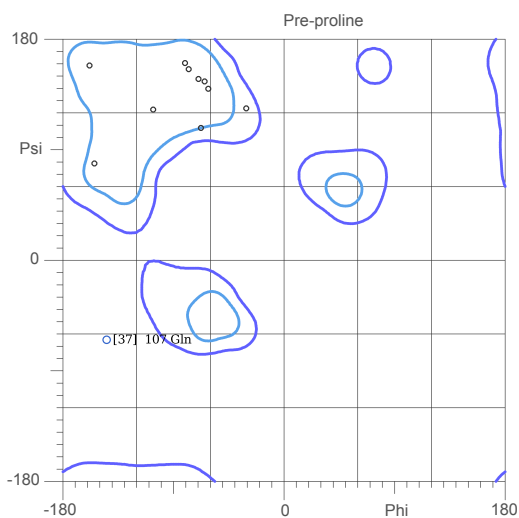
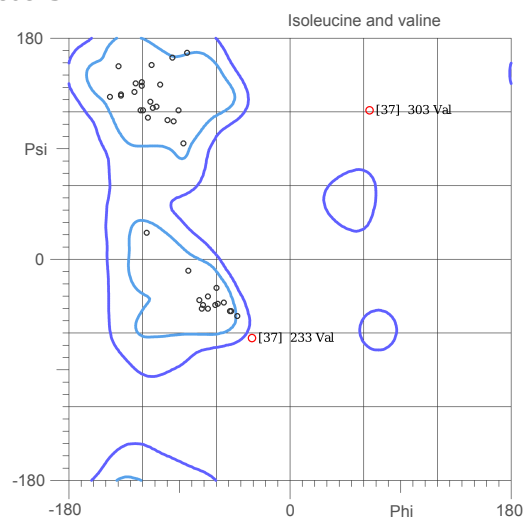
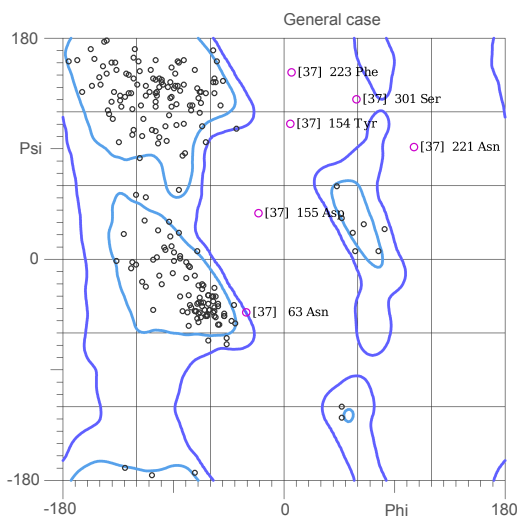
90.8% (276/304) of all residues were in favored (98%) regions.  
97.4% (296/304) of all residues were in allowed (>99.8%) regions.

There were 8 outliers (phi, psi):

[36] 45 Thr (-81.4, -109.6)  
[36] 72 Asn (1.5, -53.3)  
[36] 107 Gln (-143.3, -99.0)  
[36] 154 Tyr (52.9, 85.3)  
[36] 155 Asp (37.3, 29.3)  
[36] 300 Cys (109.8, -25.2)  
[36] 301 Ser (53.7, -178.4)  
[36] 303 Val (-41.7, 113.0)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 37



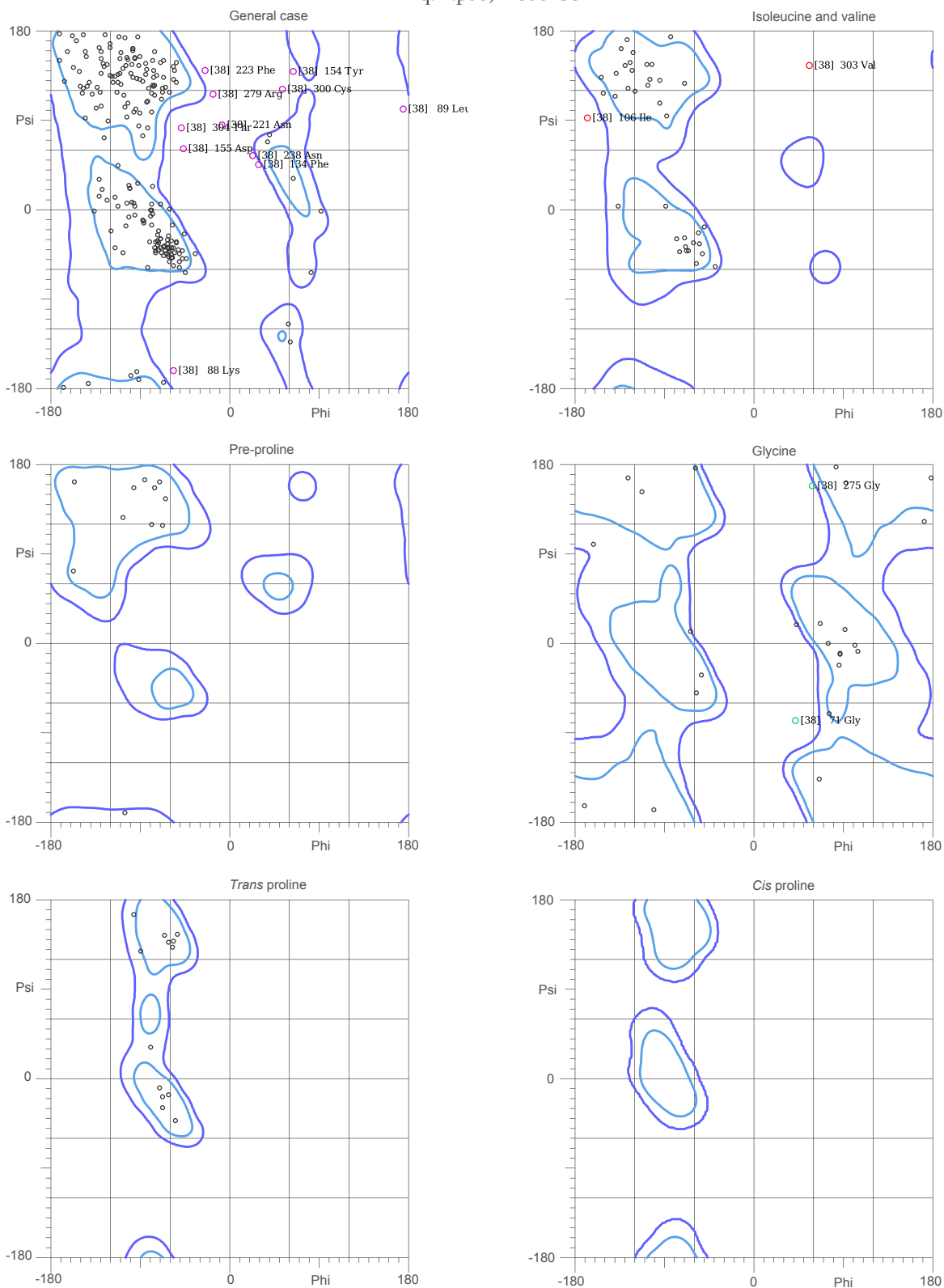
86.5% (263/304) of all residues were in favored (98%) regions.  
96.7% (294/304) of all residues were in allowed (>99.8%) regions.

There were 10 outliers (phi, psi):

|                              |                            |
|------------------------------|----------------------------|
| [37] 63 Asn (-31.0, -43.5)   | [37] 301 Ser (59.1, 131.7) |
| [37] 71 Gly (0.1, -44.2)     | [37] 303 Val (65.3, 122.1) |
| [37] 107 Gln (-145.1, -65.5) |                            |
| [37] 154 Tyr (5.7, 111.4)    |                            |
| [37] 155 Asp (-22.0, 38.5)   |                            |
| [37] 221 Asn (106.9, 93.0)   |                            |
| [37] 223 Phe (6.9, 153.1)    |                            |
| [37] 233 Val (-31.6, -64.8)  |                            |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 38



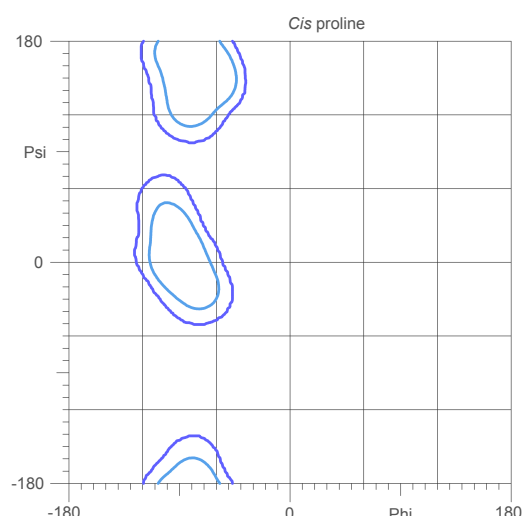
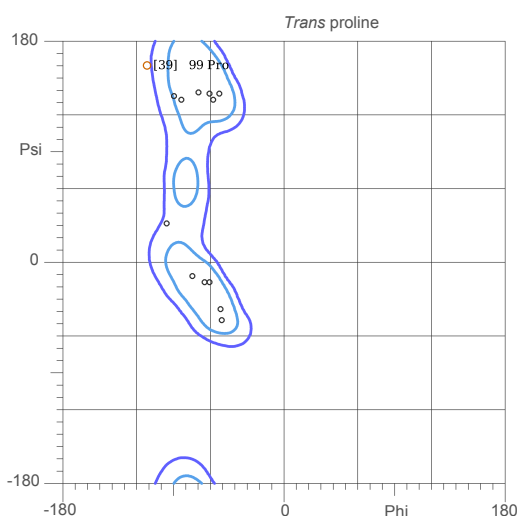
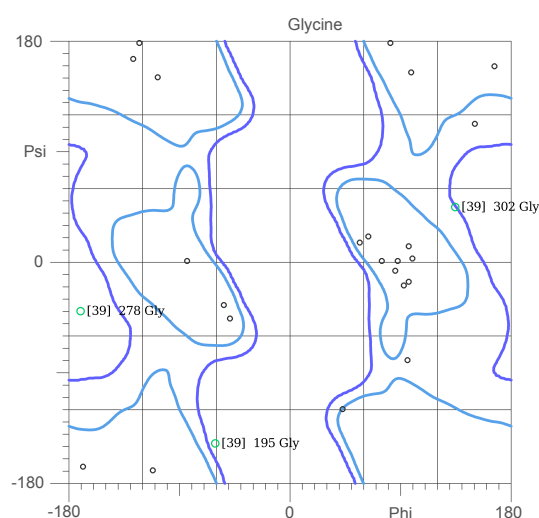
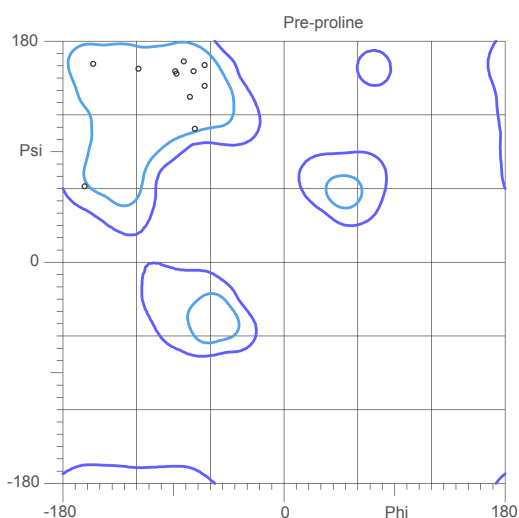
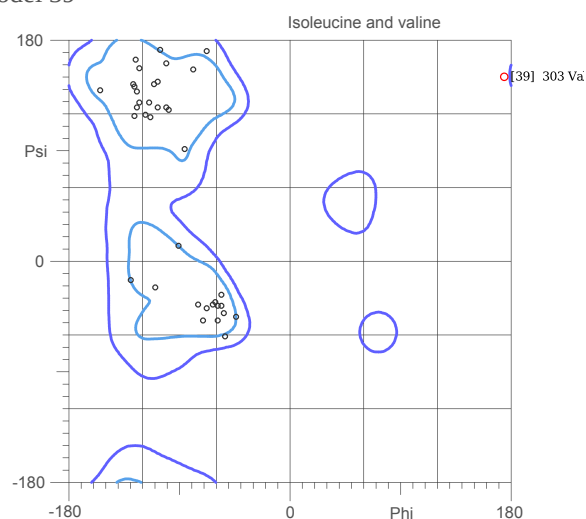
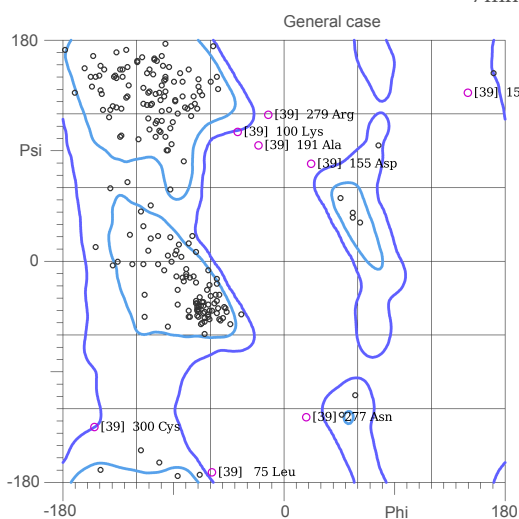
84.9% (258/304) of all residues were in favored (98%) regions.  
95.1% (289/304) of all residues were in allowed (>99.8%) regions.

There were 15 outliers (phi, psi):

|                             |                             |
|-----------------------------|-----------------------------|
| [38] 71 Gly (42.3, -78.1)   | [38] 223 Phe (-25.3, 141.2) |
| [38] 88 Lys (-57.1, -162.7) | [38] 238 Asn (23.0, 55.3)   |
| [38] 89 Leu (175.6, 102.4)  | [38] 275 Gly (59.1, 159.2)  |
| [38] 106 Ile (-168.2, 93.8) | [38] 279 Arg (-17.5, 117.8) |
| [38] 134 Phe (29.7, 46.9)   | [38] 300 Cys (53.4, 122.4)  |
| [38] 154 Tyr (64.1, 140.6)  | [38] 303 Val (56.2, 146.2)  |
| [38] 155 Asp (-47.7, 62.3)  | [38] 304 Thr (-50.0, 83.1)  |
| [38] 221 Asn (-9.0, 86.1)   |                             |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 39



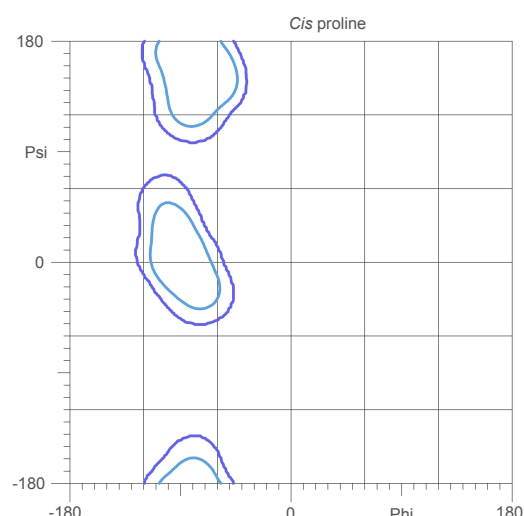
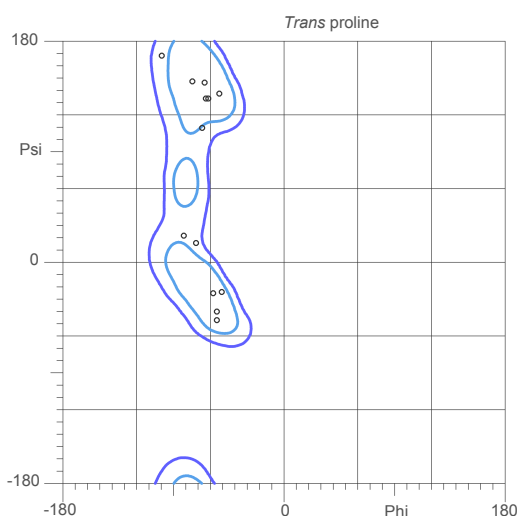
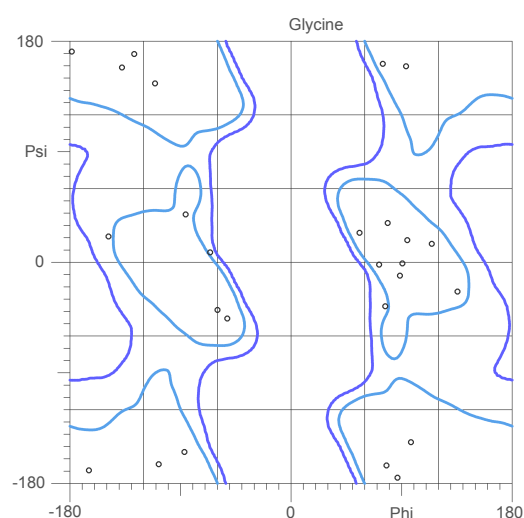
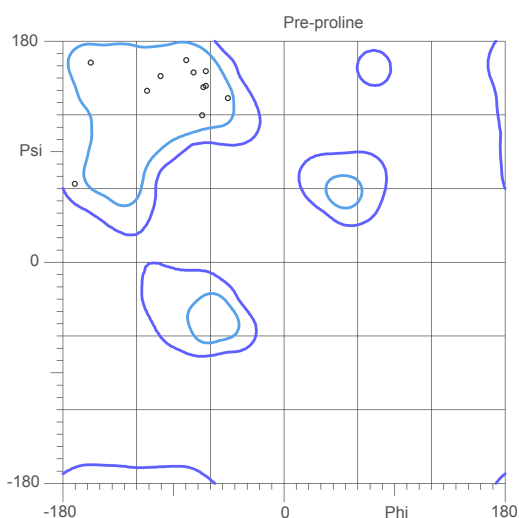
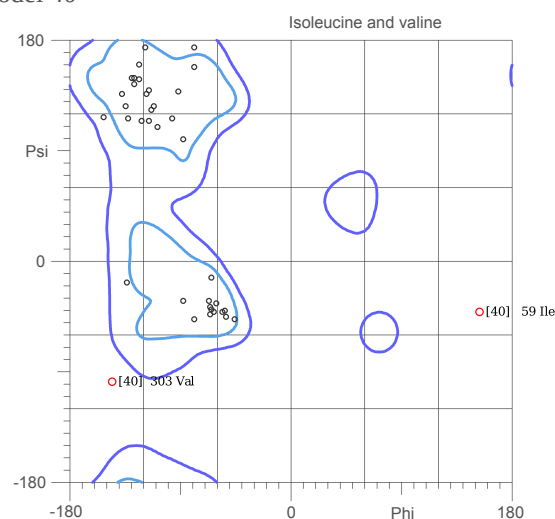
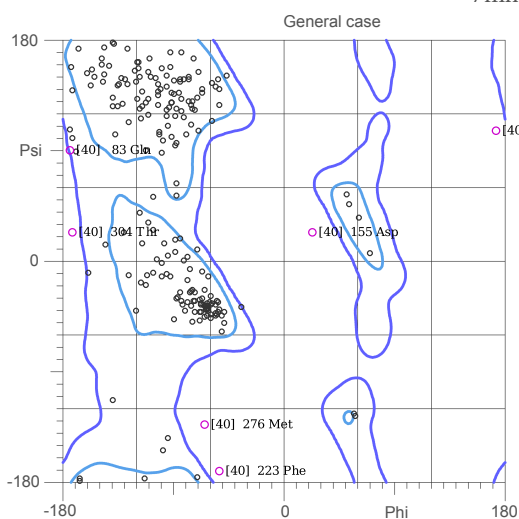
86.2% (262/304) of all residues were in favored (98%) regions.  
95.7% (291/304) of all residues were in allowed (>99.8%) regions.

There were 13 outliers (phi, psi):

|                              |                               |
|------------------------------|-------------------------------|
| [39] 75 Leu (-59.8, -172.8)  | [39] 278 Gly (-171.9, -40.5)  |
| [39] 99 Pro (-112.1, 161.4)  | [39] 279 Arg (-13.6, 121.0)   |
| [39] 100 Lys (-38.2, 106.1)  | [39] 300 Cys (-155.7, -135.6) |
| [39] 155 Asp (22.3, 80.3)    | [39] 302 Gly (135.3, 45.7)    |
| [39] 156 Cys (150.4, 138.5)  | [39] 303 Val (175.1, 151.9)   |
| [39] 191 Ala (-21.4, 95.7)   |                               |
| [39] 195 Gly (-61.9, -148.9) |                               |
| [39] 277 Asn (18.5, -127.7)  |                               |

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 40



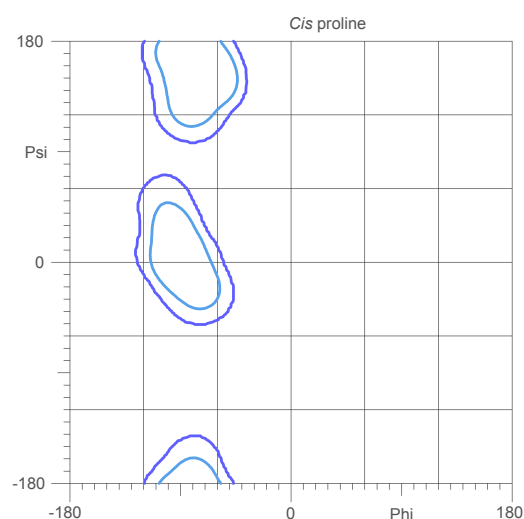
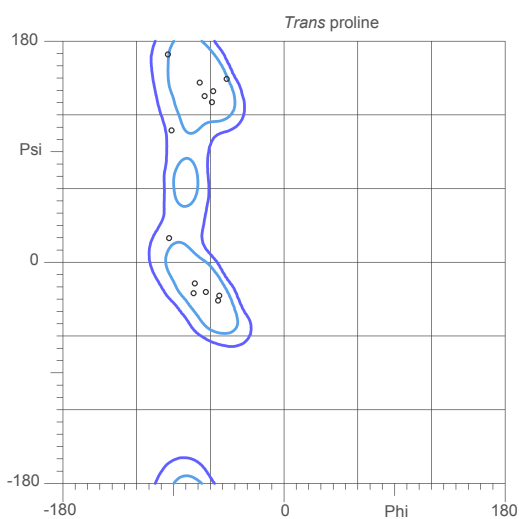
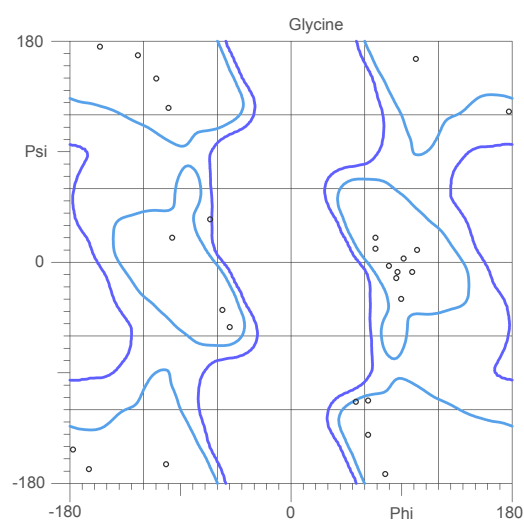
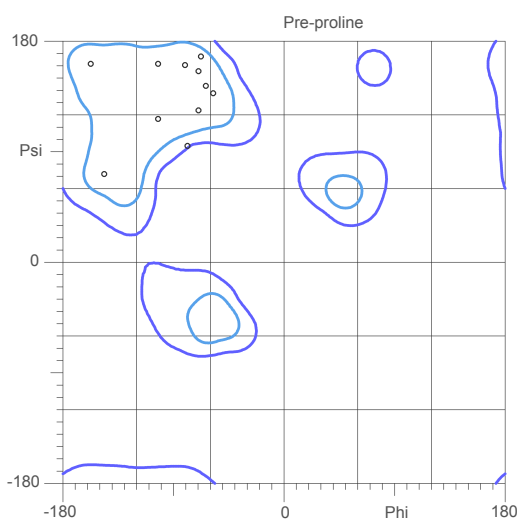
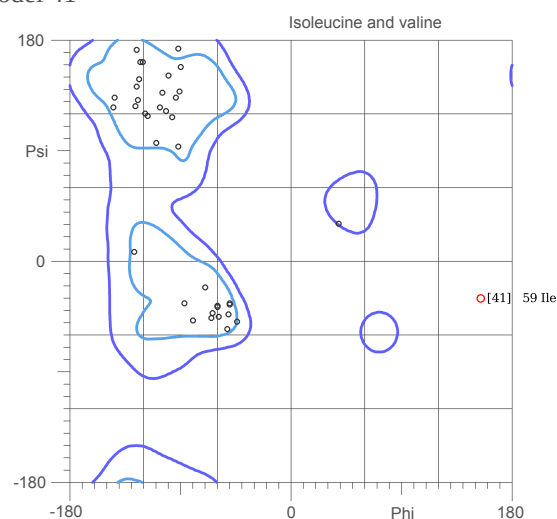
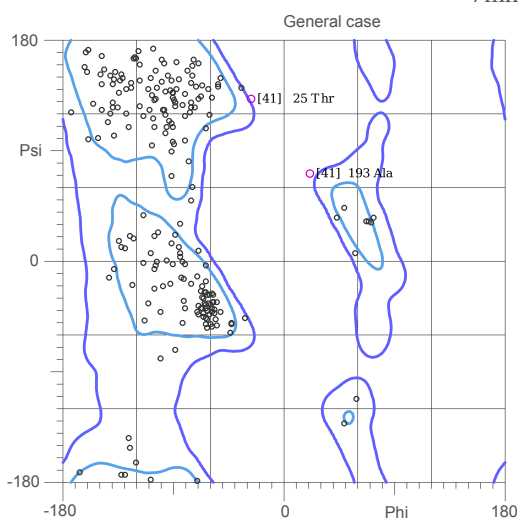
88.8% (270/304) of all residues were in favored (98%) regions.  
97.4% (296/304) of all residues were in allowed (>99.8%) regions.

There were 8 outliers (phi, psi):

- [40] 59 Ile (154.3, -41.9)
- [40] 83 Gln (-175.8, 91.4)
- [40] 155 Asp (23.1, 24.3)
- [40] 194 Ala (173.6, 107.7)
- [40] 223 Phe (-53.9, -171.9)
- [40] 276 Met (-65.3, -133.1)
- [40] 303 Val (-146.5, -98.7)
- [40] 304 Thr (-173.2, 24.4)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 41



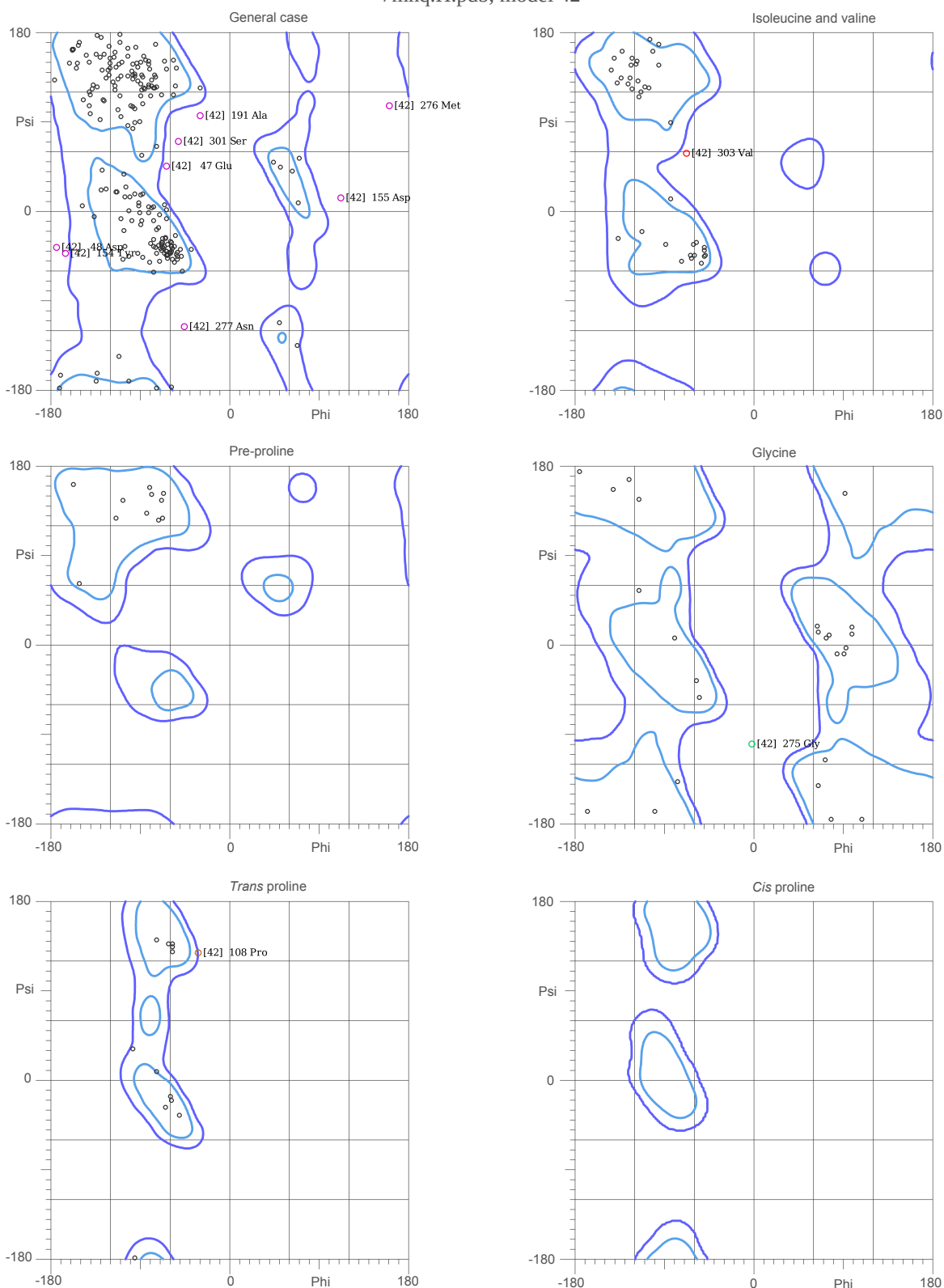
89.5% (272/304) of all residues were in favored (98%) regions.  
99.0% (301/304) of all residues were in allowed (>99.8%) regions.

There were 3 outliers (phi, psi):

[41] 25 Thr (-27.2, 133.6)  
[41] 59 Ile (155.3, -30.1)  
[41] 193 Ala (21.2, 72.2)

# MolProbity Ramachandran analysis

7mhq.H.pdb, model 42



88.2% (268/304) of all residues were in favored (98%) regions.  
96.4% (293/304) of all residues were in allowed (>99.8%) regions.

There were 11 outliers (phi, psi):

|                              |                              |
|------------------------------|------------------------------|
| [42] 47 Glu (-64.2, 46.0)    | [42] 277 Asn (-46.0, -116.2) |
| [42] 48 Asp (-175.5, -36.3)  | [42] 301 Ser (-52.8, 71.4)   |
| [42] 108 Pro (-32.2, 130.0)  | [42] 303 Val (-68.3, 59.2)   |
| [42] 154 Tyr (-166.4, -42.9) |                              |
| [42] 155 Asp (112.4, 14.7)   |                              |
| [42] 191 Ala (-30.8, 97.4)   |                              |
| [42] 275 Gly (-2.7, -100.9)  |                              |
| [42] 276 Met (161.9, 107.8)  |                              |